Modeling the Unexpected Light Curves of IMBH-TDE EP240222a

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https://li-wenkai.github.io/

W. Li et al. arXiv: 2512.02147

Inefficient Circularization, Delayed Stream–Disk Interaction and Reprocessing: A Five–Stage Model for the IMBH–TDE EP240222a

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MS-IMBH-TDE: Slow Rise
WD-IMBH-TDE: Fast Rise
Stream-Disk Interaction: Turning Point

中国科学技术大学
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Self-Introduciton

Wenkai Li (李文楷)

Research Interst:

Various Astrophysical Phenomena :D

Hobby:

Star Party, Travel, Daily Photography
All Kinds of Sports: Cue Sports, Armwrestling etc. (after my ACL injury)

USTC Undergraduate

Advisor: Ning Jiang, Ann Zabludoff

Current Research Focuses:

Tidal Disruption Events (TDEs)
Intermediate-mass Black Holes (IMBHs)
Post-Starburst Galaxies (PSBs)

Astronomy Photographer of the Year (shortlist, 2017)

JUPITER AND ITS MOONS

Kevin Li (China), aged 14
Shenzhen, Guangdong Province, China, 4 April 2017

I took this photo when Jupiter was in a favourable opposition on 4 April, on the balcony of my home in Shenzhen, China. Jupiter is one of my favourite planets, not only because of its changeable cloud band and the Great Red Spot but also because of its four famous moons. The buildings in front of my balcony were a big challenge for this photograph. My parents were also 'big challenges' for me, since I had to go to school the next day, so they wanted me to go to bed early! The seeing was great, so I used a Barlow 3X lens with a ZWO ASI120MM camera on my Sky-Watcher BKP 150/750 telescope. The Sky-Watcher EQ3-D mount can't 'go-to', so I had to find Jupiter manually. Because I used a monochrome camera, I used LRGB (luminance, red, green and blue) filters in order to show colour. For processing, I used AutoStakkert!2 to stack them. Then, I used Registax 6 to sharpen the image. After that, I used Astra Image to combine the LRGB. I also used it to wavelet sharpen and de-noise. I processed Jupiter and its moons individually. Finally, I used Adobe Photoshop to combine Jupiter and its moons. I love this photo because if you look at the picture carefully, you can see that the Jupiter and its moons are colourful!

Sky-Watcher 150 mm f/5 reflector telescope at f/15, Barlow x3 lens, Sky-Watcher EQ3-D mount, ZWO ASI120MM camera, stacked from multiple exposures

RAY CRATER LANGRENUS

Kevin Li (China), aged 14
Shenzhen, Guangdong Province, China, 1 April 2017

I took this photo during a clear dusk on the balcony of my home in Shenzhen, China. The Sun had gone down just about an hour ago. I had to take the photo so early because the Moon was about to be blocked by skyscrapers! So I prepared quickly and started to take the picture at once. The Moon is the closest celestial body to Earth. It is fantastic and mysterious! Although I have observed it many times, I will never grow tired of it. Luckily, the seeing was fantastic at that time, so I tried to use Barlow 2X and Barlow 3X lenses together on my telescope for the first time! I shot about 2,000 frames and used AutoStakkert!2 to align them, and chose about 50% to stack. Then, I used Registax 6 to sharpen the image. After that, I used Astra Image to wavelet sharpen and de-noise. In addition, Adobe Photoshop CS5 was used to lift brightness. Because the focal length was large and the seeing was good, the details of the Ray Crater Langrenus seem very impressive. I will try to take a far more detailed picture next time!

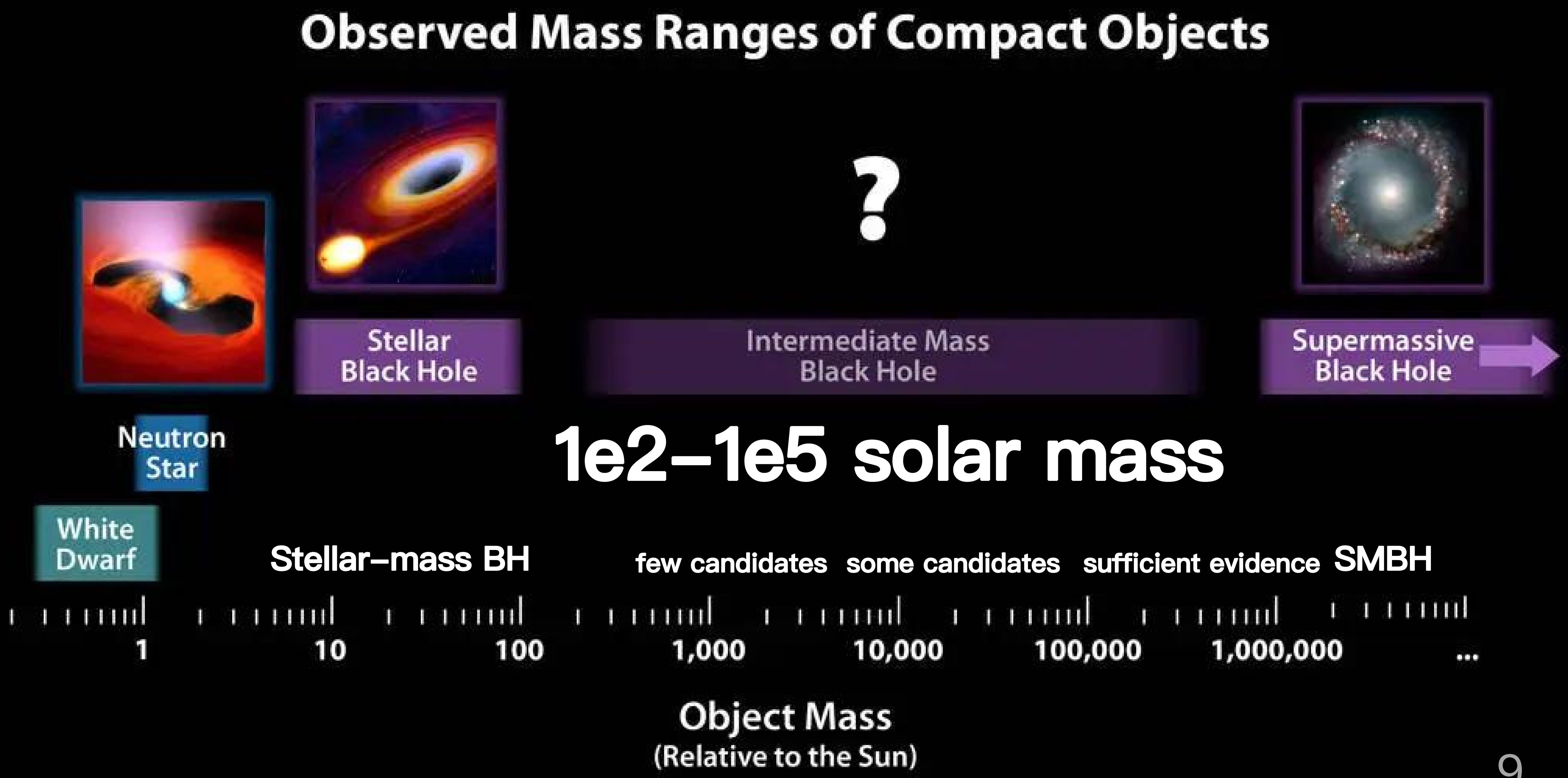
Sky-Watcher 150 mm f/5 reflector telescope at f/30, Sky-Watcher EQ3-D mount ZWO ASI120MM camera, stacked from multiple exposures

Content

- ABC for IMBHs
- ABC for TDEs
- IMBH-TDEs
- IMBH-TDE (EP240222a) Model
- Summary

ABC for IMBHs

An important and rarely found class of black holes (BHs).

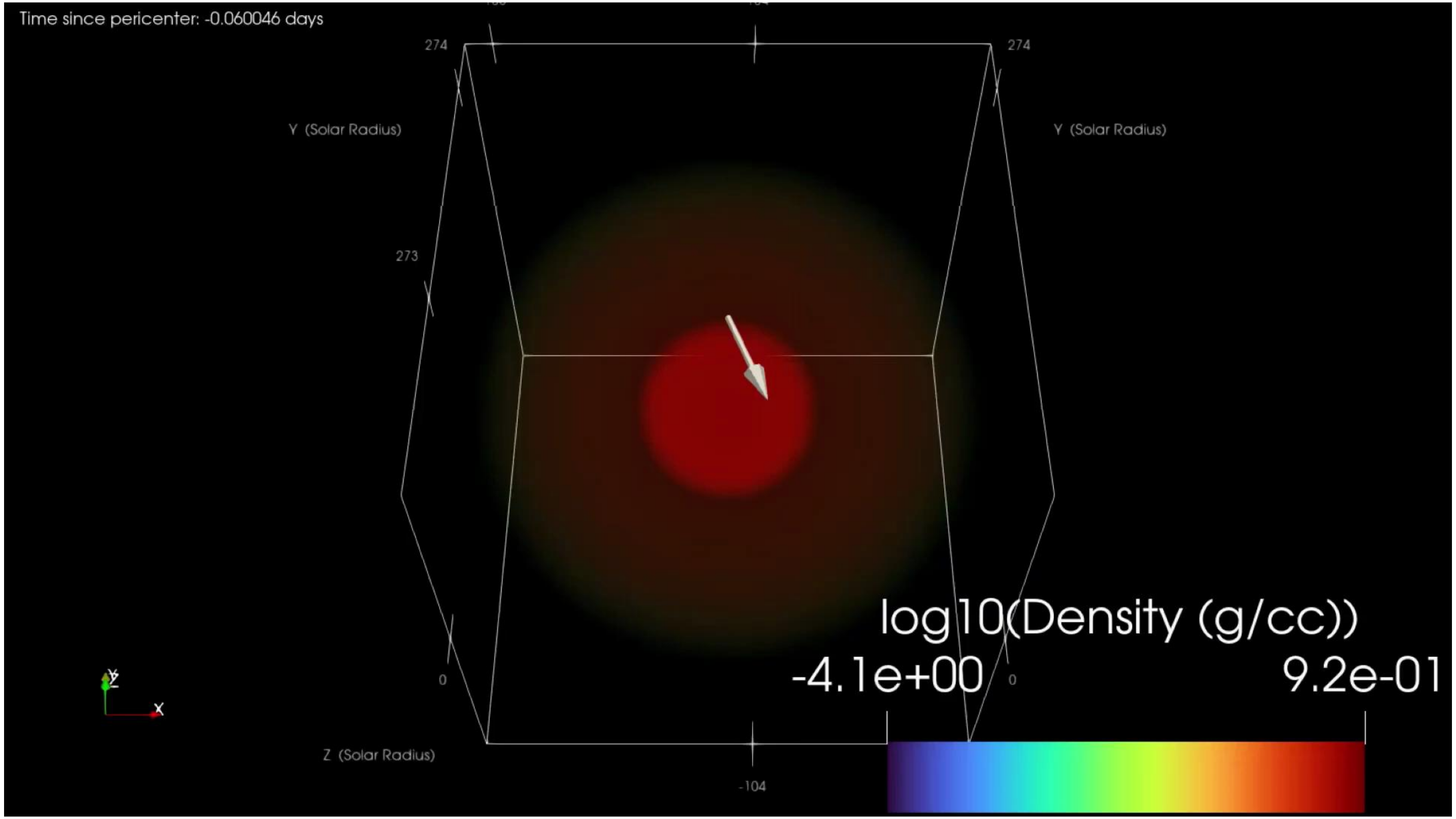


credit: NASA/JPL-Caltech

ABC for TDEs

A star disrupted by a black hole (BH).

Intuitive Simulation Animation



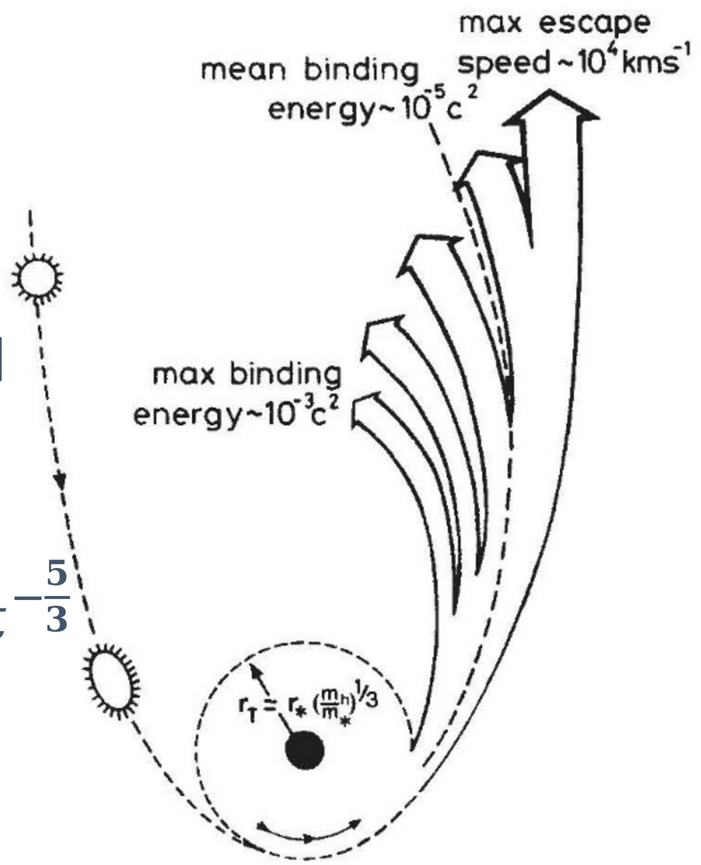
E. Steinberg & N. Stone, 2024

Classical Model

1. Stellar scattered to SMBH (loss cone)
2. Tidal stretching & compression
3. Pericenter radius < tidal radius → disruption
4. Near–parabolic → ~50% bound, ~50% unbound
5. Quick circularization? & quick accretion?
6. $L \propto \dot{M}_{acc} \propto \dot{M}_{fb}$ & $\frac{dE}{dM} = Const \rightarrow L \propto t^{-\frac{5}{3}}$

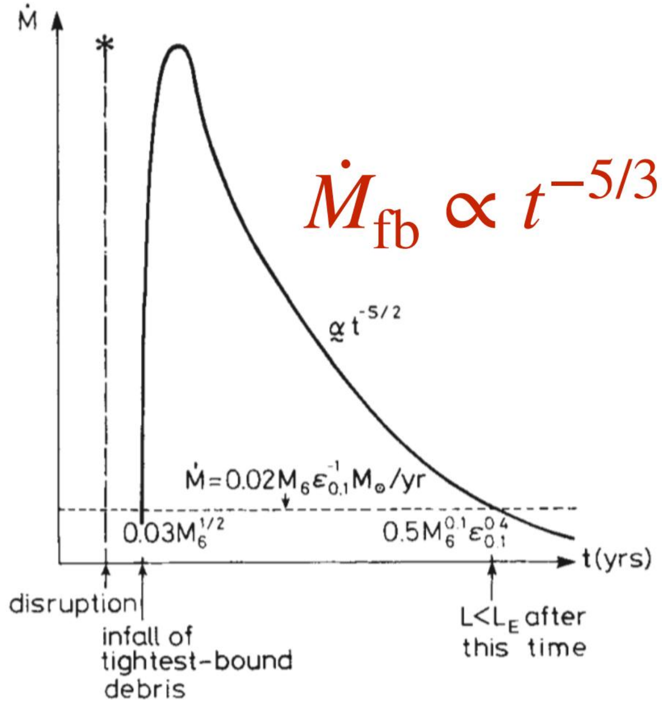
(timescale: months—years)

7. Multi–color BB disk? → (UV/soft X–ray)

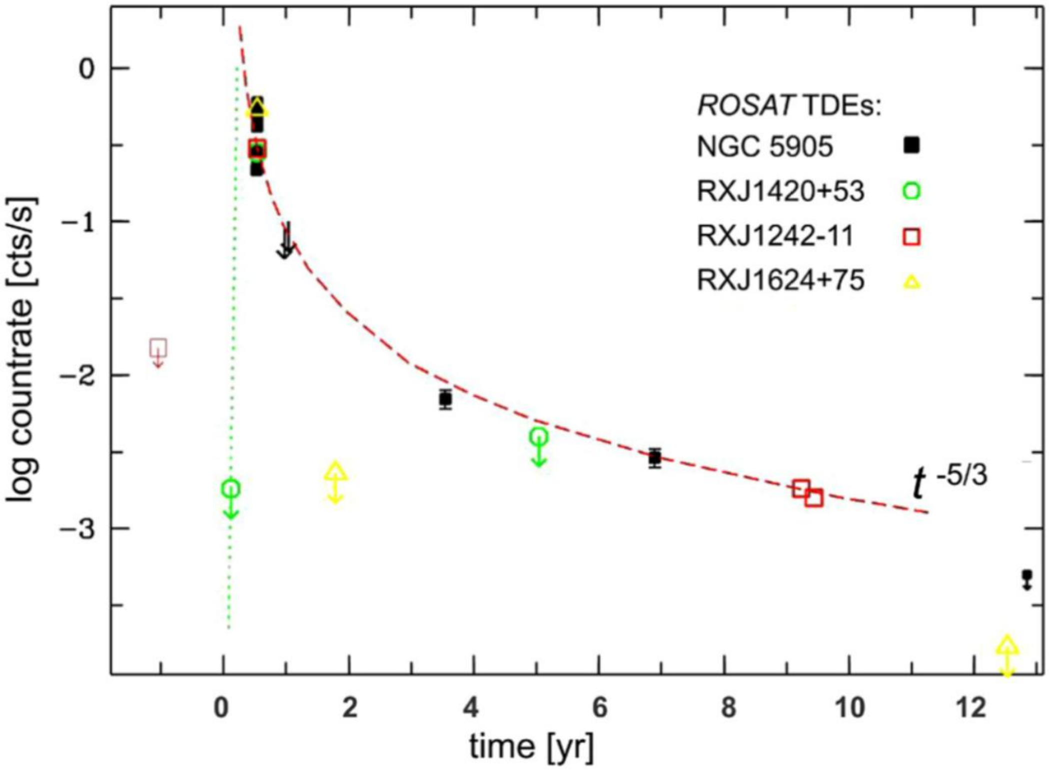


M. Rees, 1988

Classical Model

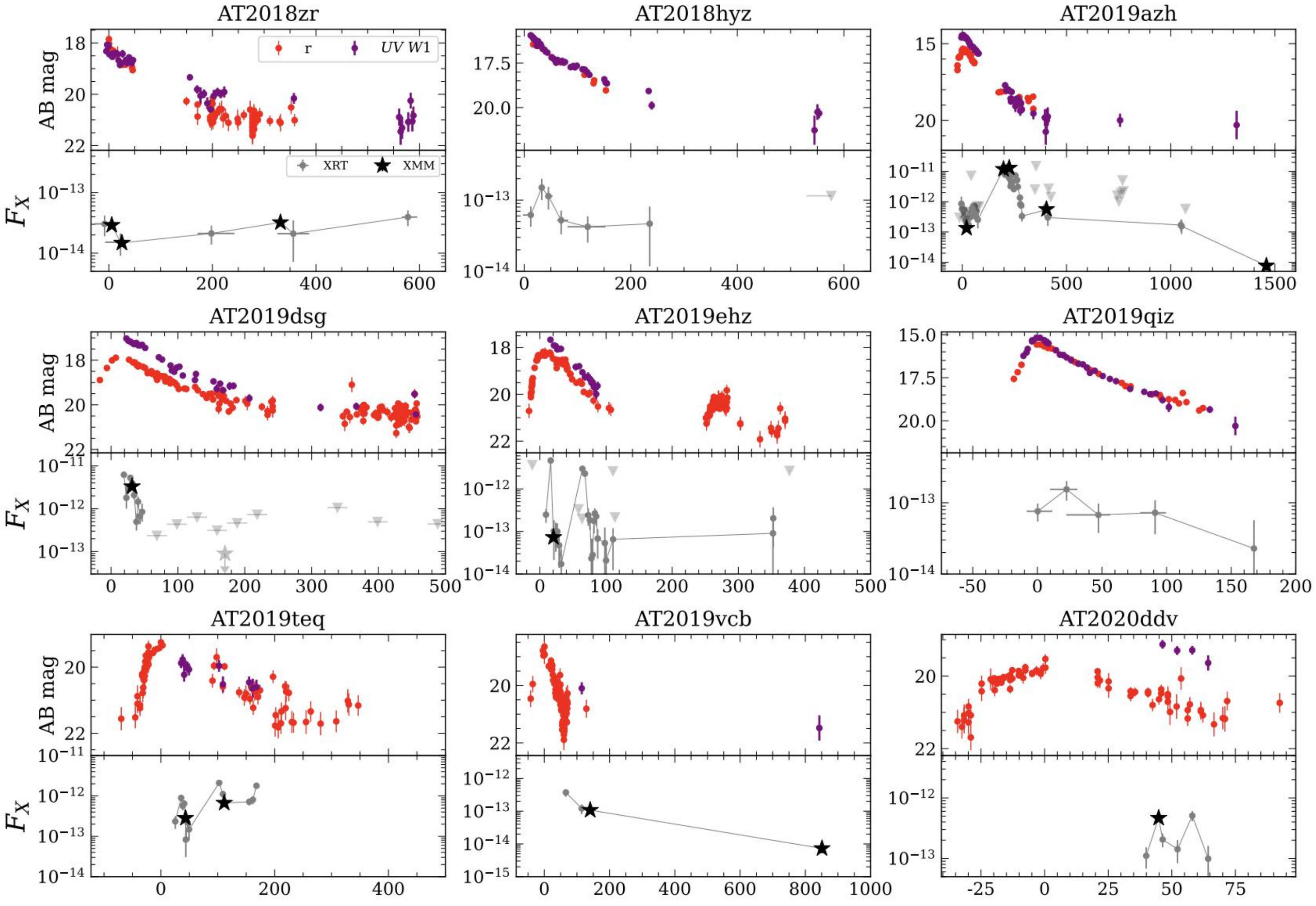


M. Rees, 1988



S. Komossa, 2015

Real Light Curves (Intuitive Impression)

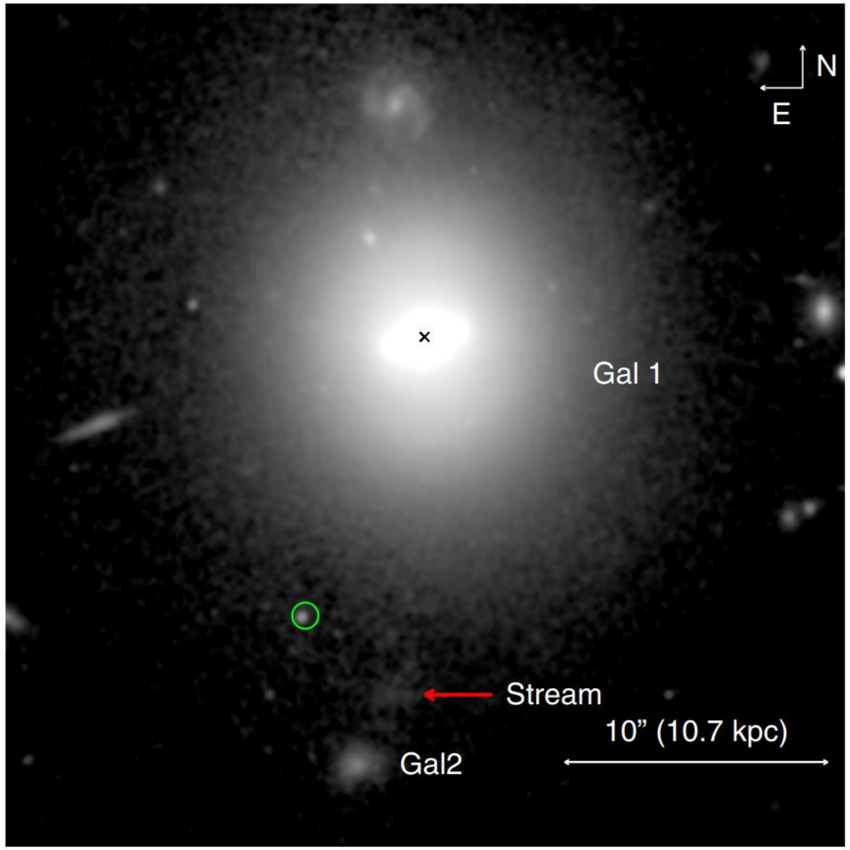


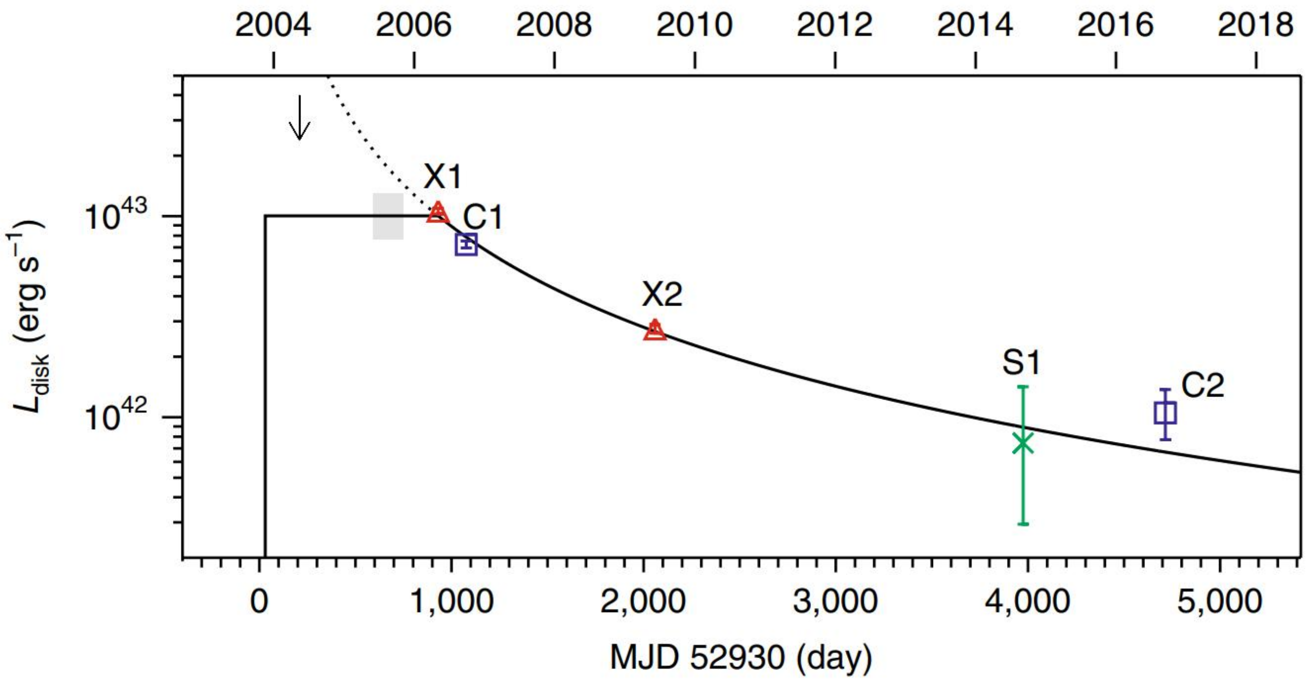
M. Guolo et al. 2024

IMBH-TDEs

A very new and interesting field.

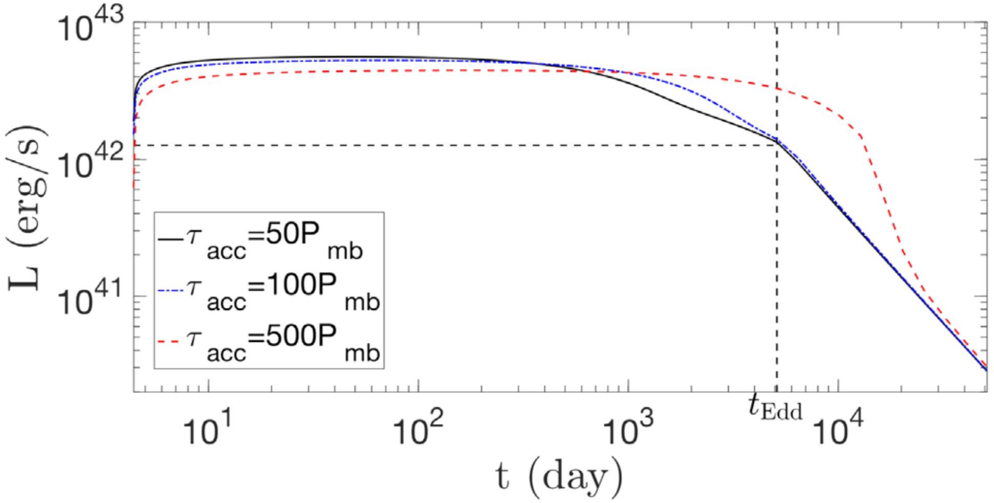
IMBH-TDE Candidate: 3XMM J215022.4−055108

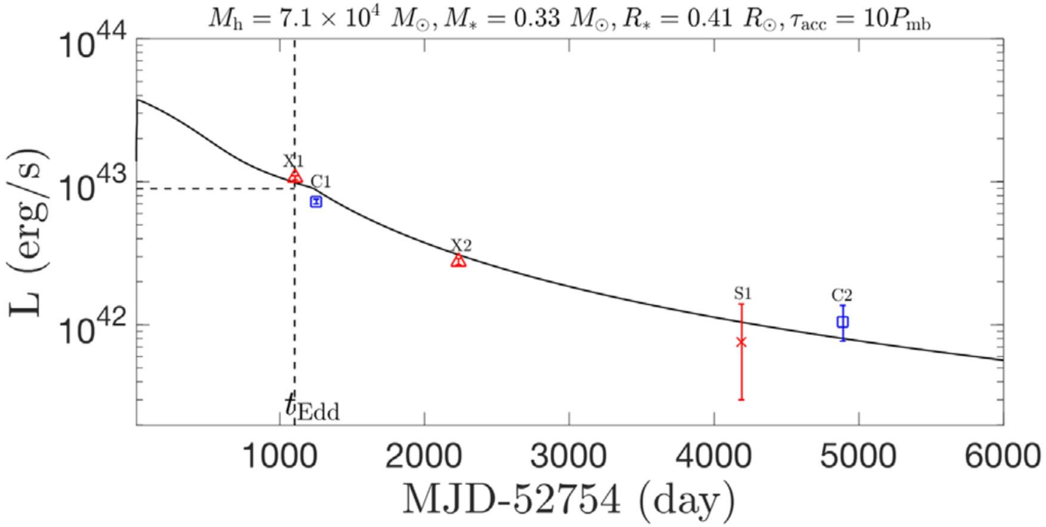




D. Lin et al. 2017

IMBH-TDE Candidate: 3XMM J215022.4−055108





$$\frac{dM_d(t)}{dt} + \dot{M}_{acc}(t) = \dot{M}_{fb}(t)$$

$$\Downarrow$$

$$\dot{M}_{acc}(t) = \frac{M_d(t)}{t_{acc}} = \frac{1}{t_{acc}}\left(e^{-t/t_{acc}} \int_{t_{fb}}^{t} e^{t'/t_{acc}} \dot{M}_{fb}(t')\, dt'\right)$$

$$L_{bol}(t) \simeq \begin{cases} \left[1 + \ln\left(\dot{M}_{acc}(t)/\dot{M}_{Edd}\right)\right] L_{Edd}, & \dot{M}_{acc}(t) \gtrsim \dot{M}_{Edd} \\ \left(\dot{M}_{acc}(t)/\dot{M}_{Edd}\right) L_{Edd}, & \dot{M}_{acc}(t) \lesssim \dot{M}_{Edd} \end{cases}$$

- **~ 200 eV**
- **Candidate from historical data**
- **No early stage light curves**
- **No spectroscopic confirmation**

J. Chen & R. Shen 2018

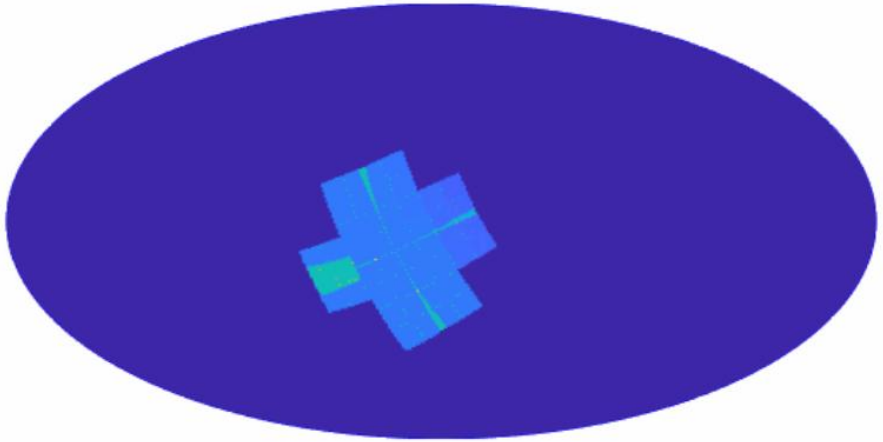
Einstein Probe: X-ray Time-domain Telescope

➢ WXT (0.5–4 keV)

- FOV: 3600 degrees
- Orbit: 600km (5800s/orbit)
- 3 pointings/orbit, ~1200s each
- ~ 1/2 sky covered in 3 orbits (~ 5 hr)
- ~6ks exposure/sky region/day

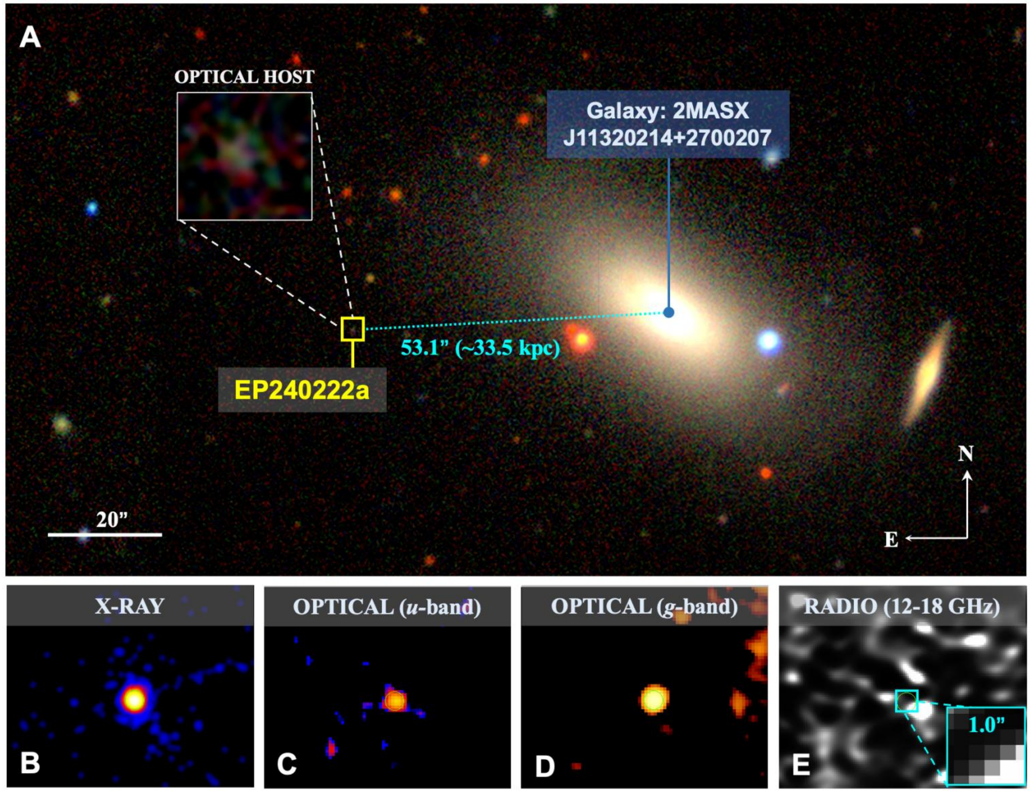
➢ FXT (0.3–10 keV)

- 2 telescopes
- Similar to eROSITA



credit: Jingwei Hu

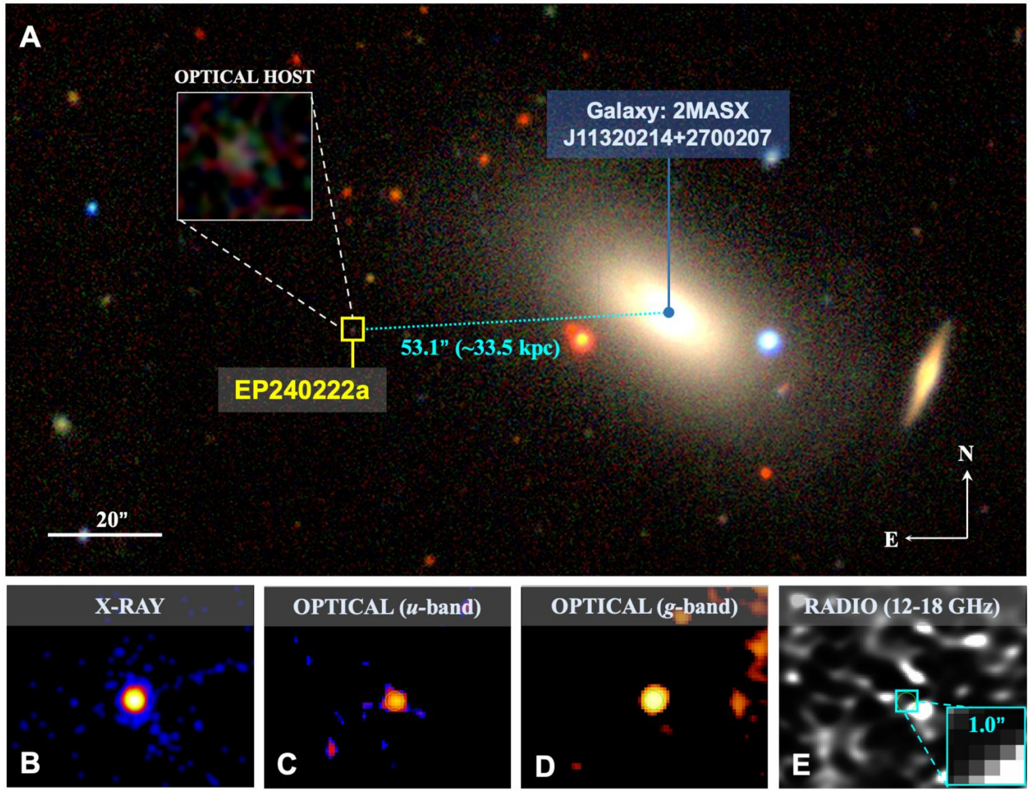
EP240222a: A Confirmed IMBH-TDE

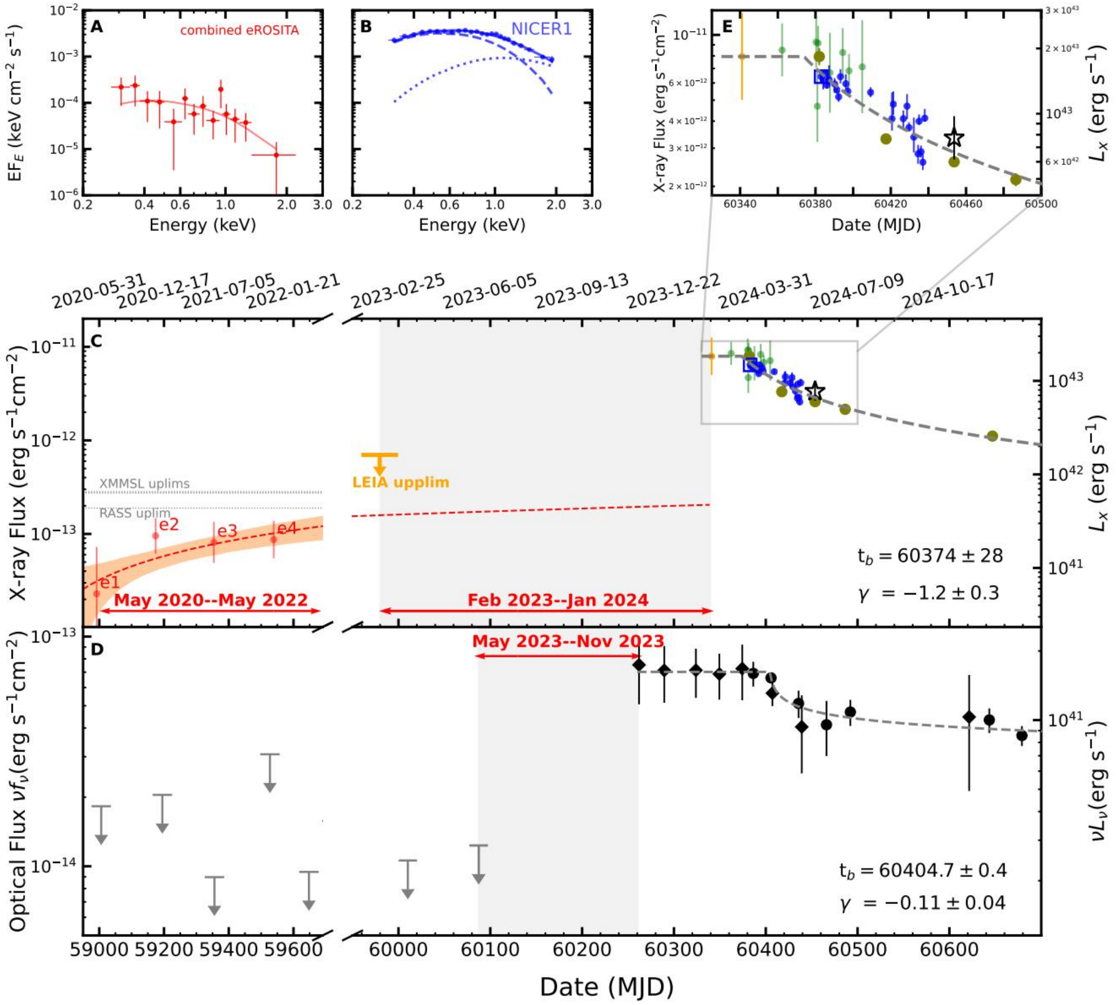


- **Real time captured**
- **Multi-wavelength observed**
- **Spectroscopic confirmed**

C. Jin et al. arXiv: 2501.09580

EP240222a: A Confirmed IMBH-TDE





C. Jin et al. arXiv: 2501.09580

Why IMBH?

- ➢ **Host (dwarf galaxy / globular cluster) Luminosity –> BH Mass**
- ➢ **X–ray SED Fitting**
  - **Blackbody temerature ~160eV**
  - **Inner disk temerature ~200eV**
- ➢ **Similar to XMMJ2150 (IMBH–TDE candidate)**
- ➢ **Long Rise (year timescale) –> Inefficient Circularization –> IMBH**

C. Jin et al. arXiv: 2501.09580

EP240222a: What Can We Learn From Light Curves?

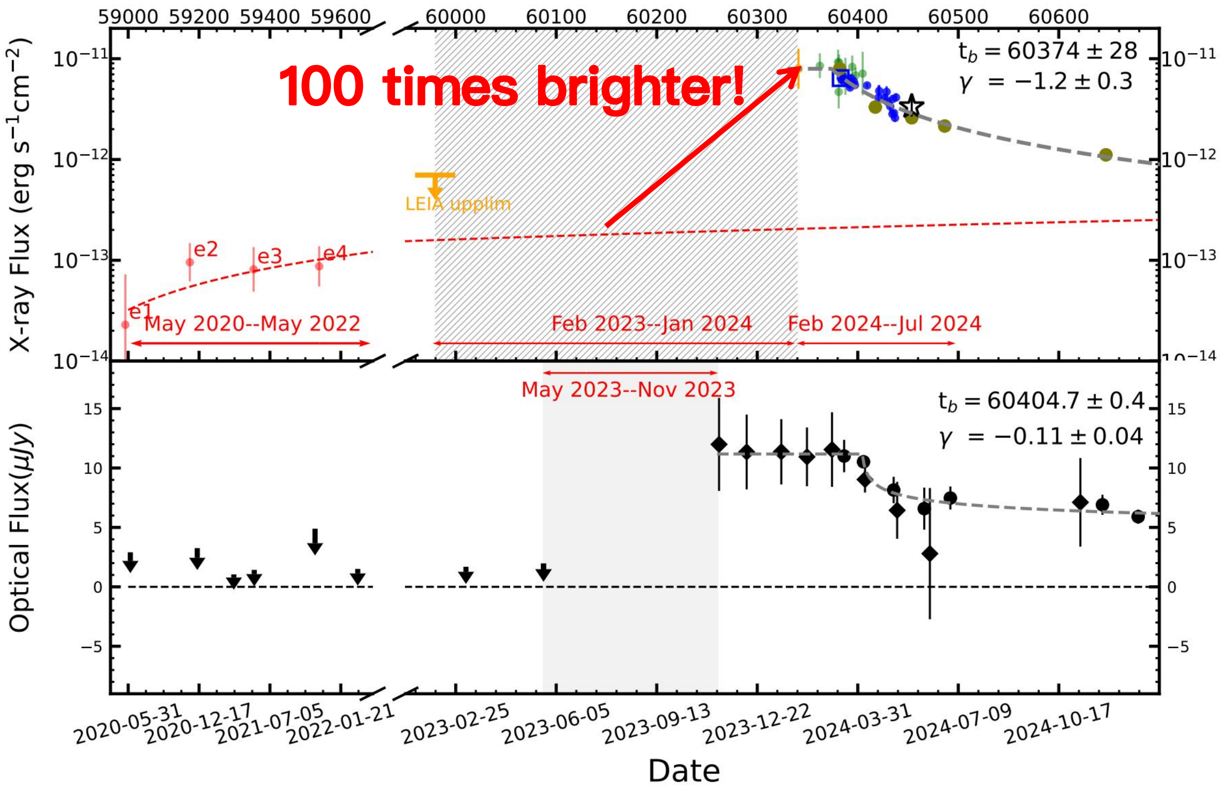
➢ **X-ray Emission**

- **Slow rising stage (~ 3 years)**
- **Turning point (slow rise -> fast rise)**
- **Plateau stage**
- **Decline stage**

➢ **Optical Emission**

- **Follow X-ray light curves**
- **200 times fainter than X-ray**
- **Delayed & slower decline compared to X-rays**

➢ **New Model Required!**



C. Jin et al. arXiv: 2501.09580

EP240222a Model

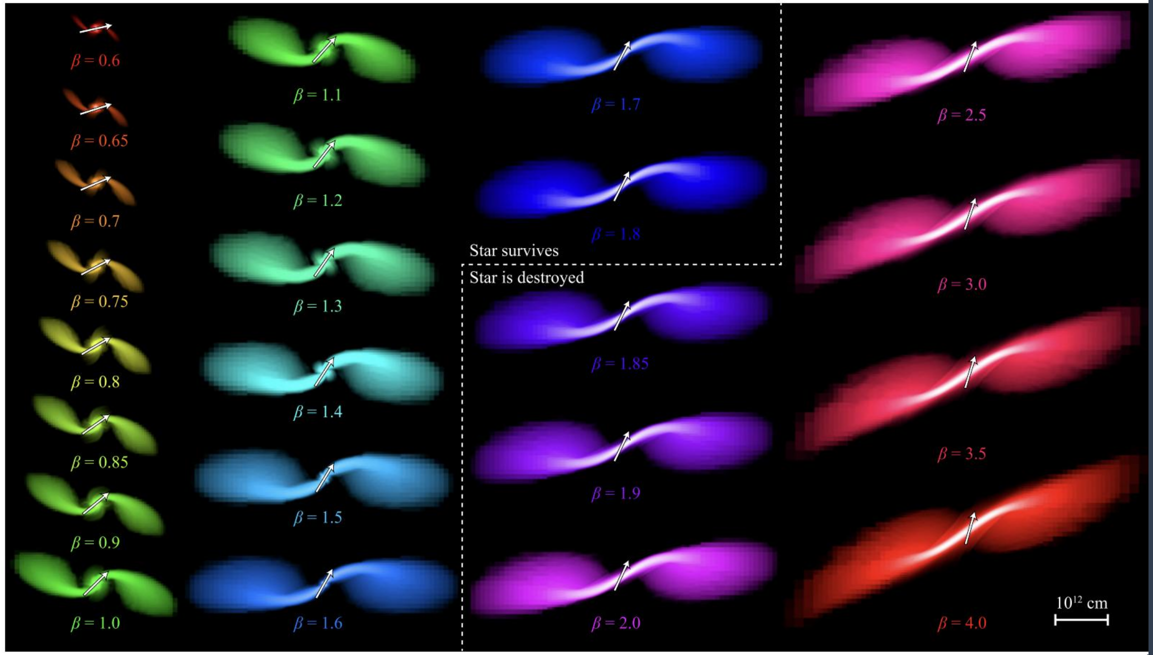
Inefficient circularization,

delayed stream-disk interaction

and reprocessing.

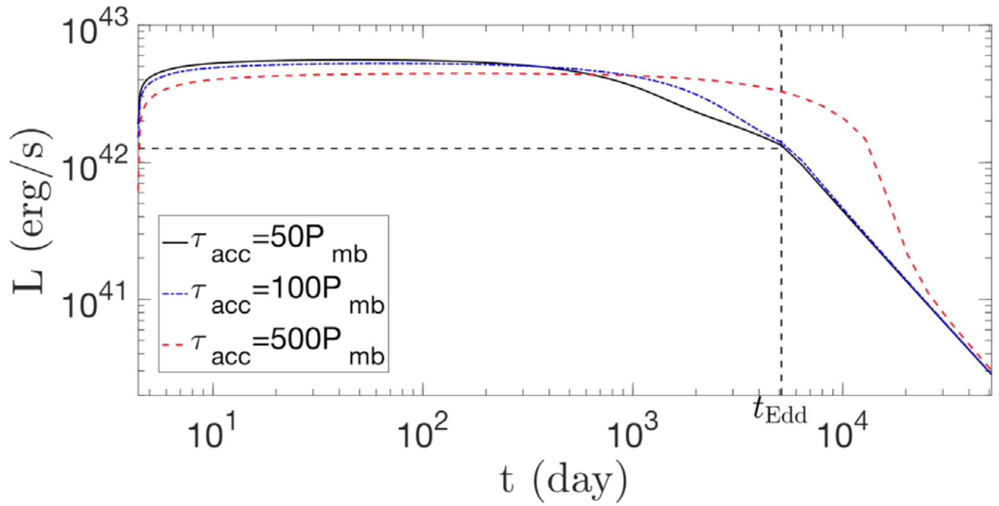
0. Disruption: Two Free Parameters

- **Stellar Parameters**
  - Stellar mass (M_*)
  - Stellar radius (R_*): M_*-R_* relation
  - Polytropic index (γ): depend on M_*
- **Orbital Parameter**
  - Penetration factor (β): R_t/R_p
- **Black Hole Parameters**
  - Black hole mass (M_{BH}): $\sim 7.7 * 10^4\ M_\odot$
  - Black hole spin (a_*): ~ 0.98



J. Guillochon & E. Ramirez-Ruiz 2013

Recall: Previous IMBH-TDE Model



$$\frac{dM_{\rm d}(t)}{dt} + \dot{M}_{\rm acc}(t) = \dot{M}_{\rm fb}(t)$$

$$L_{\rm bol}(t) \simeq \begin{cases} \left[1+\ln\left(\dot{M}_{\rm acc}(t)/\dot{M}_{\rm Edd}\right)\right] L_{\rm Edd}, & \dot{M}_{\rm acc}(t) \gtrsim \dot{M}_{\rm Edd} \\ \left(\dot{M}_{\rm acc}(t)/\dot{M}_{\rm Edd}\right) L_{\rm Edd}, & \dot{M}_{\rm acc}(t) \lesssim \dot{M}_{\rm Edd} \end{cases}$$

- **Fast Rise**
  - **Ignore circularization timescale**
- **Plateau**
  - **Super–Eddington stage**
- **Power law Decline**
  - **Trace fallback rate**

J. Chen & R. Shen 2018

Timescales

$$t_{\mathrm{fb}} = 2\pi\sqrt{\frac{a_0^3}{GM_{\mathrm{BH}}}} \simeq 13\ M_5^{1/2} r_*^{3/2} m_*^{-1}\ \mathrm{days} \times \begin{cases} \beta^{-3}, & \beta \lesssim \beta_{\mathrm{d}} \\ 1, & \beta \gtrsim \beta_{\mathrm{d}} \end{cases}$$

$$t_{\mathrm{cir}} = \frac{2\epsilon_0 t_{\mathrm{fb}} e_0^2}{\Delta\epsilon_0} \simeq 328\ M_6^{-7/6} m_*^{-4/3} r_*^{7/2}\ \mathrm{days} \times \begin{cases} \beta^{-4}, & \beta \lesssim \beta_{\mathrm{d}} \\ \beta^{-3}, & \beta > \beta_{\mathrm{d}} \end{cases}$$

$$t_{\mathrm{acc}} \simeq \left(\frac{H}{R}\right)_{\mathrm{d}}^{-2} \alpha^{-1} t_{\mathrm{dyn,c}} \simeq 13 \left(\frac{(H/R)_{\mathrm{d}}}{0.2}\right)^{-2} \left(\frac{\alpha}{0.1}\right)^{-1} \beta^{-3/2} r_*^{3/2} m_*^{-1/2}\ \mathrm{days}$$

J. Chen, L. Dou, & R. Shen 2022

W. Li et al. arXiv: 2512.02147

Timescales

$$t_{\rm fb} = 2\pi\sqrt{\frac{a_0^3}{GM_{\rm BH}}} \simeq 13\ M_5^{1/2} r_*^{3/2} m_*^{-1}\ \text{days} \times \begin{cases} \beta^{-3}, & \beta \lesssim \beta_{\rm d} \\ 1, & \beta \gtrsim \beta_{\rm d} \end{cases}$$

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Be Careful !

- **thin stream**
- **j conserved**
- **100% eff**

$$t_{\rm acc} \simeq \left(\frac{H}{R}\right)_{\rm d}^{-2} \alpha^{-1} t_{\rm dyn,c} \simeq 13 \left(\frac{(H/R)_{\rm d}}{0.2}\right)^{-2} \left(\frac{\alpha}{0.1}\right)^{-1} \beta^{-3/2} r_*^{3/2} m_*^{-1/2}\ \text{days}$$

J. Chen, L. Dou, & R. Shen 2022

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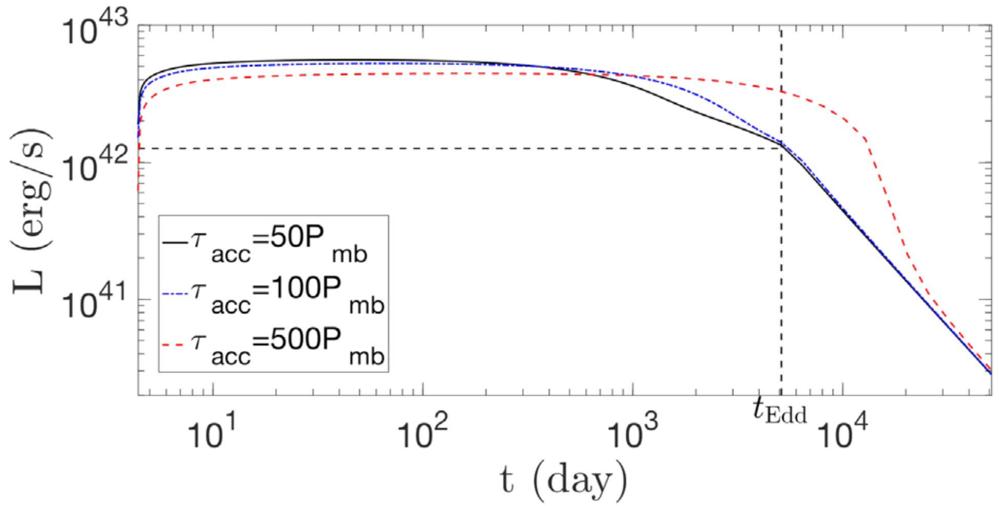
Recall: Previous IMBH-TDE Model



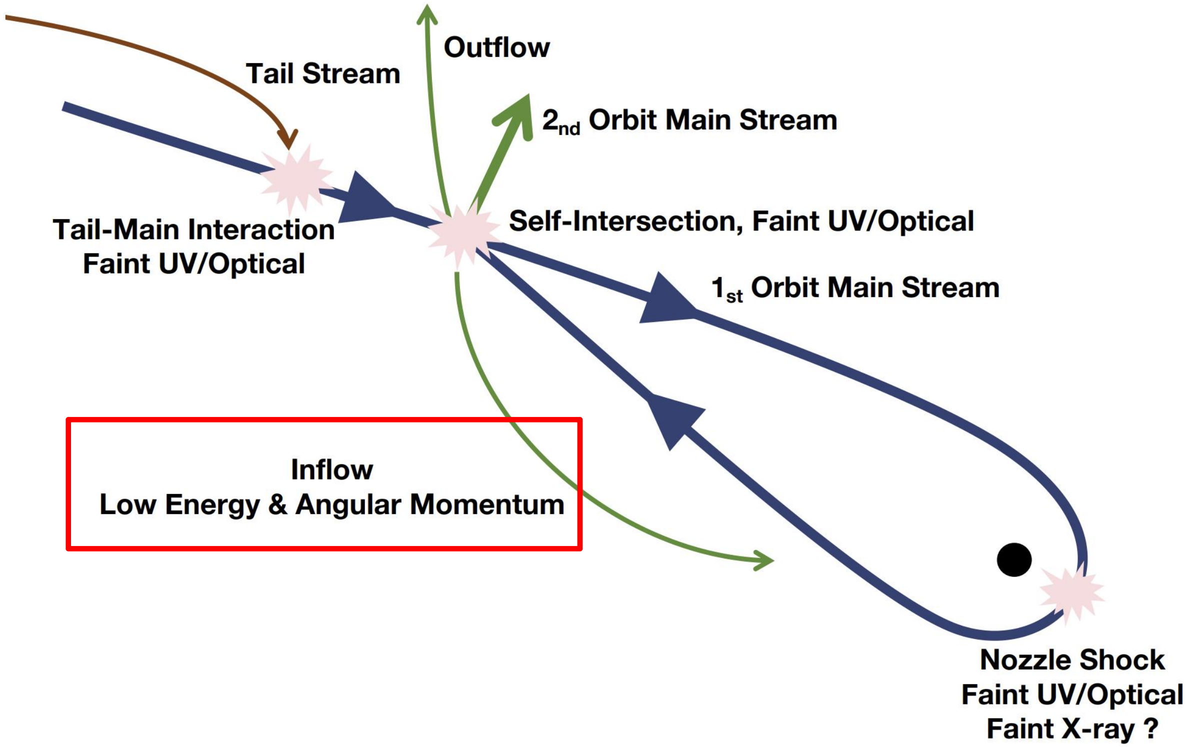
$$\frac{dM_{\rm d}(t)}{dt} + \dot{M}_{\rm acc}(t) = \boxed{\dot{M}_{\rm fb}(t)} \rightarrow \dot{M}_{\rm sup}(t)$$

$$L_{\rm bol}(t) \simeq \begin{cases} \left[1 + \ln\left(\dot{M}_{\rm acc}(t)/\dot{M}_{\rm Edd}\right)\right] L_{\rm Edd}, & \dot{M}_{\rm acc}(t) \gtrsim \dot{M}_{\rm Edd} \\ \left(\dot{M}_{\rm acc}(t)/\dot{M}_{\rm Edd}\right) L_{\rm Edd}, & \dot{M}_{\rm acc}(t) \lesssim \dot{M}_{\rm Edd} \end{cases}$$

- ➢ **Fast Rise**
  - **Ignore circularization timescale**
- ➢ **Plateau**
  - **Super–Eddington stage**
- ➢ **Power law Decline**
  - **Trace fallback rate**

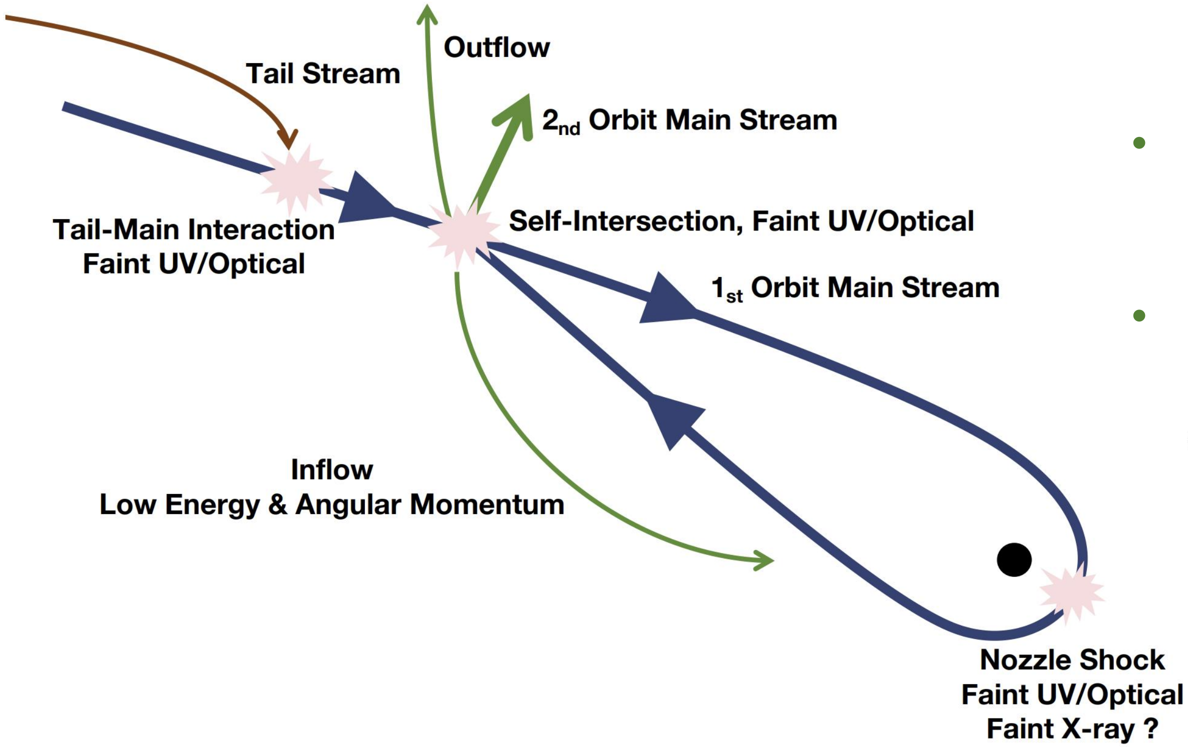
J. Chen & R. Shen 2018

1. Initial Stage: Inefficient Circularization



W. Li et al. arXiv: 2512.02147

1. Initial Stage: Inefficient Circularization



- **Dissipation channels（inefficient for IMBHs!）**

- **nozzle shock**

$$\epsilon_{\rm noz} \simeq \frac{\frac{1}{2}v_{\rm z,max}^2}{E_{\rm c}} \simeq 9.3\times10^{-6}\,\beta M_5^{-2/3} m_*^{2/3} \ll 1$$

- **self-intersection of the main stream**

$$\epsilon_{\rm self} \simeq \frac{\Delta E_0}{E_{\rm c}} \simeq 4.6\times10^{-4}\, e_0^2 \beta^2 M_5^{4/3} r_*^{-2} m_*^{2/3} \ll 1$$

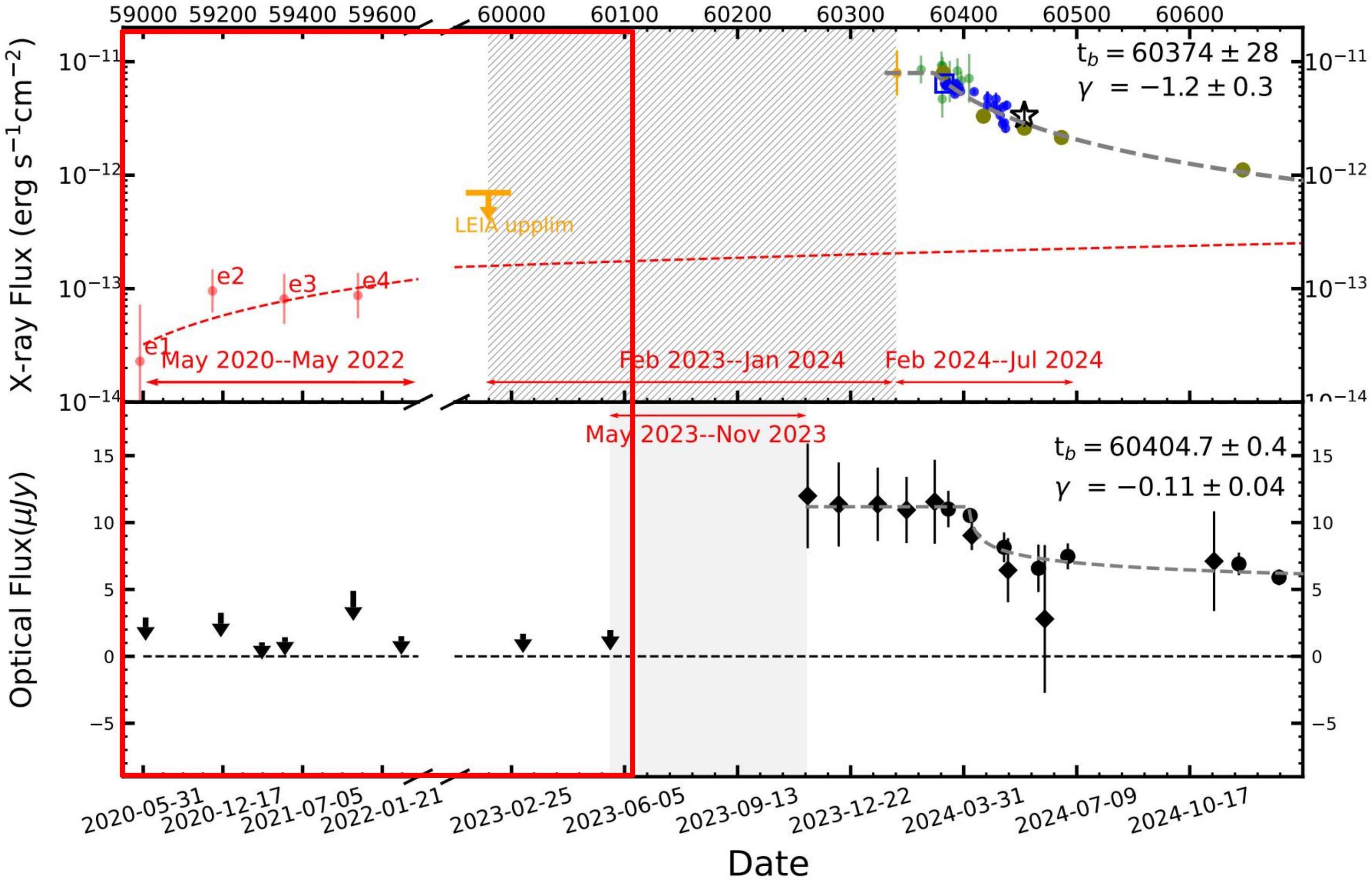
- **tail stream–main stream interaction**

$$\epsilon_{\rm tm}(t) \simeq \frac{\Delta E_{\rm tm}}{E_{\rm c}} \simeq \frac{\dot{M}_{\rm fb}(t) t_{\rm fb} E_0}{M_{\rm s}(t) E_{\rm c}}$$
$$\simeq 4.3\times10^{-2}\,\left(\frac{t}{t_{\rm fb}}\right)^{-n}\left[1-\left(\frac{t}{t_{\rm fb}}\right)^{1-n}\right]^{-1}(n-1)(1+e_0)$$
$$\times M_5^{-1/3} m_*^{1/3} \times \begin{cases}\beta, & \beta \lesssim \beta_{\rm d} \\ \beta^{-1}, & \beta \gtrsim \beta_{\rm d}\end{cases} \ll 1$$

- **Self-intersection –> inflow & outflow**
- **Timescale: t_fb**

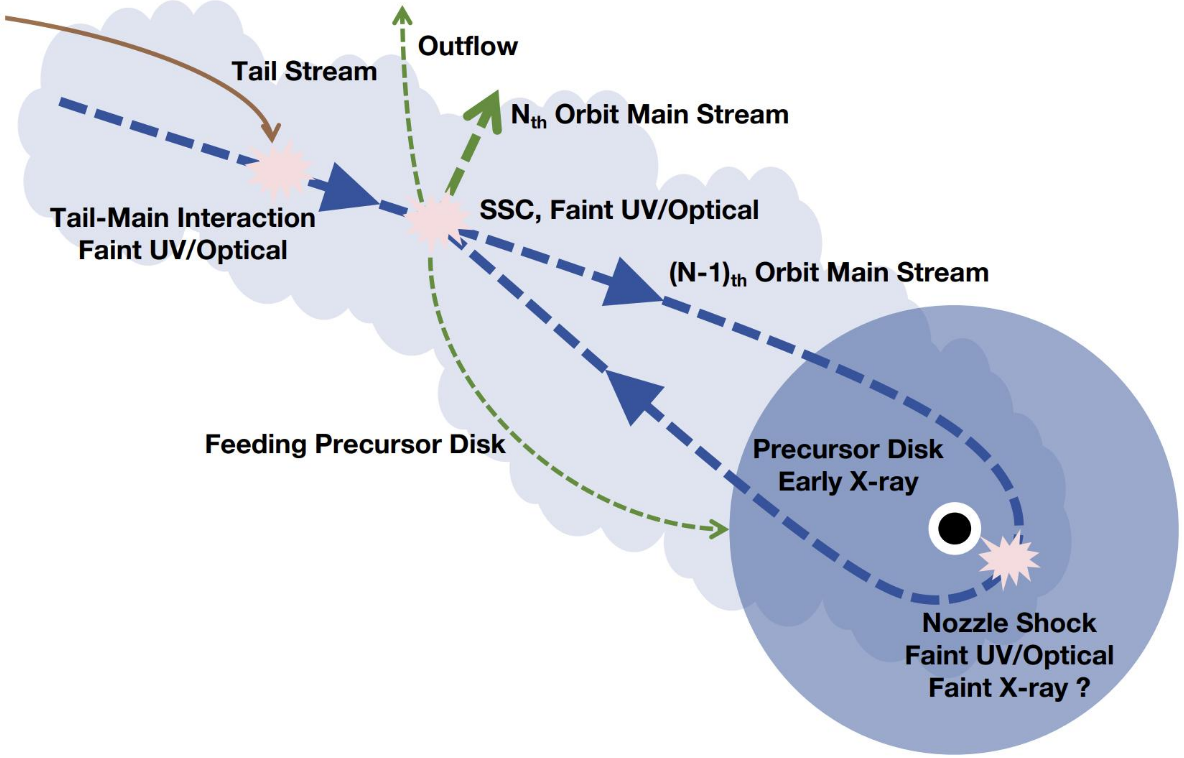
W. Li et al. arXiv: 2512.02147

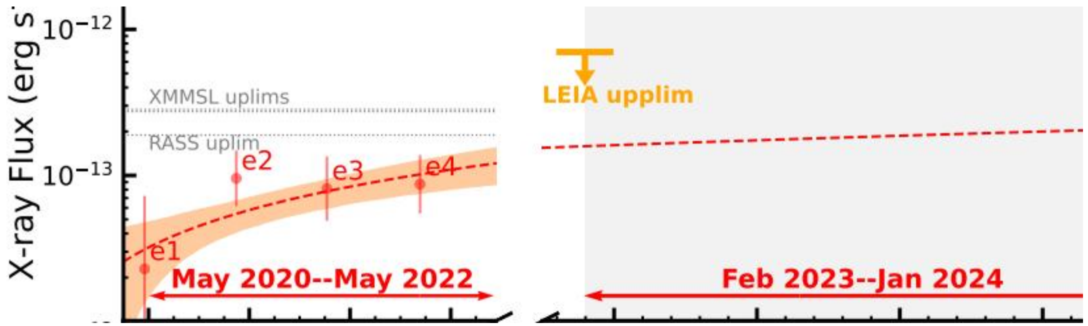
2. Slow-Rising Stage



C. Jin et al. arXiv: 2501.09580

2. Slow-Rising Stage: SSC-Feeding Precursor Disk





- Photon trapped –> thick main stream
- Dissipation continue, dominated by SSC (a succession of self–crossings)
- Inflow & outflow continue
- Precursor disk forms –> early X–rays
- Timescale: years

W. Li et al. arXiv: 2512.02147

Basic Equations for Accretion Disk: Slow-Rising Stage

$$\frac{dM_{\rm d}(t)}{dt} = \boxed{\dot{M}_{\rm sup}(t)} - \dot{M}_{\rm acc}(t)$$

$$\boxed{\dot{M}_{\rm sup}(t)} \simeq \eta_{\rm sup}(t) M_{\rm s}(t) \frac{\boxed{\dot{E}(t)}}{E_{\rm c} - E_0}$$

$$\dot{M}_{\rm acc}(t) = M_{\rm d}(t)/t_{\rm acc,early}$$

$$L_{\rm bol}(t) \simeq \eta_{\rm mis} \dot{M}_{\rm acc}(t) c^2$$

- Key equation: mass supply rate
- Early accretion timescale
  - can be longer than late time
  - MRI not fully developed –> smaller alpha
  - low accretion rate –> tinner disk

$$t_{\rm acc} \simeq \left(\frac{H}{R}\right)^{-2} \alpha^{-1} t_{\rm dyn,c}$$

- Misaligned disk

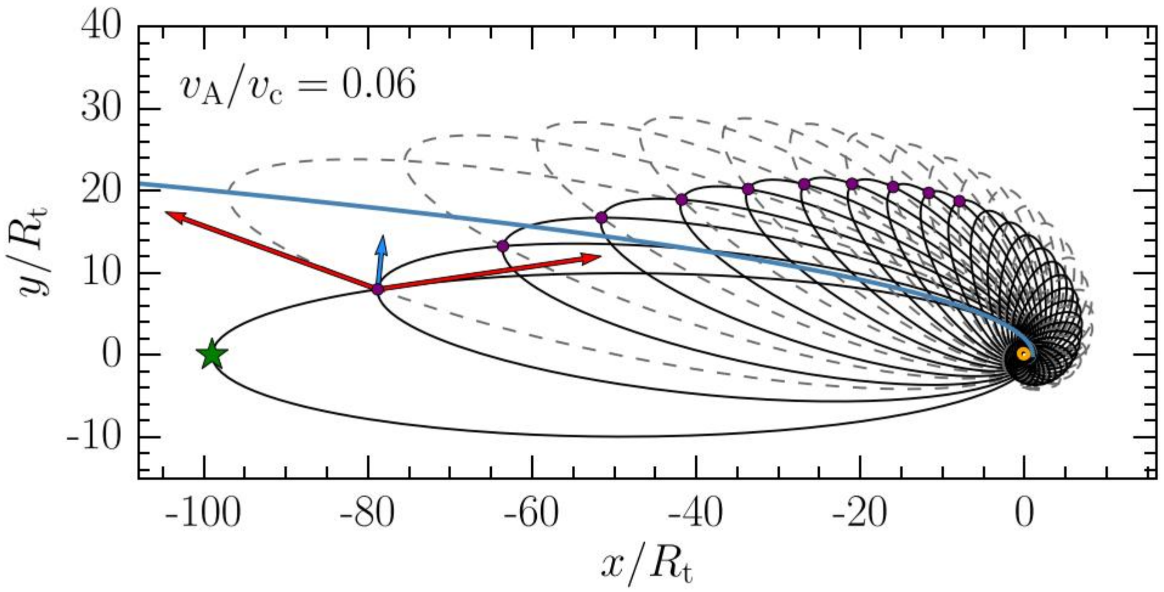
$$\eta_{\rm retro} < \eta_{\rm mis} < \eta_{\rm align}$$

W. Li et al. arXiv: 2512.02147

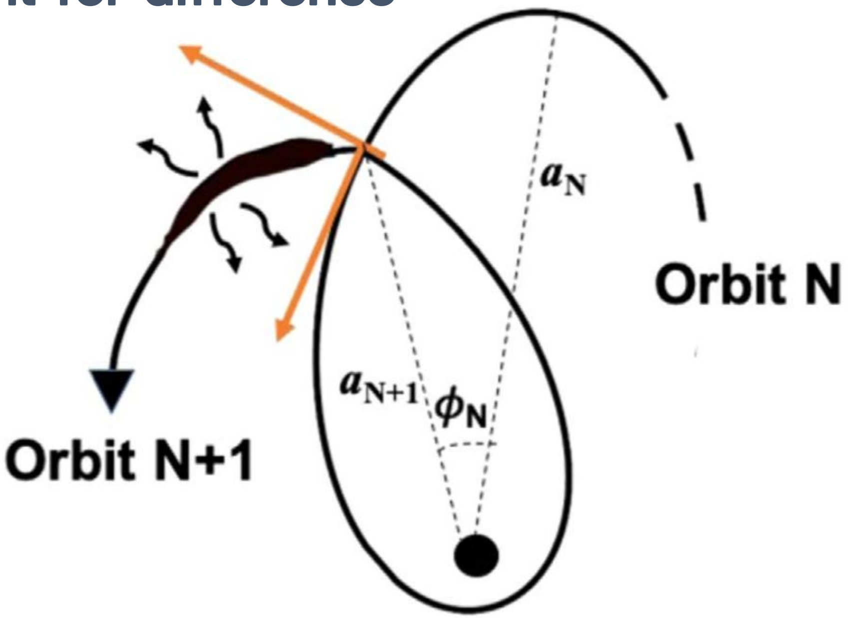
SSC Model with Correction

- **Assumption: thin main stream**
- **Here: thick main stream**
- **Free parameter eta_self(t) (can >1) to account for difference**

$$\boxed{\dot{E}_{\rm ideal}(t)} \simeq \frac{\Delta E_0}{t_{\rm fb}} \frac{1}{e_0^2} \left(1 - \frac{E_{\rm ideal}(t)}{E_{\rm c}}\right) \left(\frac{E_{\rm ideal}(t)}{E_0}\right)^{3/2}$$



C. Bonnerot et al. 2017



J. Chen & R. Shen 2021

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- **Misaligned disk**

$$\eta_{\rm retro} < \eta_{\rm mis} < \eta_{\rm align}$$

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Basic Equations for Accretion Disk: Slow-Rising Stage

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Basic Equations for Accretion Disk: Slow-Rising Stage

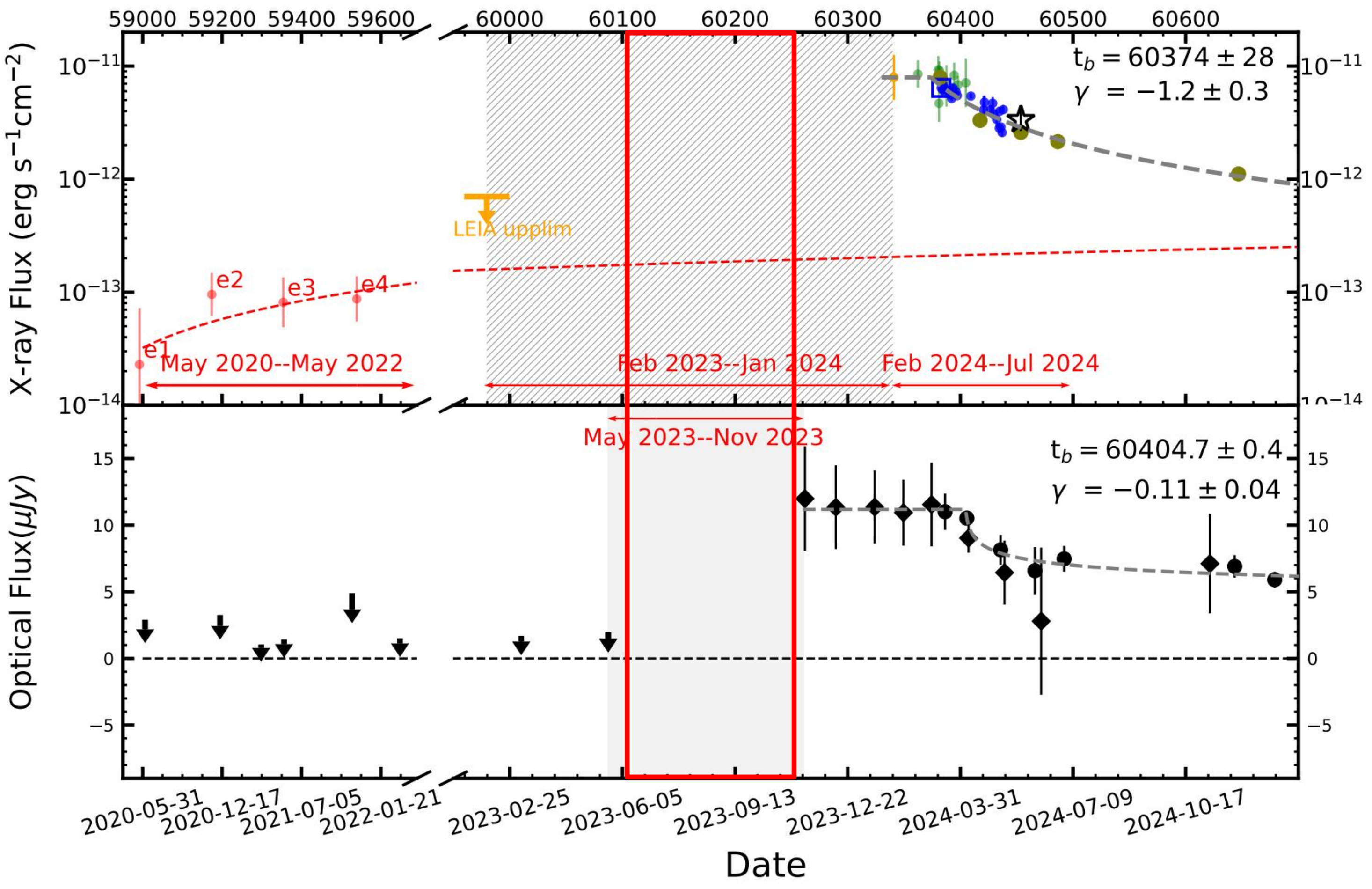
$$\dot{M}_{\rm sup}(t) \simeq \eta_{\rm sup}(t) M_{\rm s}(t) \frac{\dot{E}(t)}{E_{\rm c} - E_0} \simeq \eta_{\rm self}(t)\eta_{\rm sup}(t) M_{\rm s}(t) \frac{\dot{E}_{\rm ideal}(t)}{E_{\rm c} - E_0}$$

$$\simeq \overline{\eta_{\rm self}\eta_{\rm sup}} M_{\rm s}(t) \frac{\dot{E}_{\rm ideal}(t)}{E_{\rm c} - E_0}$$

W. Li et al. arXiv: 2512.02147

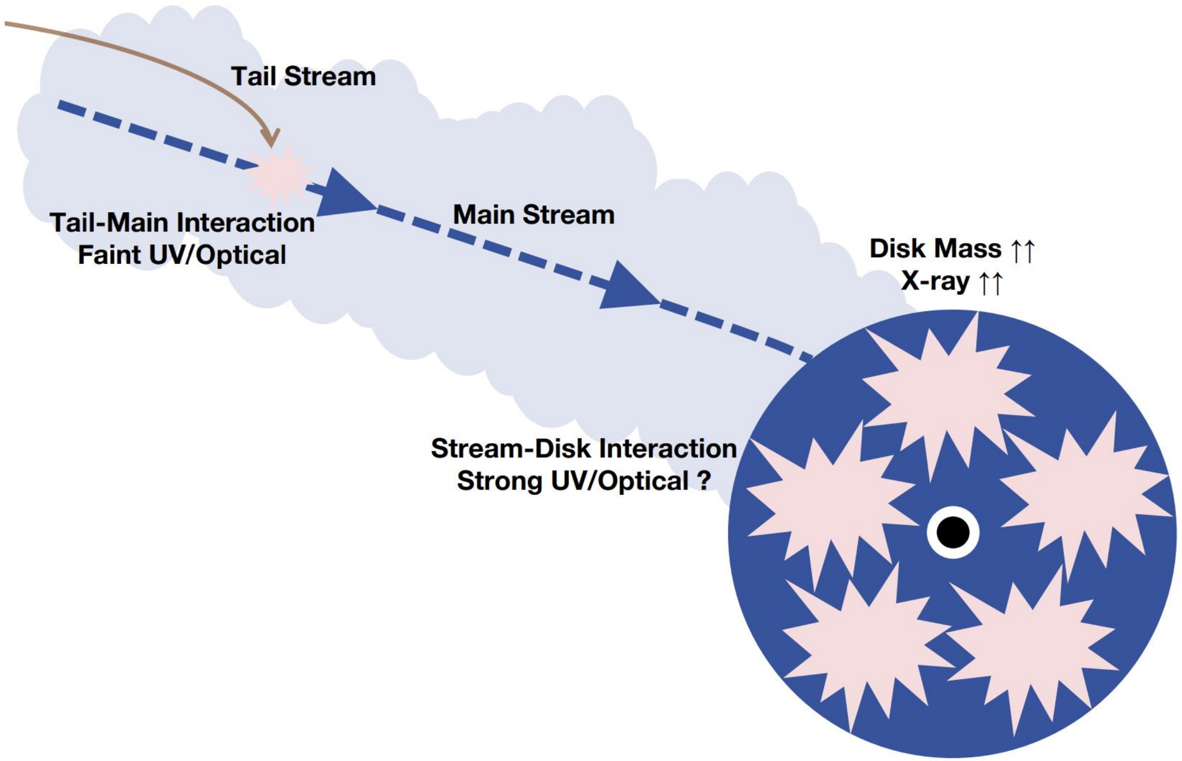
EP240222a Model

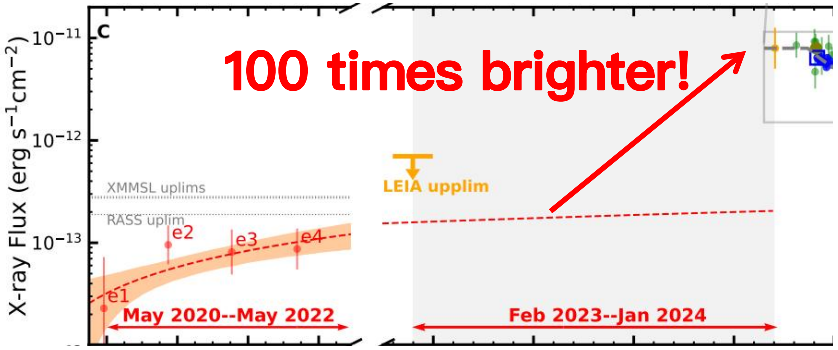
3. Fast-Rising Stage



C. Jin et al. arXiv: 2501.09580

3. Fast-Rising Stage: Stream-Disk Interaction

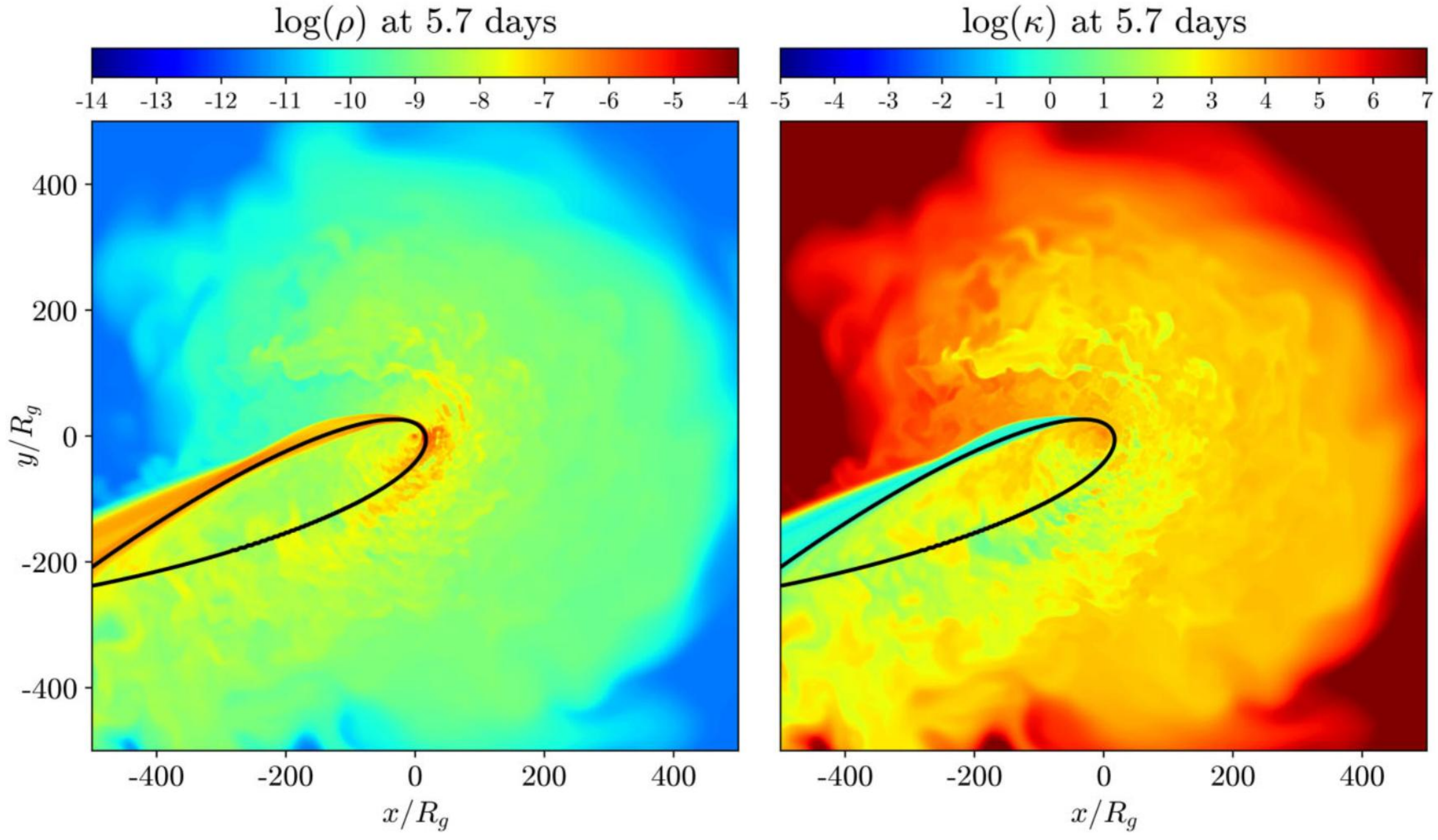




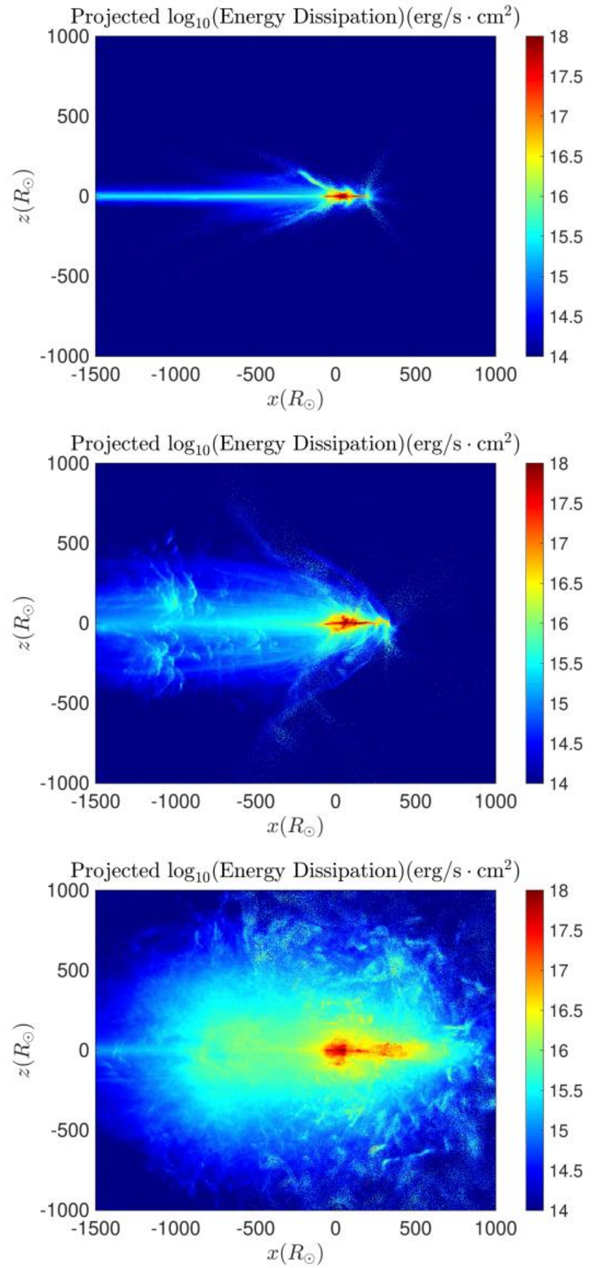
- Momentum flux matching
- Stream–disk interaction
- Runaway circularization (~t_fb)
- Massive disk –> sharp X–ray rise
- Potential strong UV/optical flares
- Disk aligned with the BH's spin
- Timescale: < t_fb or t_fb

W. Li et al. arXiv: 2512.02147

Stream-Disk Interaction



Z. Andalman et al. 2022



E. Steinberg & N. Stone 2024

Basic Equations for Accretion Disk: Fast-Rising Stage

$$\frac{dM_{\rm d}(t)}{dt} = \boxed{\dot{M}_{\rm sup}(t)} - \dot{M}_{\rm acc}(t)$$

$$\boxed{\dot{M}_{\rm sup}(t)} \simeq \frac{M_{\rm s}(t)}{t_{\rm s,N}} + \dot{M}_{\rm fb}(t)$$

$$\dot{M}_{\rm acc}(t) = M_{\rm d}(t)/t_{\rm acc,late}$$

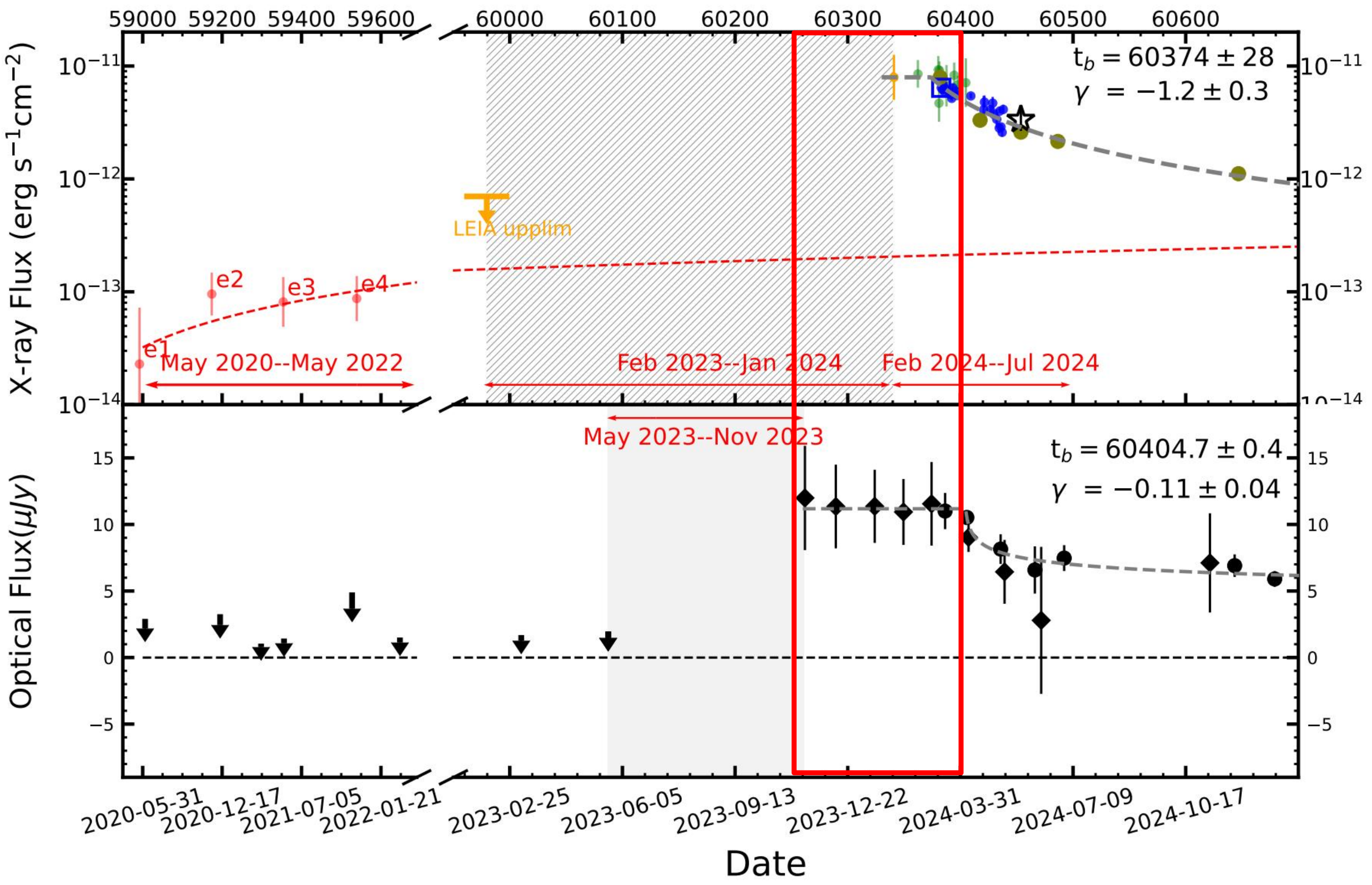
$$L_{\rm bol}(t) \simeq \begin{cases} \left[1+\ln\left(\dot{M}_{\rm acc}(t)/\dot{M}_{\rm Edd}\right)\right] L_{\rm Edd}, & \dot{M}_{\rm acc}(t) \gtrsim \dot{M}_{\rm Edd} \\ \left(\dot{M}_{\rm acc}(t)/\dot{M}_{\rm Edd}\right) L_{\rm Edd}, & \dot{M}_{\rm acc}(t) \lesssim \dot{M}_{\rm Edd} \end{cases}$$

- **Mass supply dominated by main stream (stream–disk interaction)**
- **Late accretion timescale (shorter)**
- **Accretion rate >> Eddington rate**
- **Lumonosity is “throttled” –> L_bol ~ L_edd**
  - **outflow & advection**
- **Accretion efficiency = eta_align**

W. Li et al. arXiv: 2512.02147

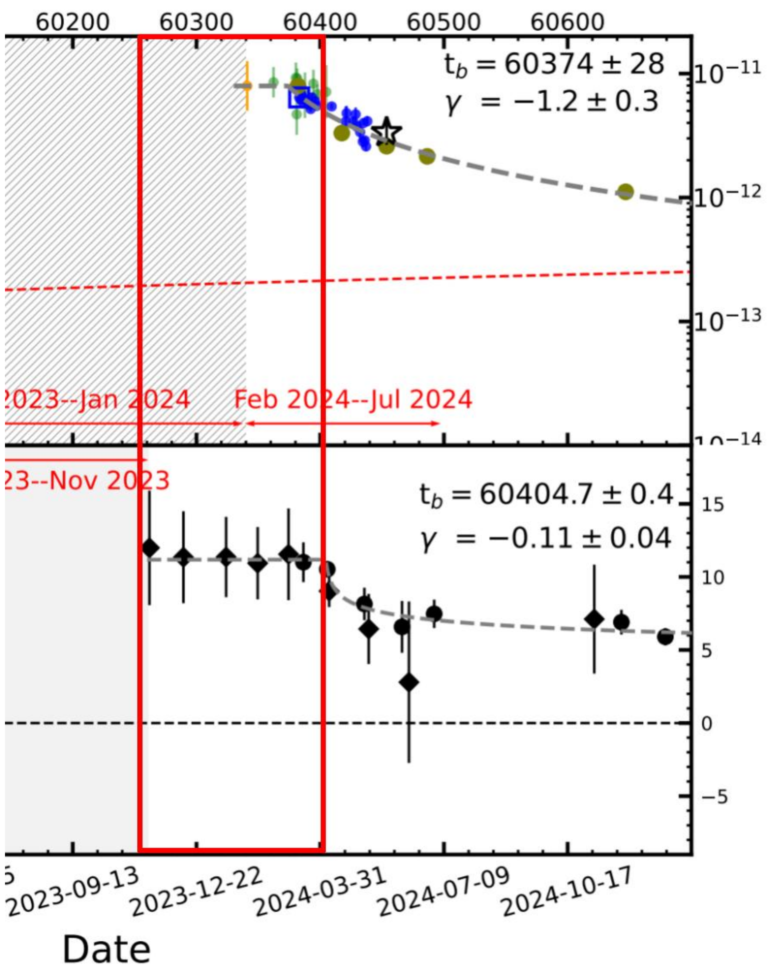
EP240222a Model

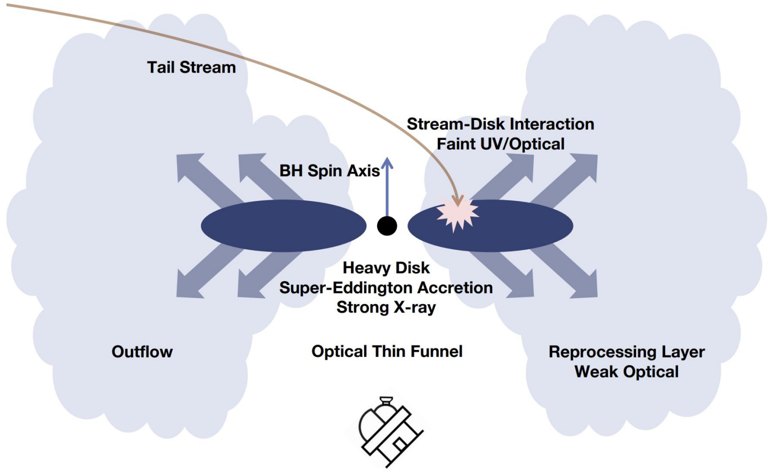
4. Plateau Stage



C. Jin et al. arXiv: 2501.09580

4. Plateau Stage: Super-Eddington Accretion

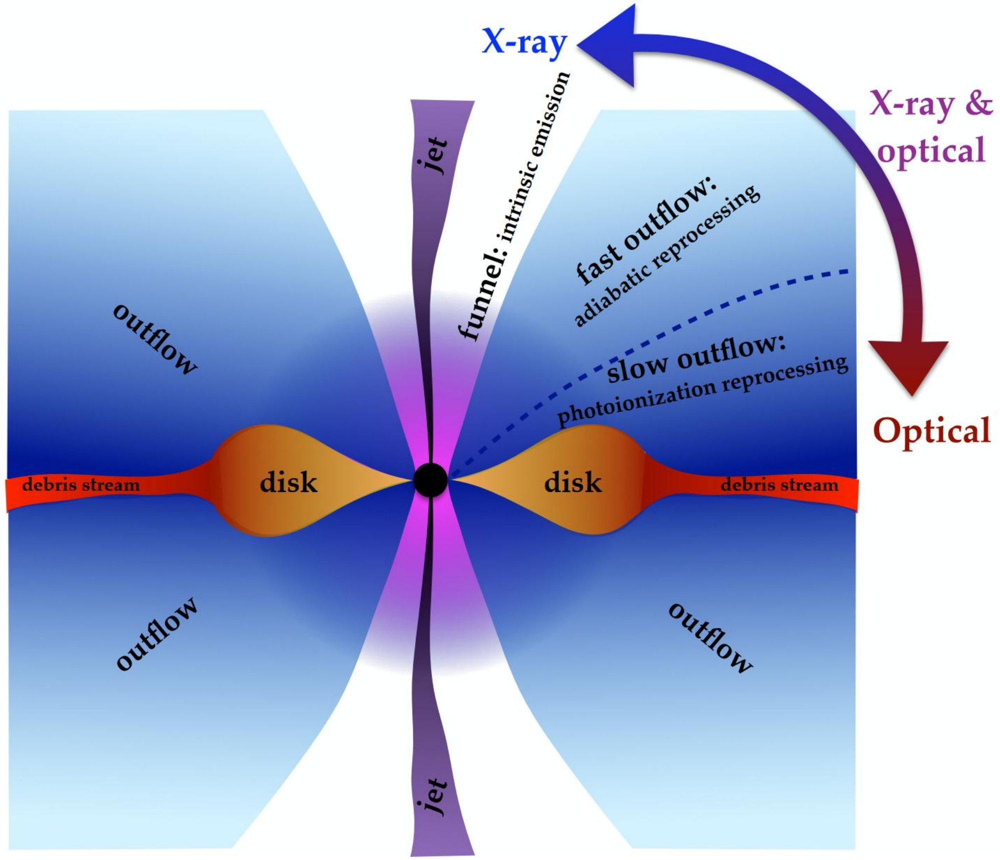




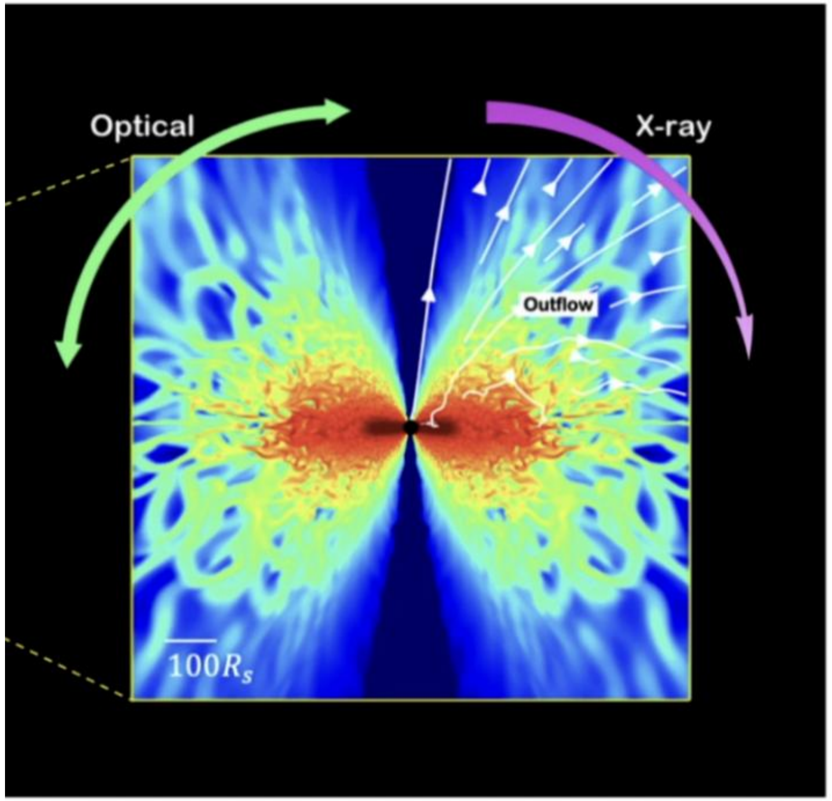
- Massive disk –> super–Eddington
- L_bol ~ L_Edd (outflow & advection)
- Polar funnel –> line of sight to X–rays
- Weak optical (reprocessed X–rays)
- Timescale: t_acc,late

W. Li et al. arXiv: 2512.02147

Unified Model for TDEs



L. Dai et al. 2018



E. Qiao et al. 2025

Basic Equations for Accretion Disk: Plateau Stage

$$\frac{dM_{\rm d}(t)}{dt} = \boxed{\dot{M}_{\rm sup}(t)} - \dot{M}_{\rm acc}(t)$$

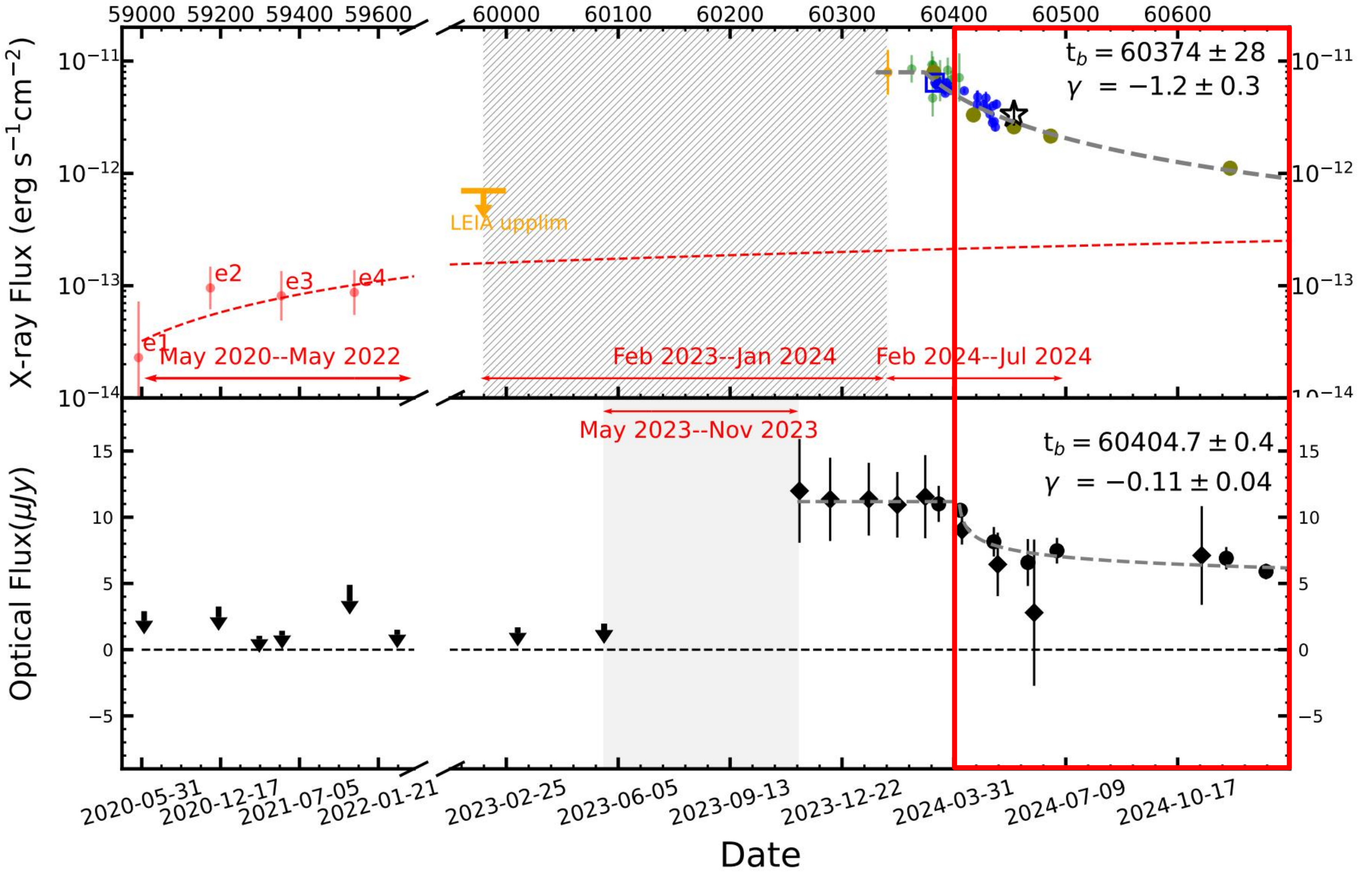
$$\boxed{\dot{M}_{\rm sup}(t)} \simeq \dot{M}_{\rm fb}(t)$$

$$\dot{M}_{\rm acc}(t) = M_{\rm d}(t)/t_{\rm acc,late}$$

$$L_{\rm bol}(t) \simeq \begin{cases} \left[1+\ln\left(\dot{M}_{\rm acc}(t)/\dot{M}_{\rm Edd}\right)\right] L_{\rm Edd}, & \dot{M}_{\rm acc}(t) \gtrsim \dot{M}_{\rm Edd} \\ \left(\dot{M}_{\rm acc}(t)/\dot{M}_{\rm Edd}\right) L_{\rm Edd}, & \dot{M}_{\rm acc}(t) \lesssim \dot{M}_{\rm Edd} \end{cases}$$

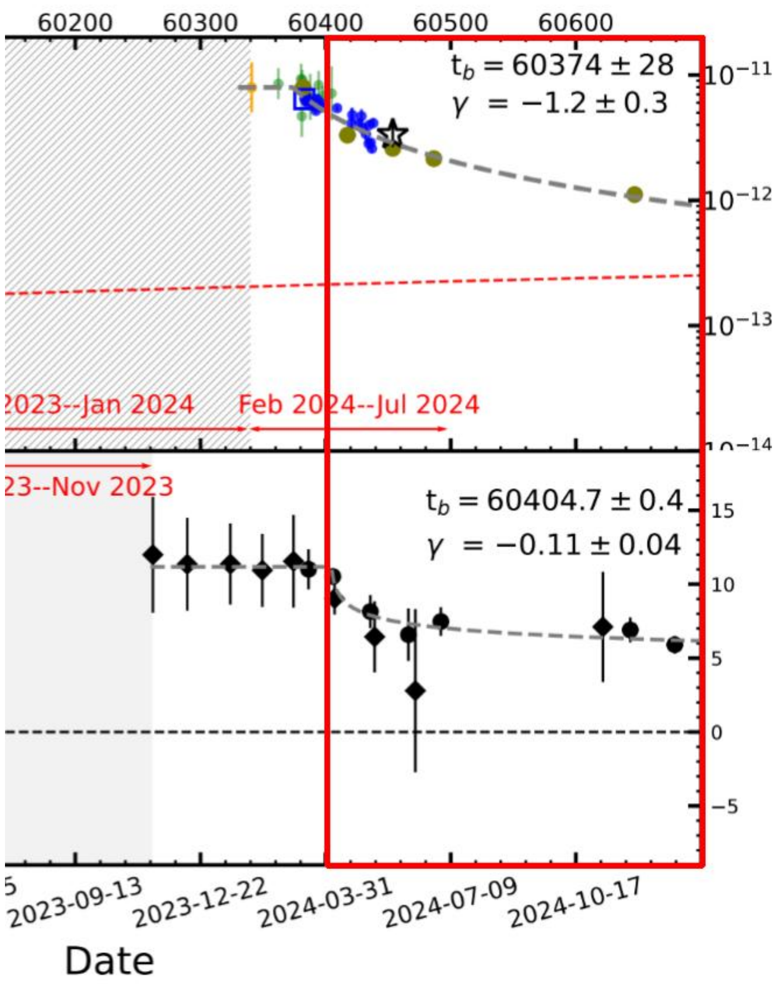
W. Li et al. arXiv: 2512.02147

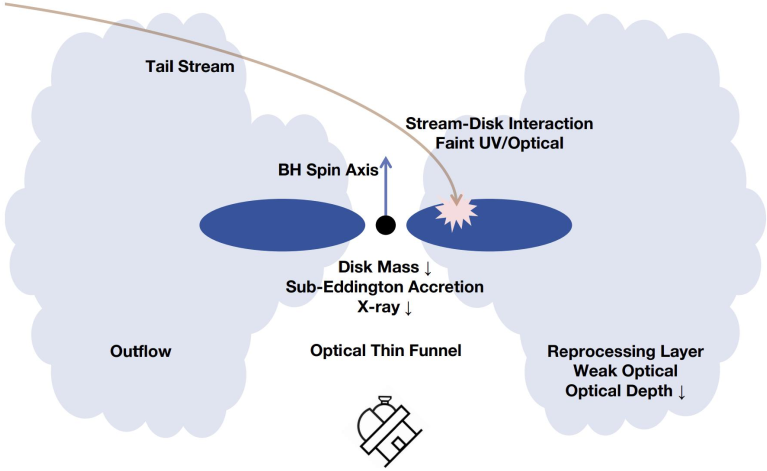
5. Decline Stage



C. Jin et al. arXiv: 2501.09580

5. Decline Stage: Sub-Eddington Accretion





- **Delayed & slower optical decline –> reprocess**
- **Disk mass depletes –> sub–Eddington acc**
- **Outflow ceases**
- **Optically thin layer (not yet reached)**
- **Timescale: years**

W. Li et al. arXiv: 2512.02147

Fitting

W. Li et al. arXiv: 2512.02147

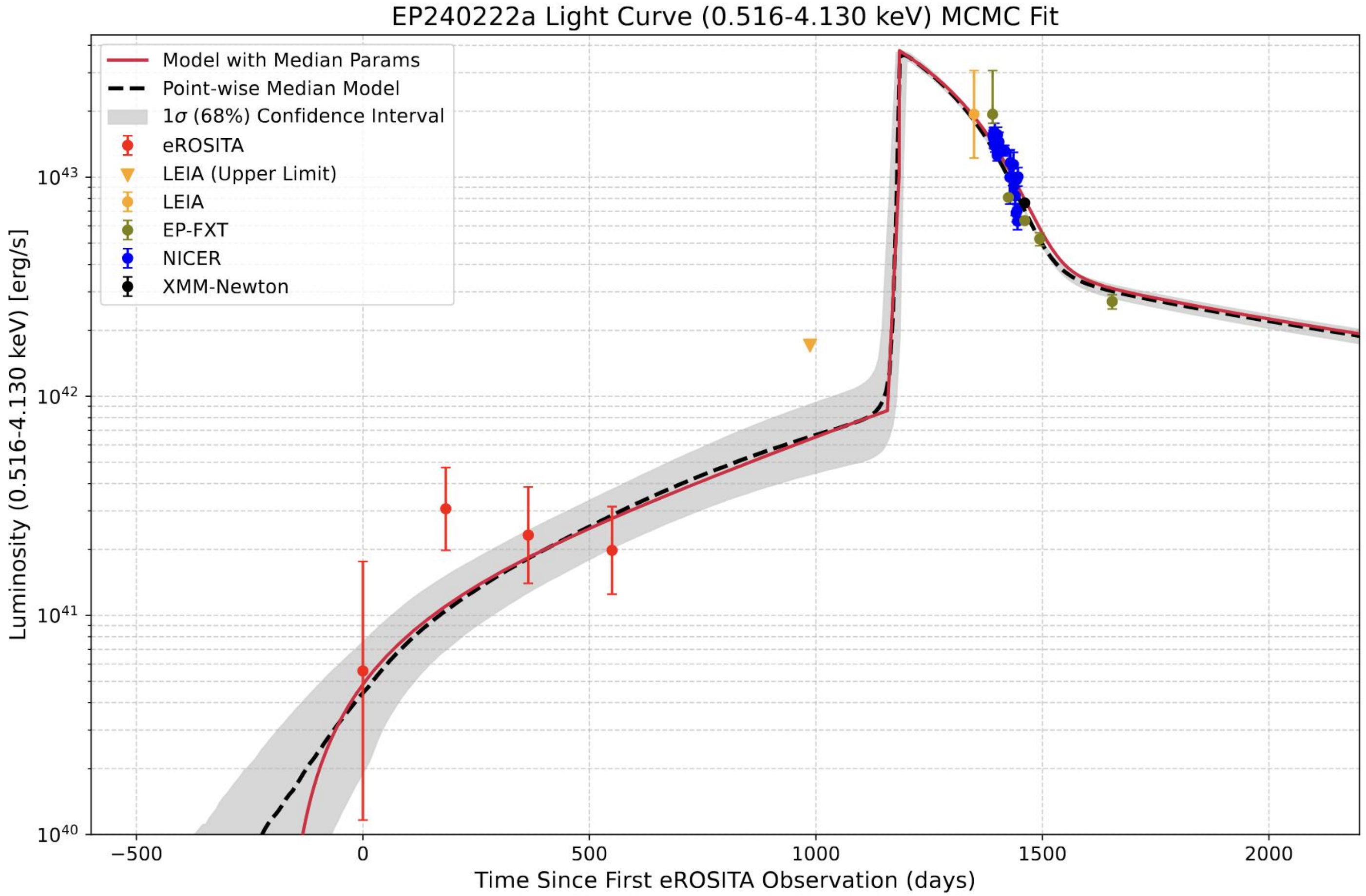
Free Parameters

1. **Stellar Mass,** $\log_{10}(M_*/M_\odot)$: $[-1.097, 0.477]$.
2. **Penetration Factor,** $\log_{10}(\beta)$: $[-0.301, 0.477]$.
3. **Efficiency Factor,** $\eta \equiv \overline{\eta_{\rm int}\eta_{\rm sup}}\eta_{\rm mis}$: $[-6, 1]$.
4. **Early Accretion Timescale,** $\log_{10}(t_{\rm acc,early}/t_{\rm dyn,c})$: $[0, 9]$.
5. **Late Accretion Timescale,** $\log_{10}(t_{\rm acc,late}/t_{\rm dyn,c})$: $[0, 4]$.
6. **Critical Time,** $t_{\rm crit} - t_0 \in [950, 1300]$ days.
7. **Time Offset,** $t_0 \in [0, 1000]$ days

W. Li et al. arXiv: 2512.02147

EP240222a Model

Fitting Results (X-rays)



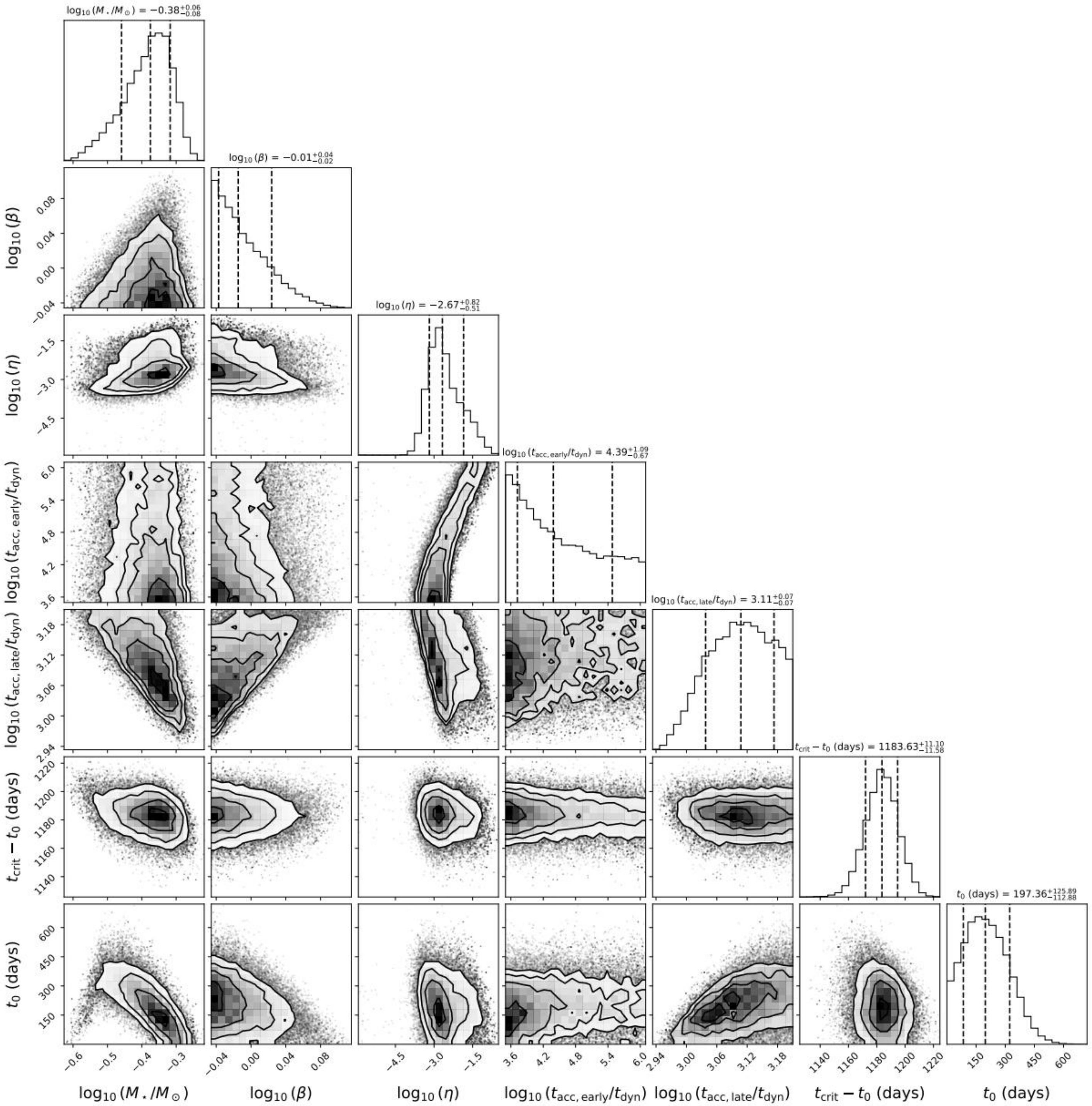
W. Li et al. arXiv: 2512.02147

EP240222a Model

Corner Plot

➢ **Best Model**

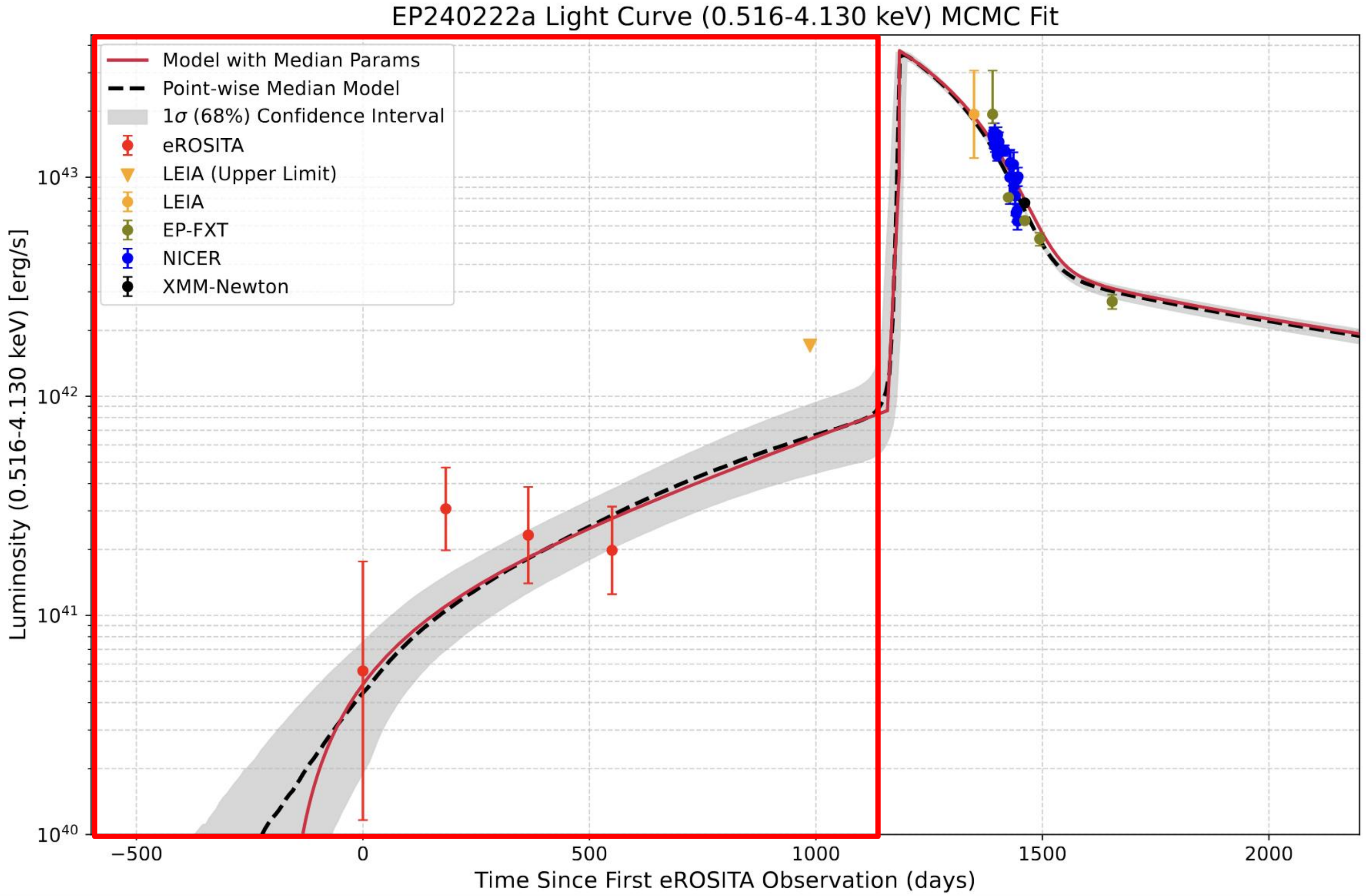
- **M_star ≈ 0.4 M_sun**
- **β ≈ 1.0**



W. Li et al. arXiv: 2512.02147

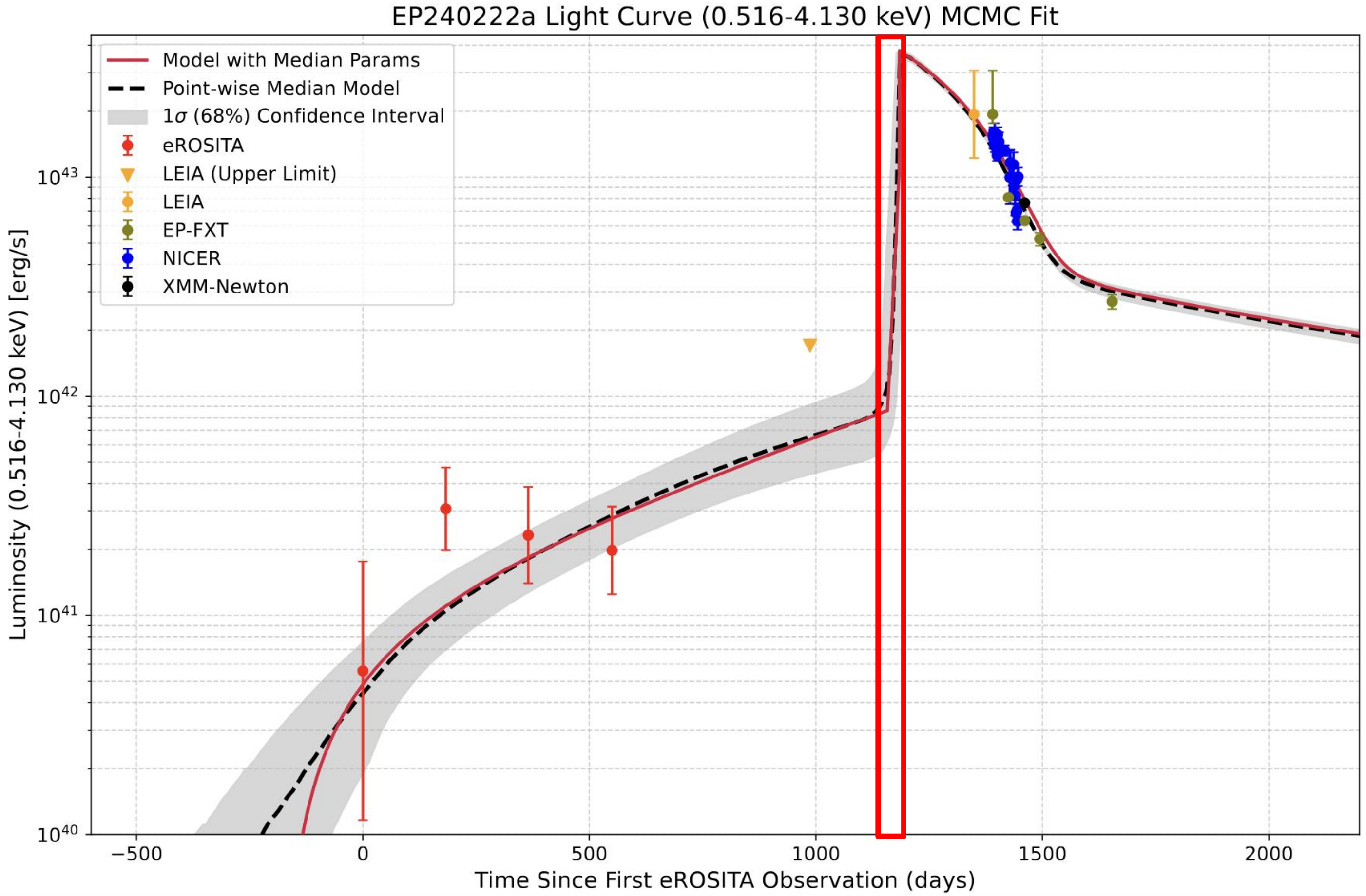
EP240222a Model

Slow Rise (years)



W. Li et al. arXiv: 2512.02147

Fast Rise (<t_fb or t_fb)

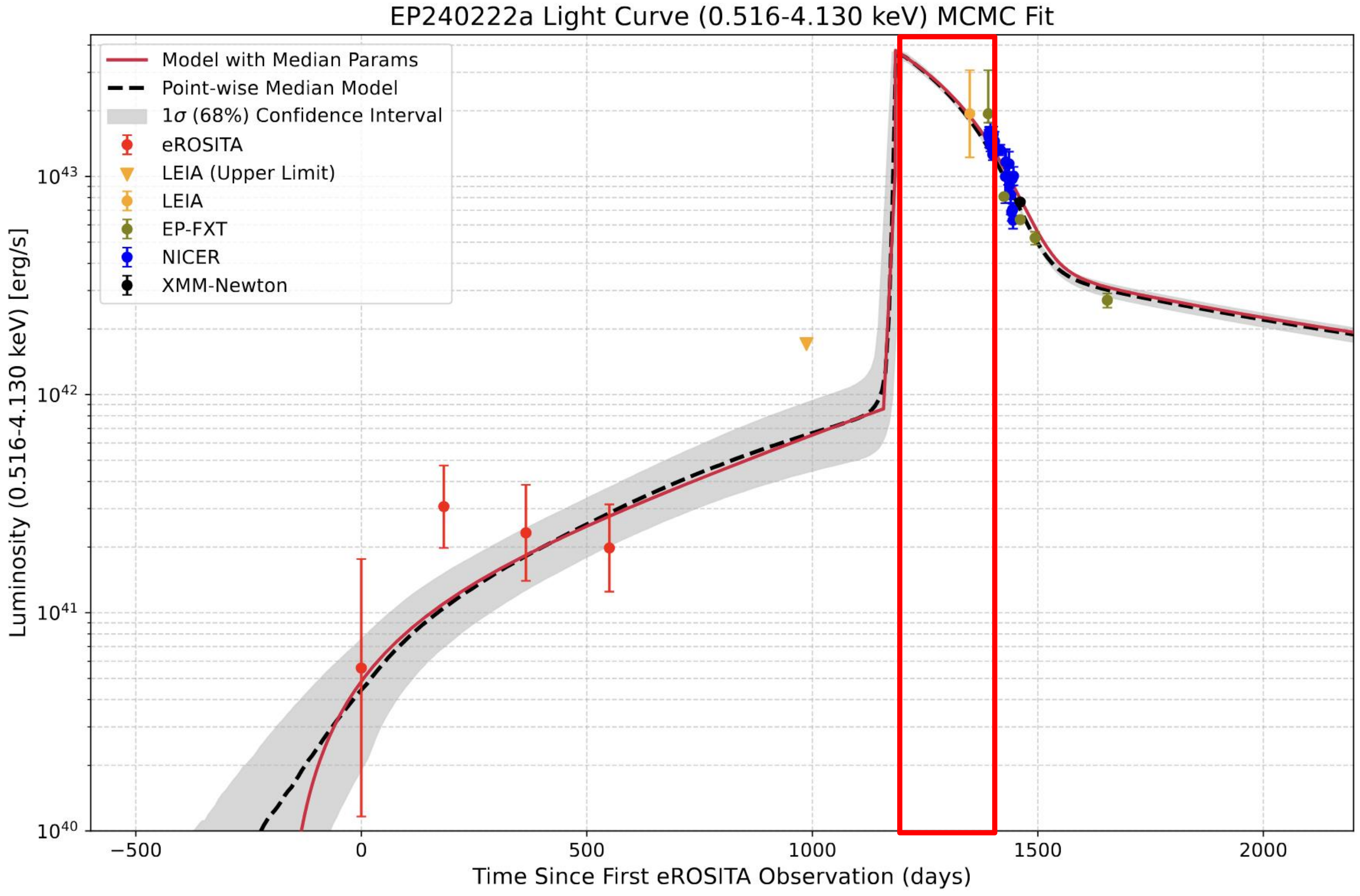


W. Li et al. arXiv: 2512.02147

EP240222a Model

Plateau

$$L_{\rm bol}(t) \simeq \left[1 + \ln\left(\frac{M_{\rm fb}}{\dot{M}_{\rm Edd} t_{\rm acc,late}}\right) - \frac{t}{t_{\rm acc,late}}\right] L_{\rm Edd}$$

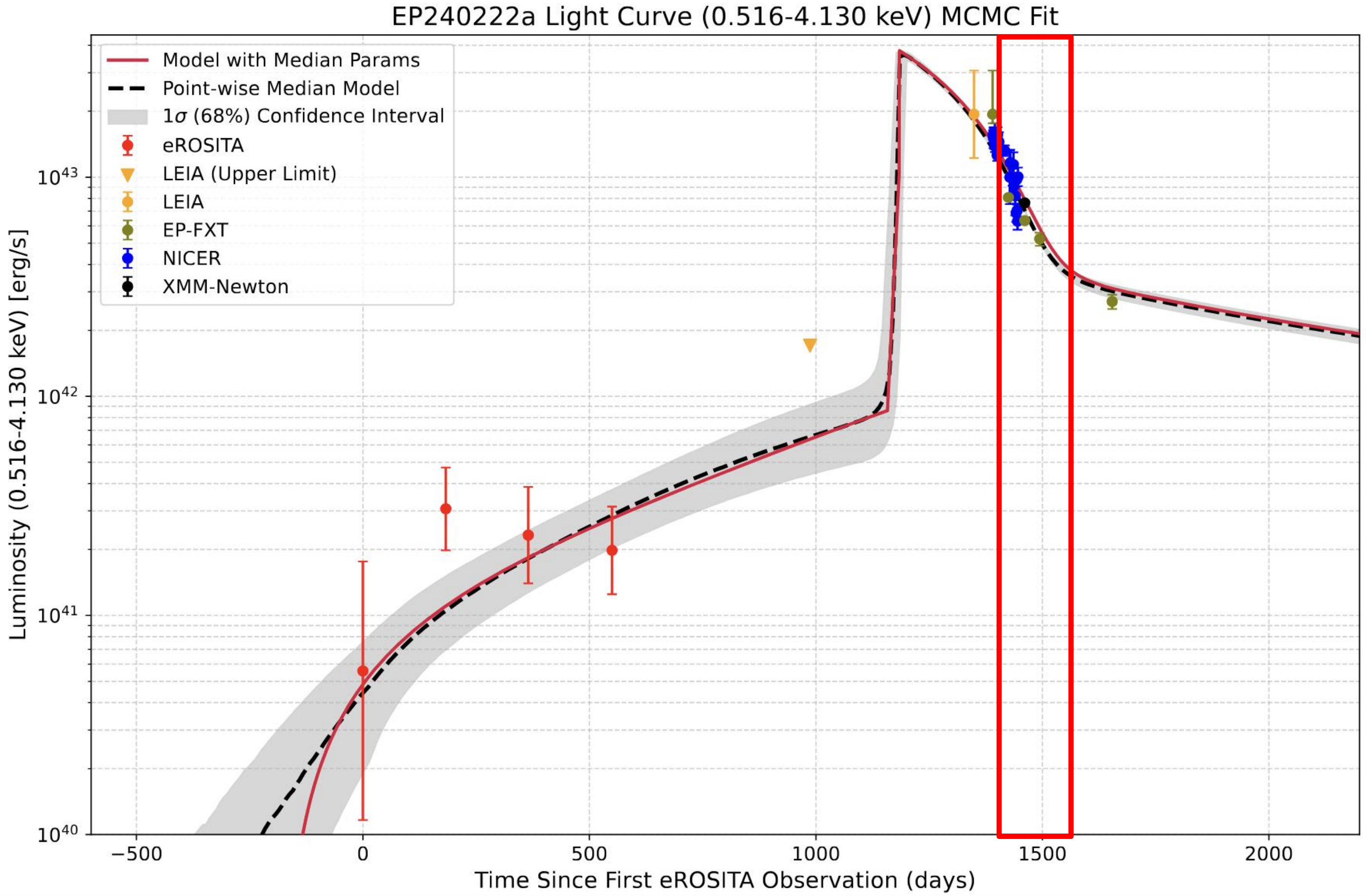


W. Li et al. arXiv: 2512.02147

EP240222a Model

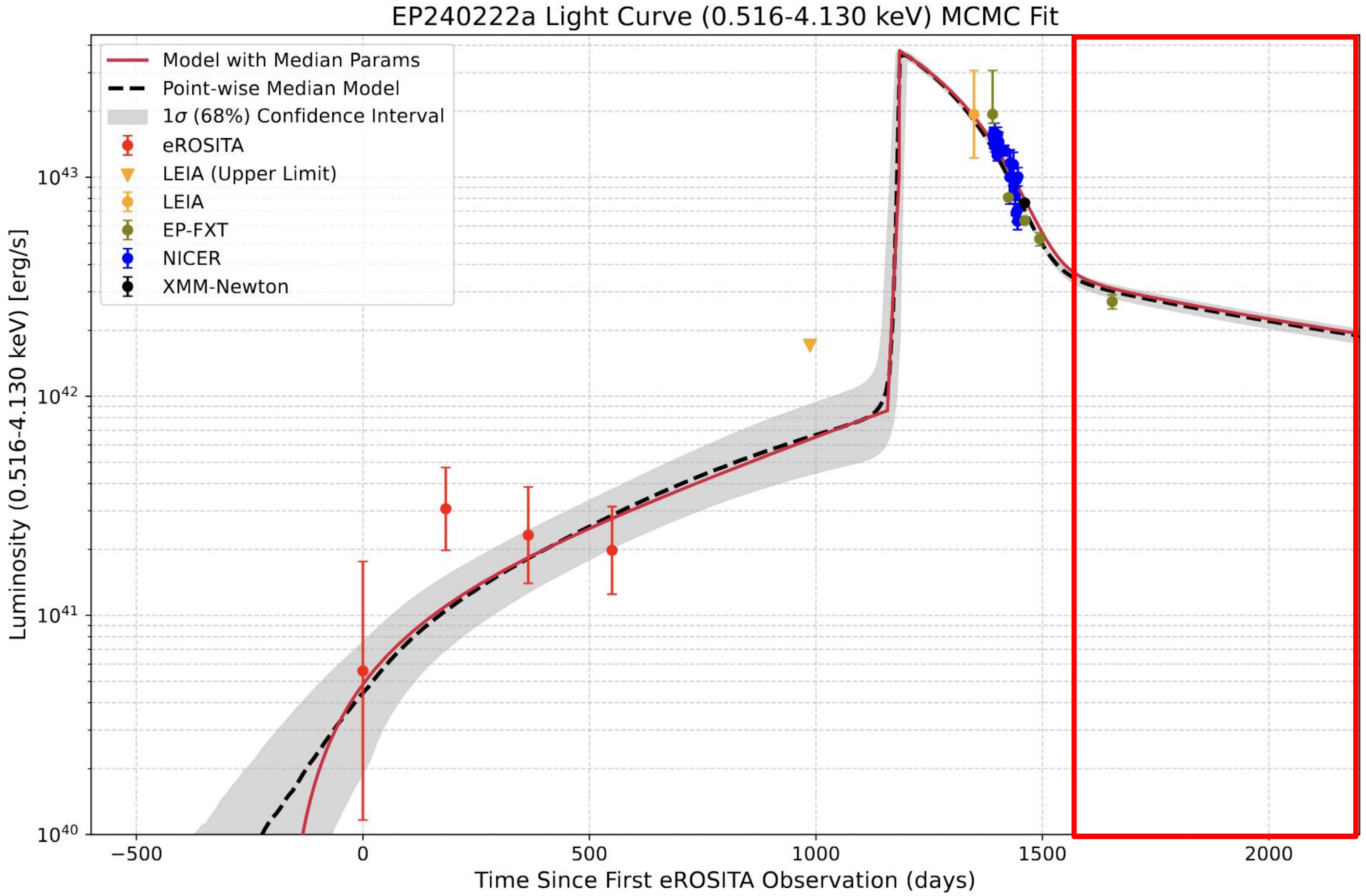
Decline

$$L_{\rm bol}(t) \simeq \left[1+\ln\left(\frac{M_{\rm fb}}{\dot{M}_{\rm Edd}t_{\rm acc,late}}\right)-\frac{t}{t_{\rm acc,late}}\right]L_{\rm Edd}$$



W. Li et al. arXiv: 2512.02147

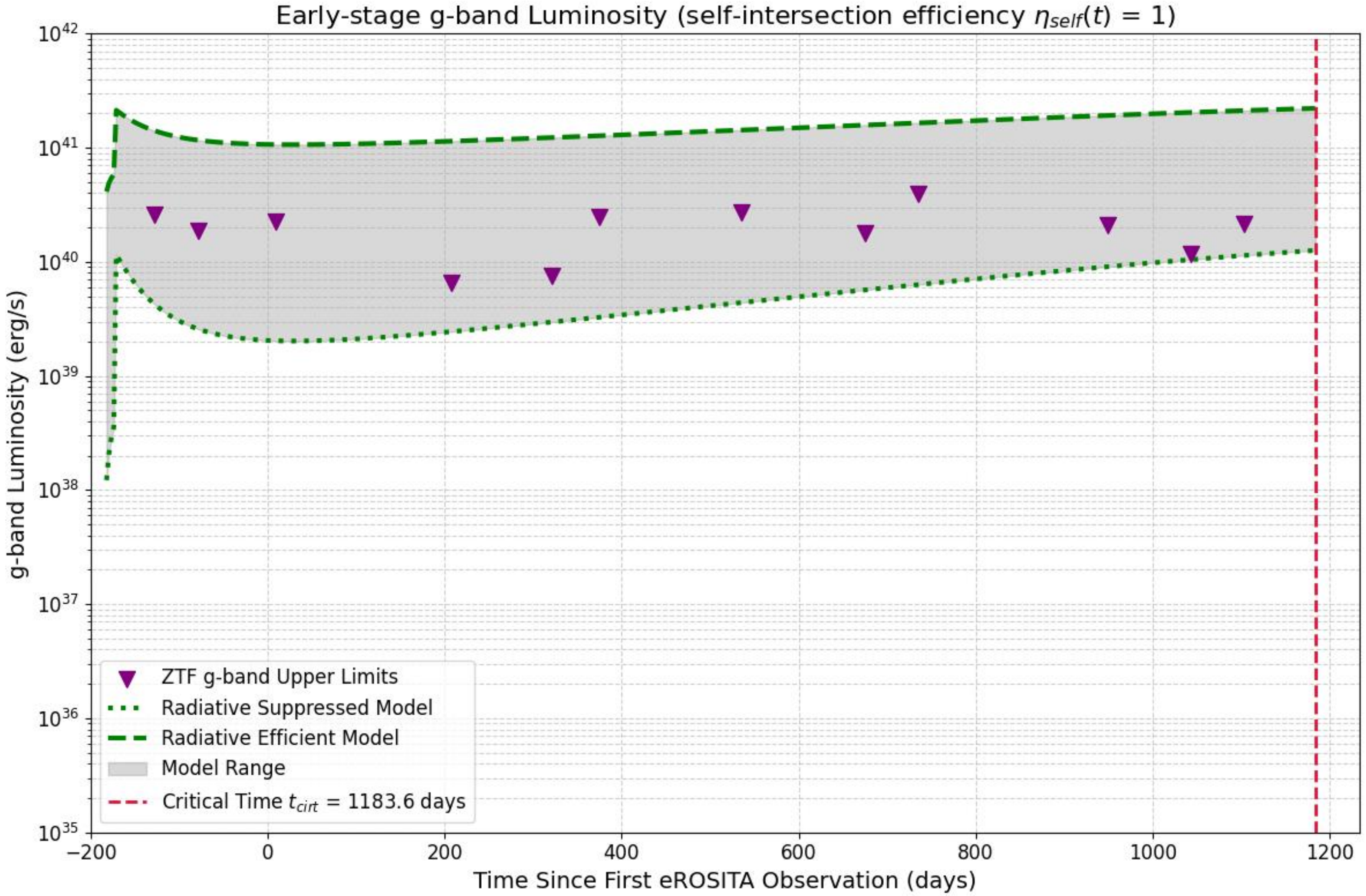
Decline: Power Law



W. Li et al. arXiv: 2512.02147

Discussion

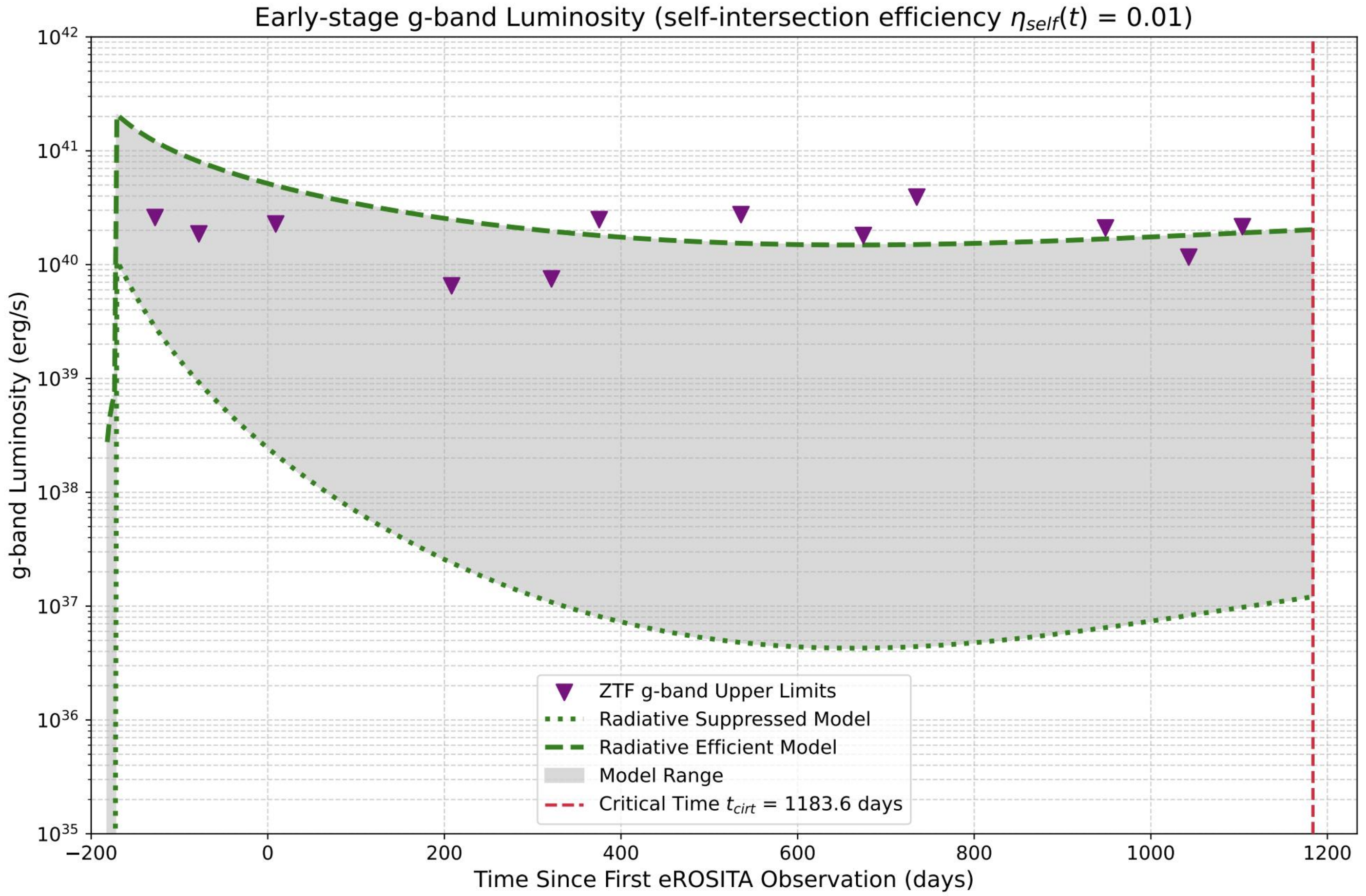
Early optical: $L_{\rm g,early}(t) \simeq \frac{M_{\rm s}(t)\dot{E}(t)+\dot{M}_{\rm fb}(t)\,[E_0-E_{\rm fb}(t)]}{K_{\rm g}(t)}$



W. Li et al. arXiv: 2512.02147

EP240222a Model

Early optical: $L_{g,\mathrm{early}}(t) \simeq \frac{M_s(t)\dot{E}(t) + \dot{M}_{fb}(t)\left[E_0 - E_{fb}(t)\right]}{K_g(t)}$

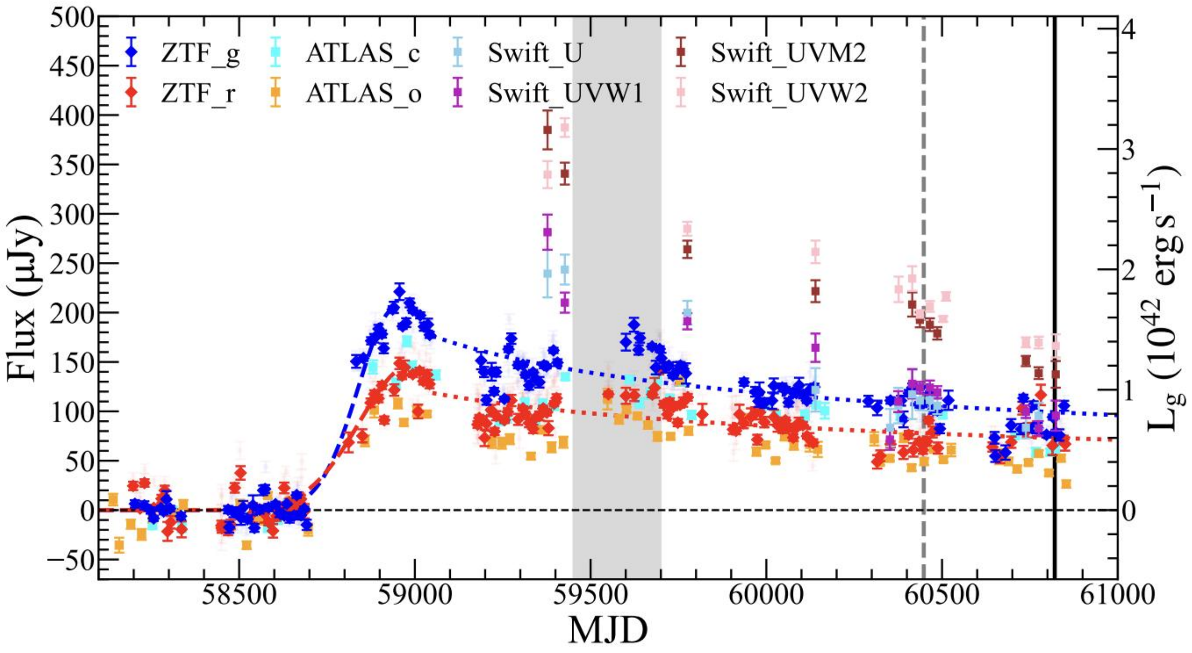


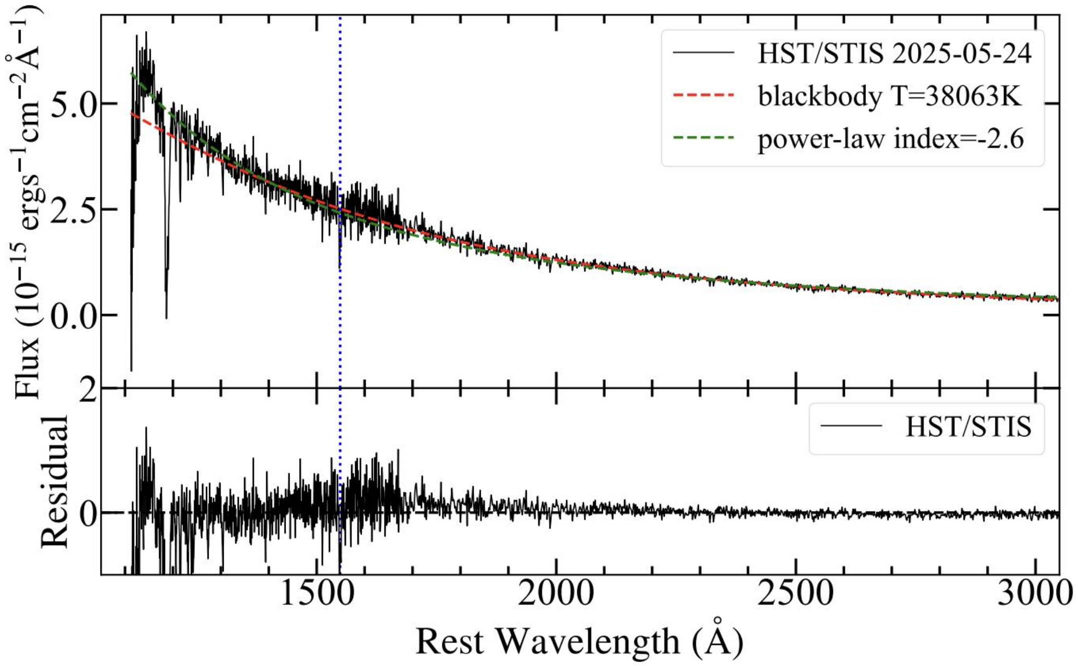
W. Li et al. arXiv: 2512.02147

EP240222a Model

Ansky (most extreme QPE) from an TDE

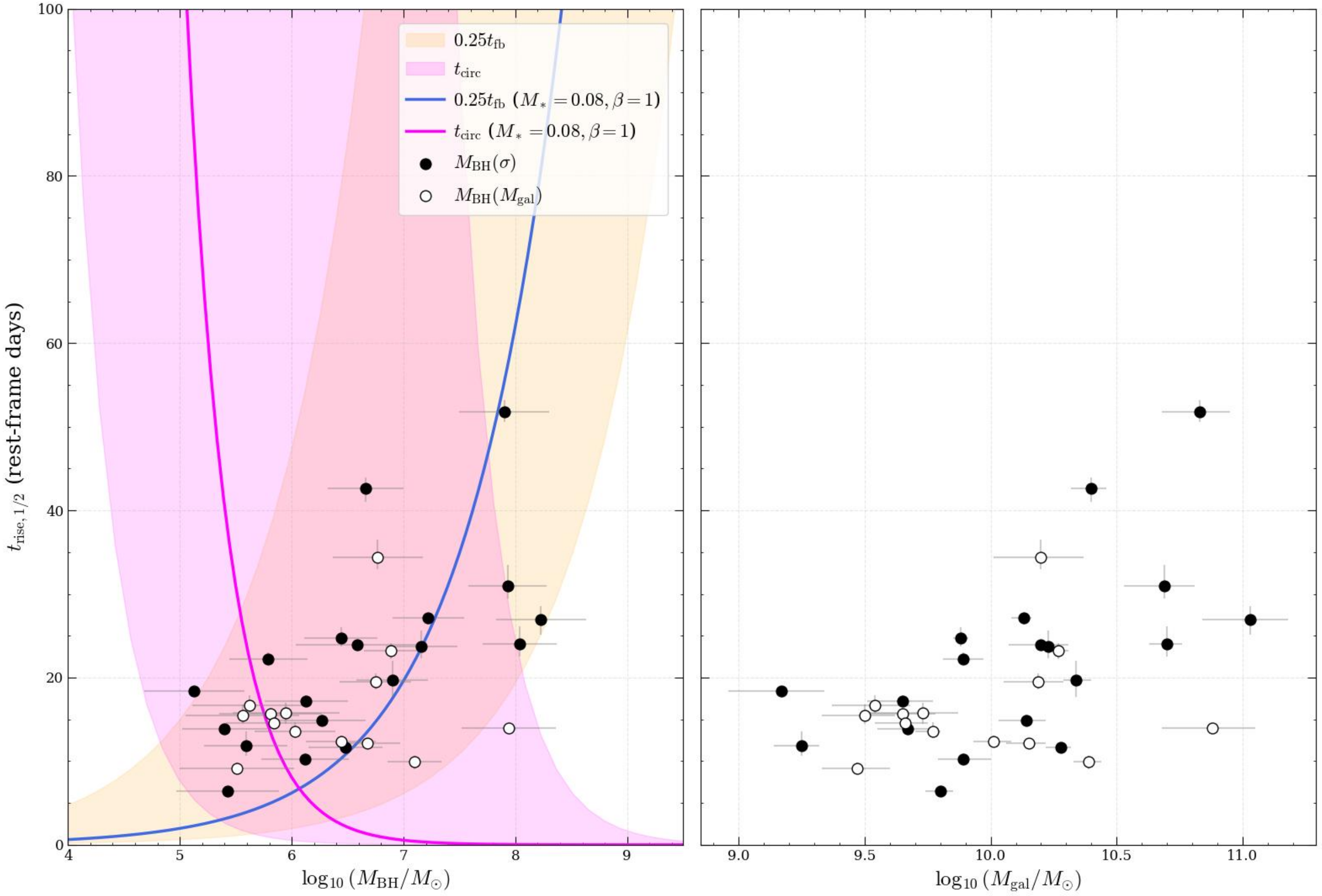
- **IMBH-TDE?**
- **Post-main sequence star SMBH-TDE**





J. Zhu et al. 2025

EP240222a: Far Away in the Upper Left (draft)



W. Li et al. arXiv: 2512.02147

WD-IMBH-TDEs: Fast Rise (~t_fb)

➢ **Dissipation channels（efficient for WD–IMBH–TDEs!）**

- **nozzle shock**

$$\epsilon_{\rm noz} \simeq \frac{\frac{1}{2}v_{\rm z,max}^2}{E_{\rm c}} \simeq 9.3 \times 10^{-6}\ \beta M_5^{-2/3} m_*^{2/3} \ll 1$$

- **self–intersection of the main stream**

$$\epsilon_{\rm self} \simeq \frac{\Delta E_0}{E_{\rm c}} \simeq 4.6 \times 10^{-4}\ e_0^2 \beta^2 M_5^{4/3} r_*^{-2} m_*^{2/3}$$

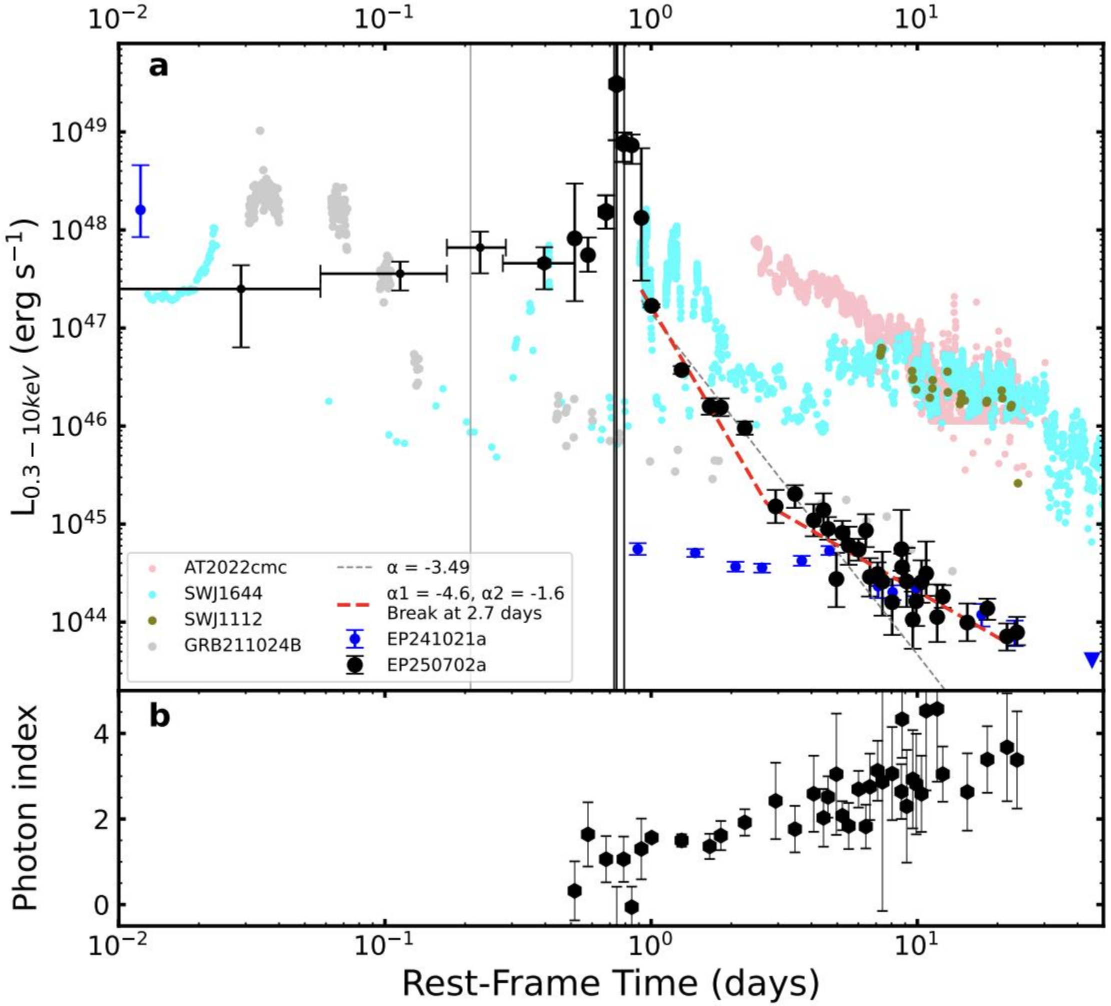
- **tail stream–main stream interaction**

$$\epsilon_{\rm tm}(t) \simeq \frac{\Delta E_{\rm tm}}{E_{\rm c}} \simeq \frac{\dot{M}_{\rm fb}(t) t_{\rm fb} E_0}{M_{\rm s}(t) E_{\rm c}}$$
$$\simeq 4.3 \times 10^{-2} \left(\frac{t}{t_{\rm fb}}\right)^{-n} \left[1 - \left(\frac{t}{t_{\rm fb}}\right)^{1-n}\right]^{-1} (n-1)(1+e_0)$$
$$\times M_5^{-1/3} m_*^{1/3} \times \begin{cases} \beta, & \beta \lesssim \beta_{\rm d} \\ \beta^{-1}, & \beta \gtrsim \beta_{\rm d} \end{cases} \ll 1$$

W. Li et al. arXiv: 2512.02147

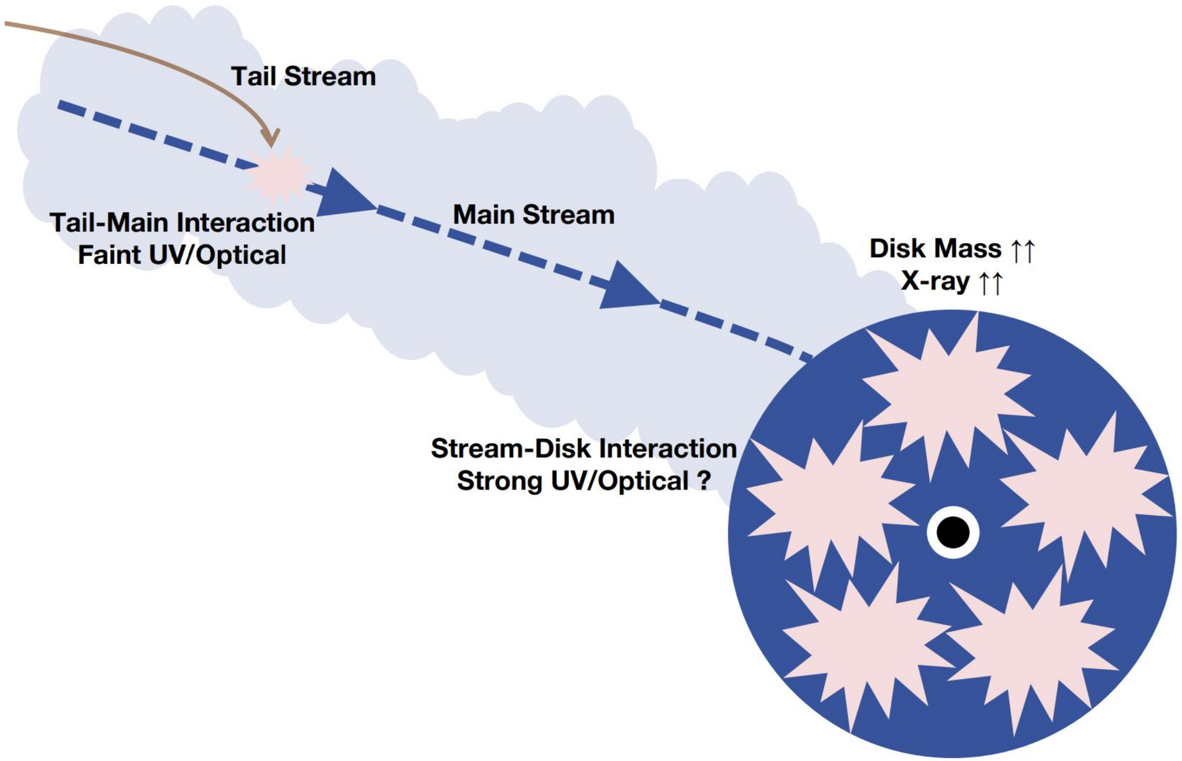
EP240222a Model

Jetted WD-IMBH-TDE Candidate



D. Li et al. (preprints)

Critical Moment Check



- **Momentum flux matching**
- **Stream–disk interaction**
- **Runaway circularization**

$$\sqrt{2}\dot{M}_{\rm s}(t_{\rm crit})\left(\frac{H}{R}\right)_{\rm d,early} \simeq \dot{M}_{\rm d}(t_{\rm crit})$$

$$\left(\frac{H}{R}\right)^3_{\rm d,early} \alpha_{\rm early} \simeq \frac{t_{\rm s,N}(t_{\rm crit})L_{\rm bol}(t_{\rm crit})}{2\sqrt{2}\pi\eta_{\rm mis}M_{\rm s}(t_{\rm crit})c^2}$$

$\log_{10}(t_{\rm acc,early}/t_{\rm dyn,c})$ posterior to $[3.5, 6.1]$

W. Li et al. arXiv: 2512.02147

Summary & Take Away Massage

- EP240222a is the first IMBH-TDE captured in real-time with multi-wavelength observations and spectroscopic confirmation.
- We reveals inefficient circularization, delayed stream-disk interaction and reprocessing in EP240222a.
- Our results indicate EP240222a was the disruption of a main-sequence star (M_star ≈ 0.4 M_sun; β ≈ 1.0).
- MS-IMBH-TDE (optical, X-rays): slow rise and/or quasi plateau.
- WD-IMBH-TDE (optical, X-rays): fast rise.

W. Li et al. arXiv: 2512.02147

Thanks!

中国科学技术大学
University of Science and Technology of China

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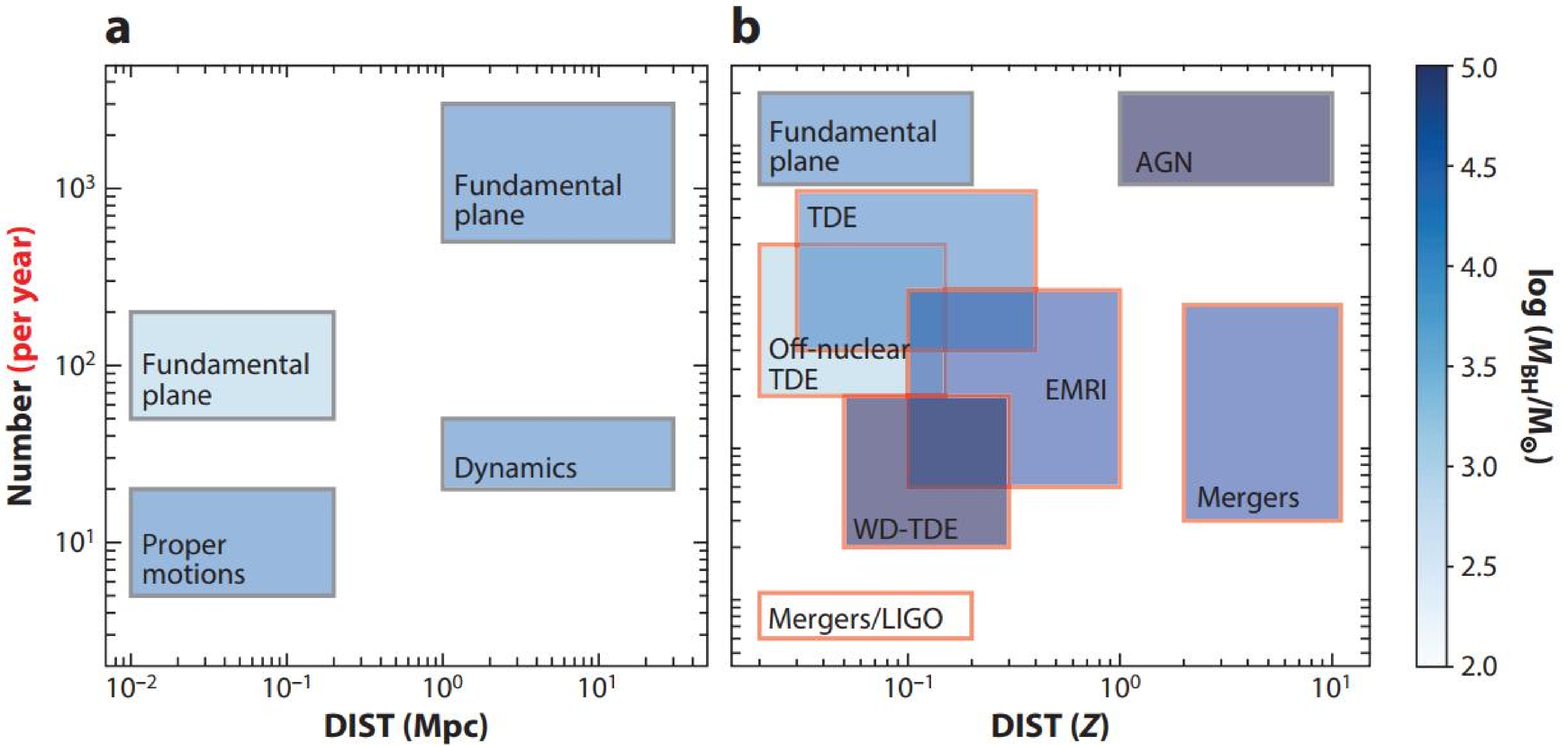
W. Li et al. arXiv: 2512.02147

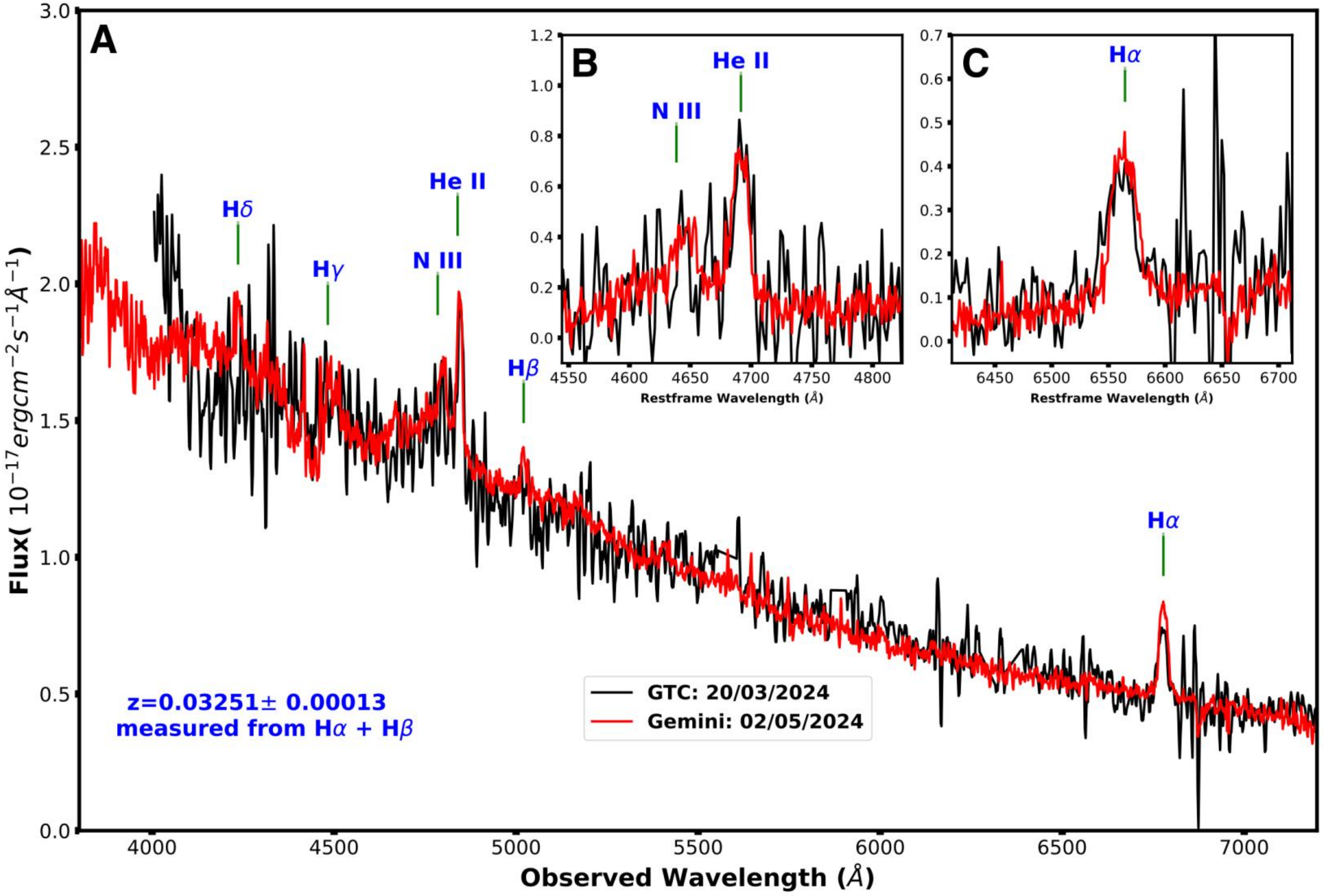
Backup

Using TDE to Find IMBH



J. Greene et al. 2020

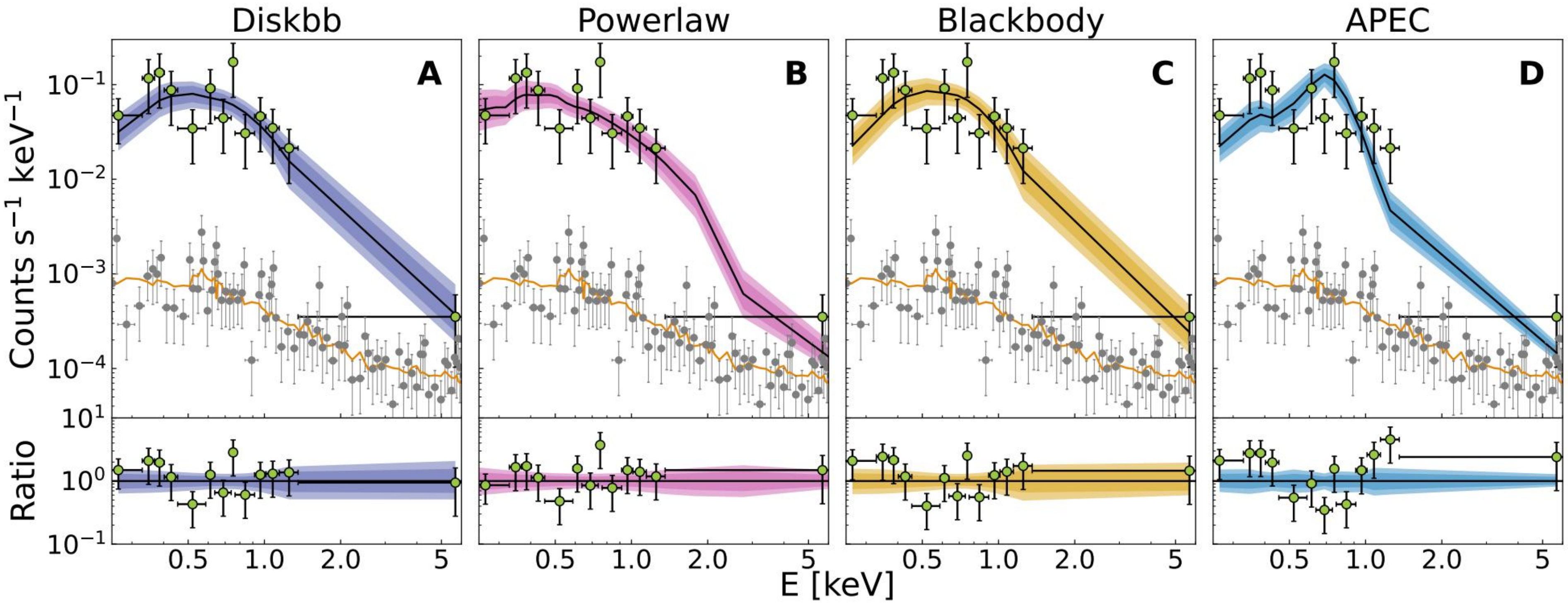
Spectrum: Not as broad as typical TDE (1e4 km/s)



- **z = 0.03251**
- **Typical TDE Lines**
- **No [O III] 4959/5007**
  - AGN ×
- **Kinematics?**
  - SMBH–TDE ×?
  - IMBH–TDE ?
- **Electron Scattering?**
  - SMBH–TDE ✓?
  - IMBH–TDE ?

C. Jin et al. arXiv: 2501.09580

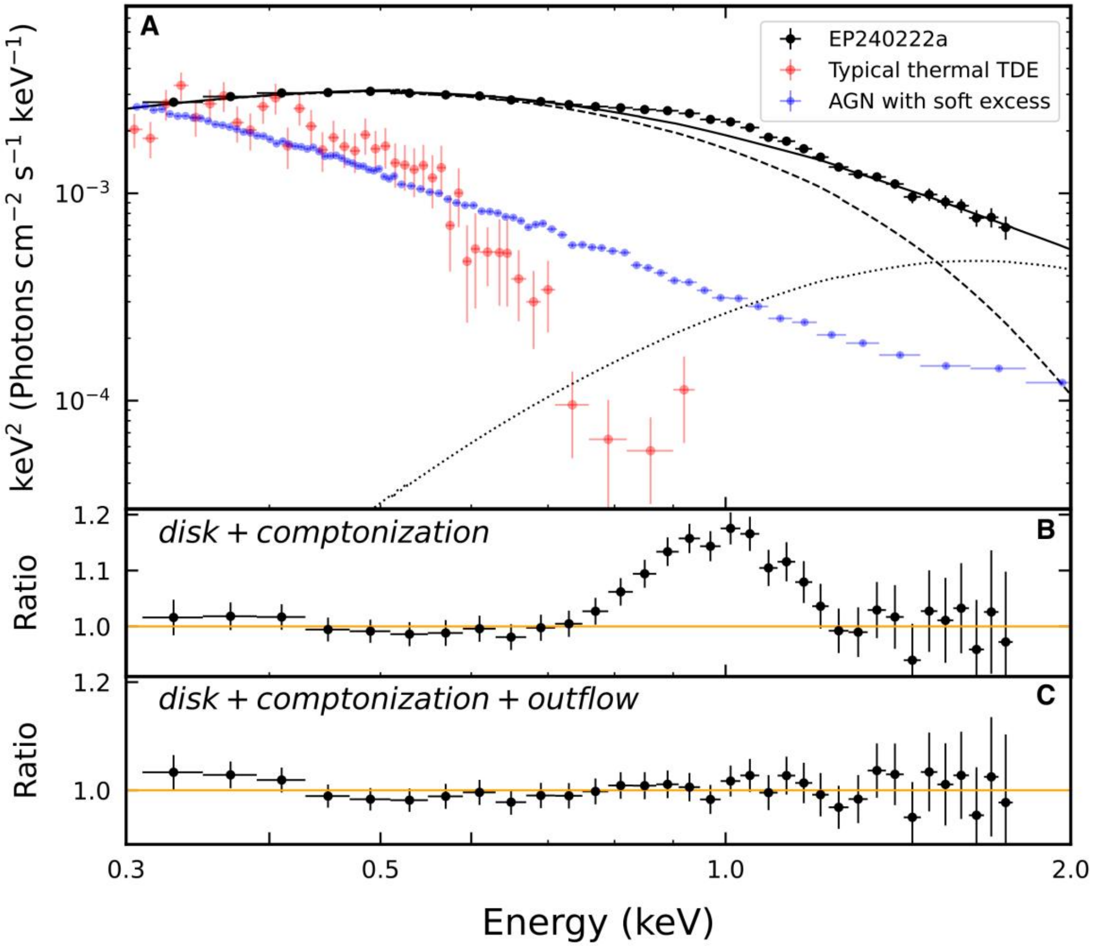
Early X-ray SED Fitting



C. Jin et al. arXiv: 2501.09580

EP240222a Model

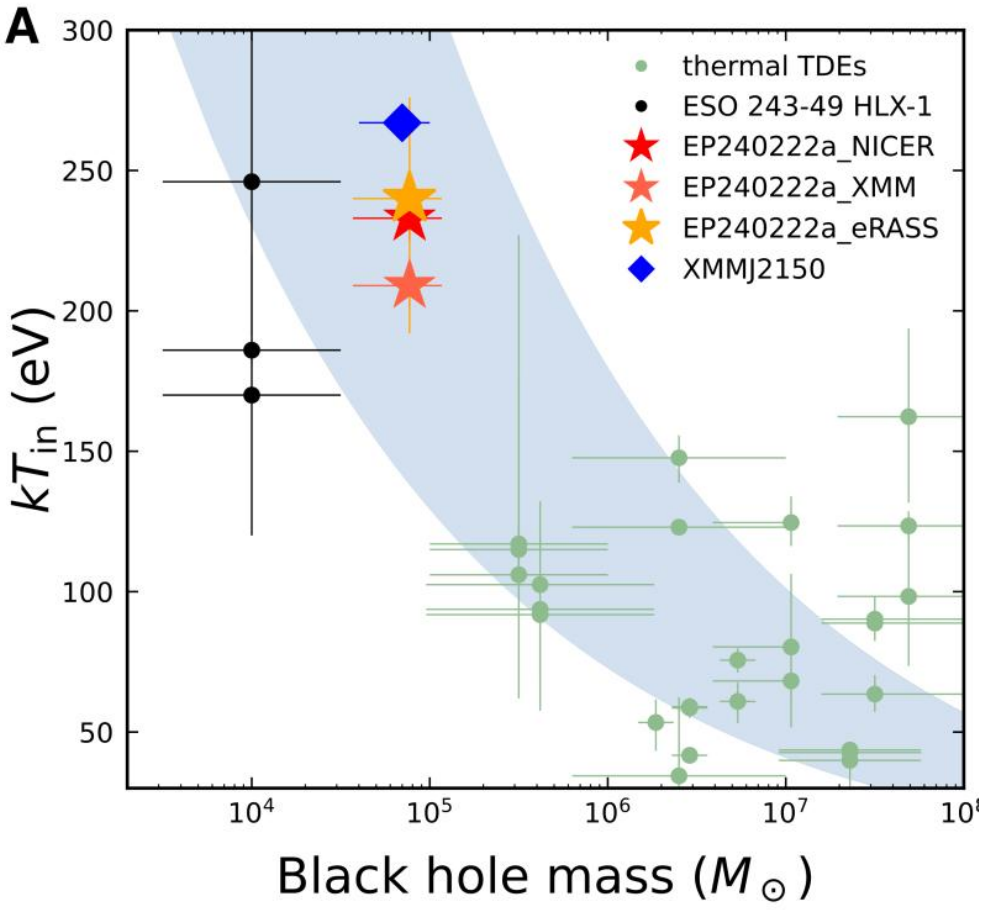
X-ray SED

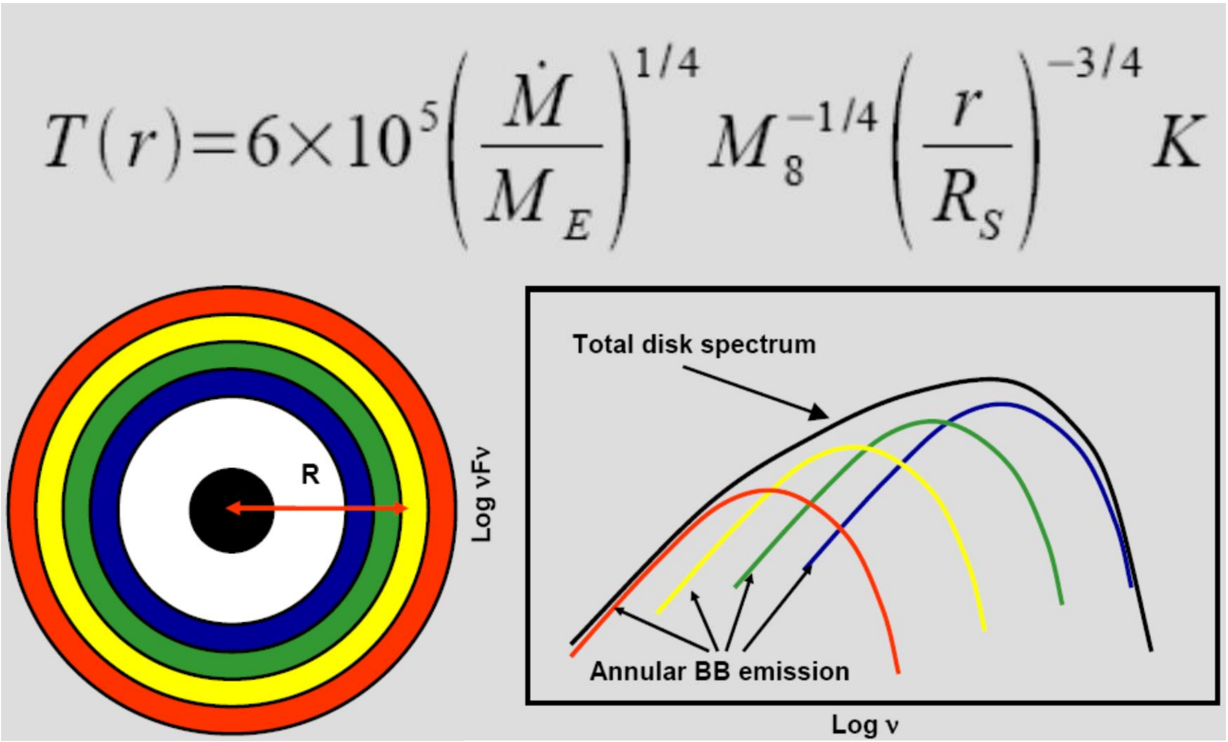


- ➢**Harder than typical termal TDE**
  - **IMBH-TDE**
- ➢**Different from AGN**
- ➢**Outflow signature @ 1keV**
  - **0.34c (very fast)**
  - **why no radio?**

C. Jin et al. arXiv: 2501.09580

BH Mass vs Inner Disk Temerature

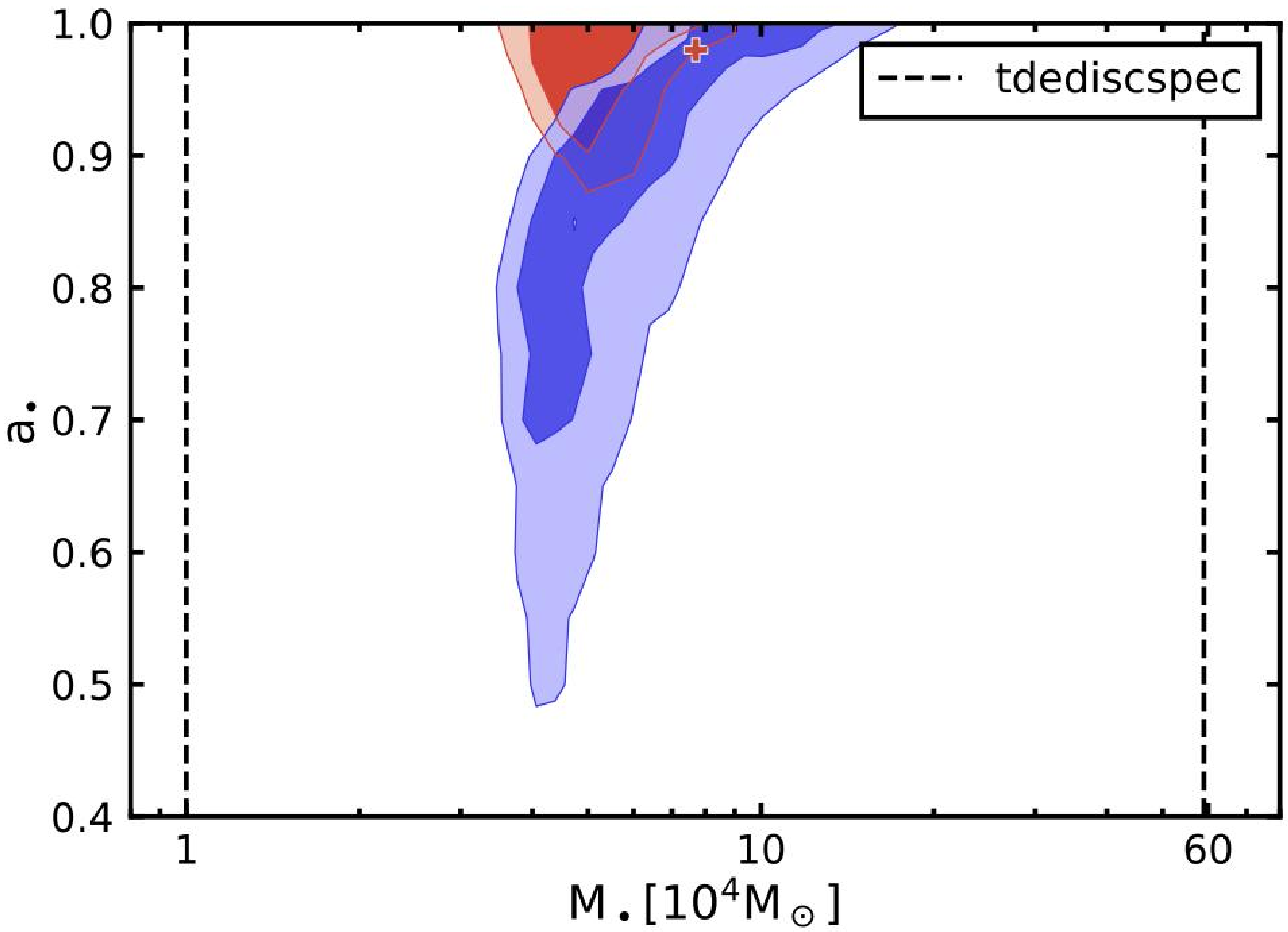




C. Jin et al. arXiv: 2501.09580

credit: J. Wang

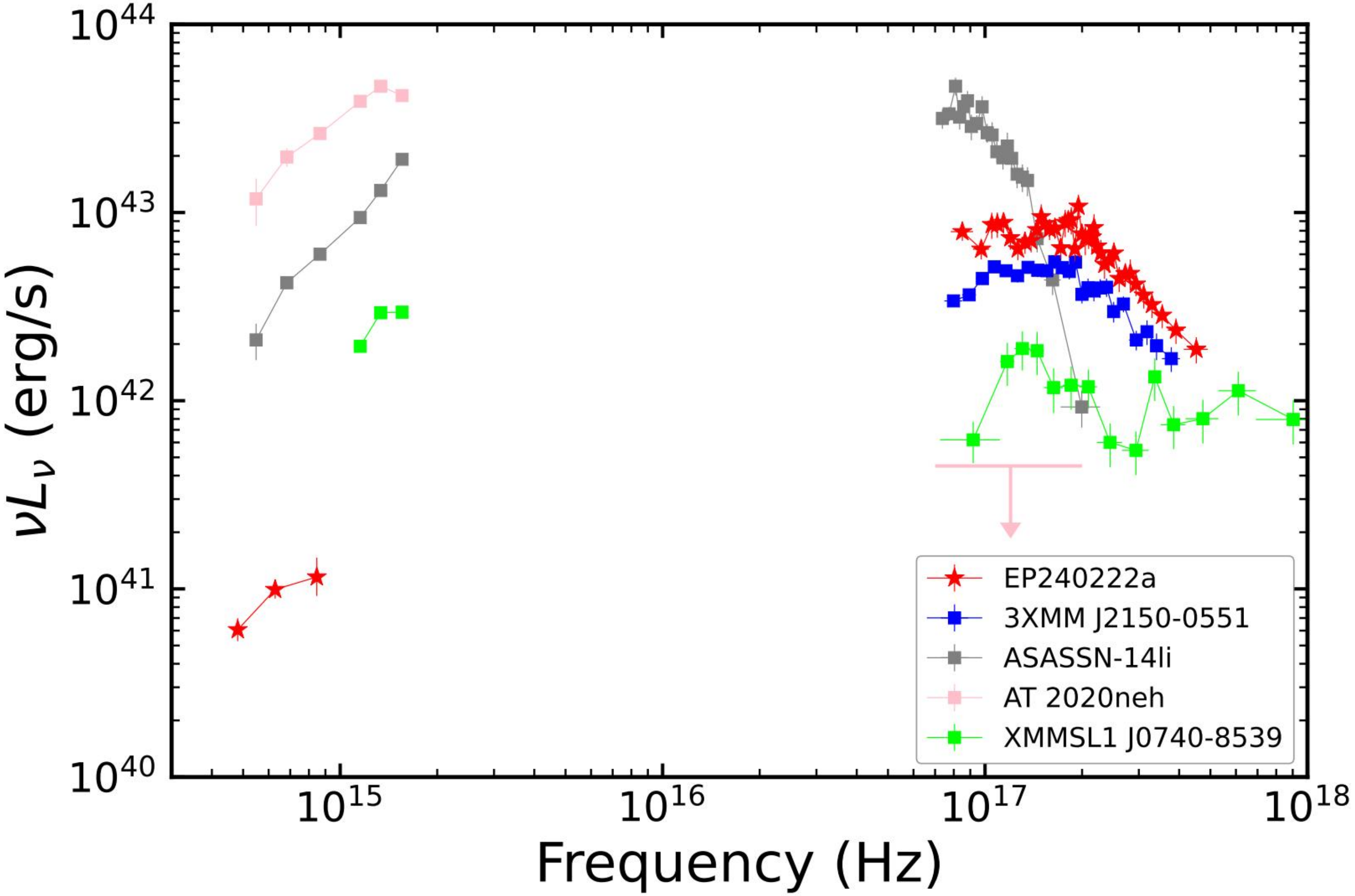
BH Mass & Spin (Sixiang Wen)



- ➢ **BH mass**
  - **luminosity**
  - **spectrum shape**
- ✓ **BH spin –> ISCO**
  - **spectrum shape**
- ➢ **Multi-epoch –> better**

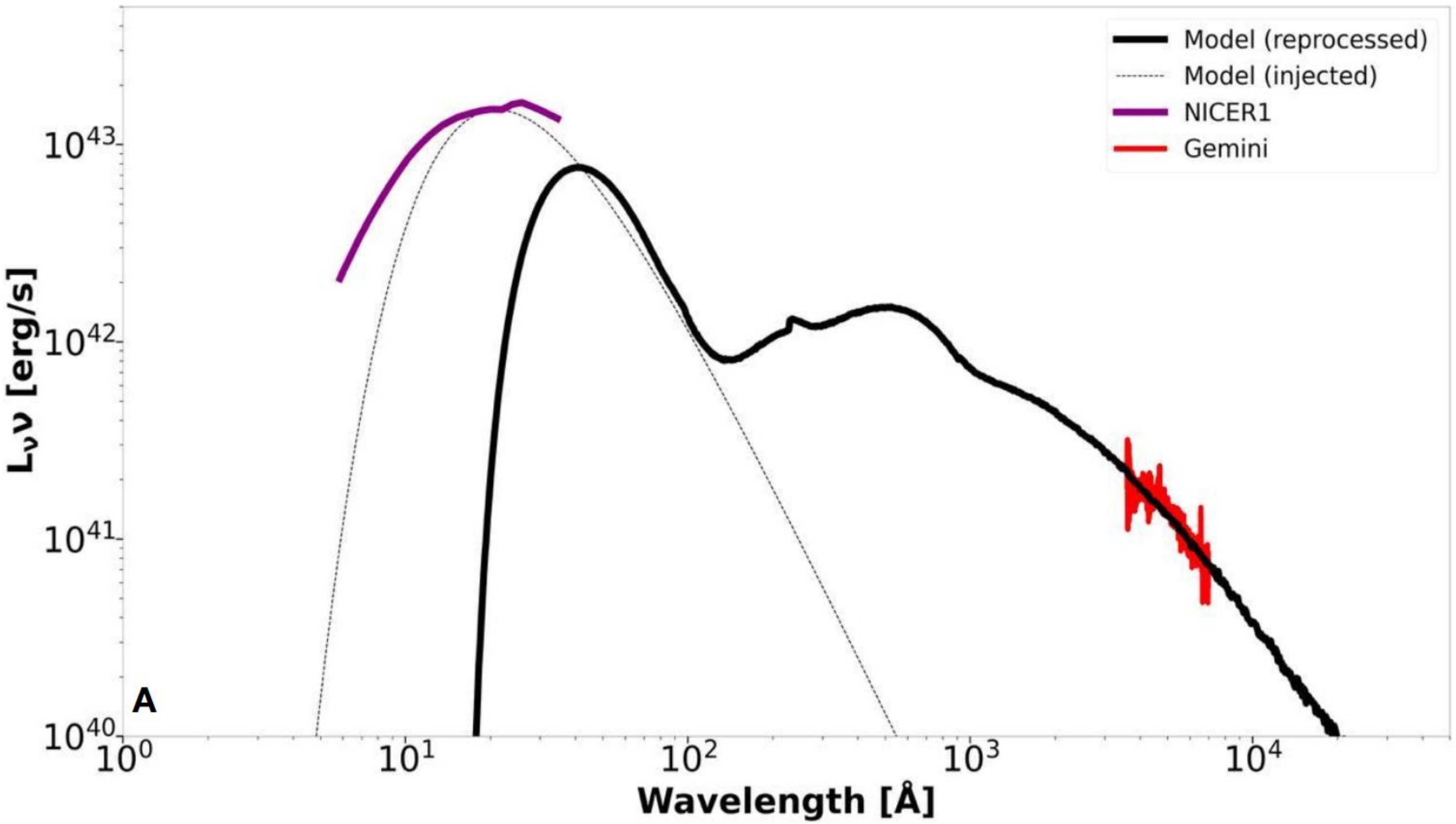
C. Jin et al. arXiv: 2501.09580

Broad Band SED



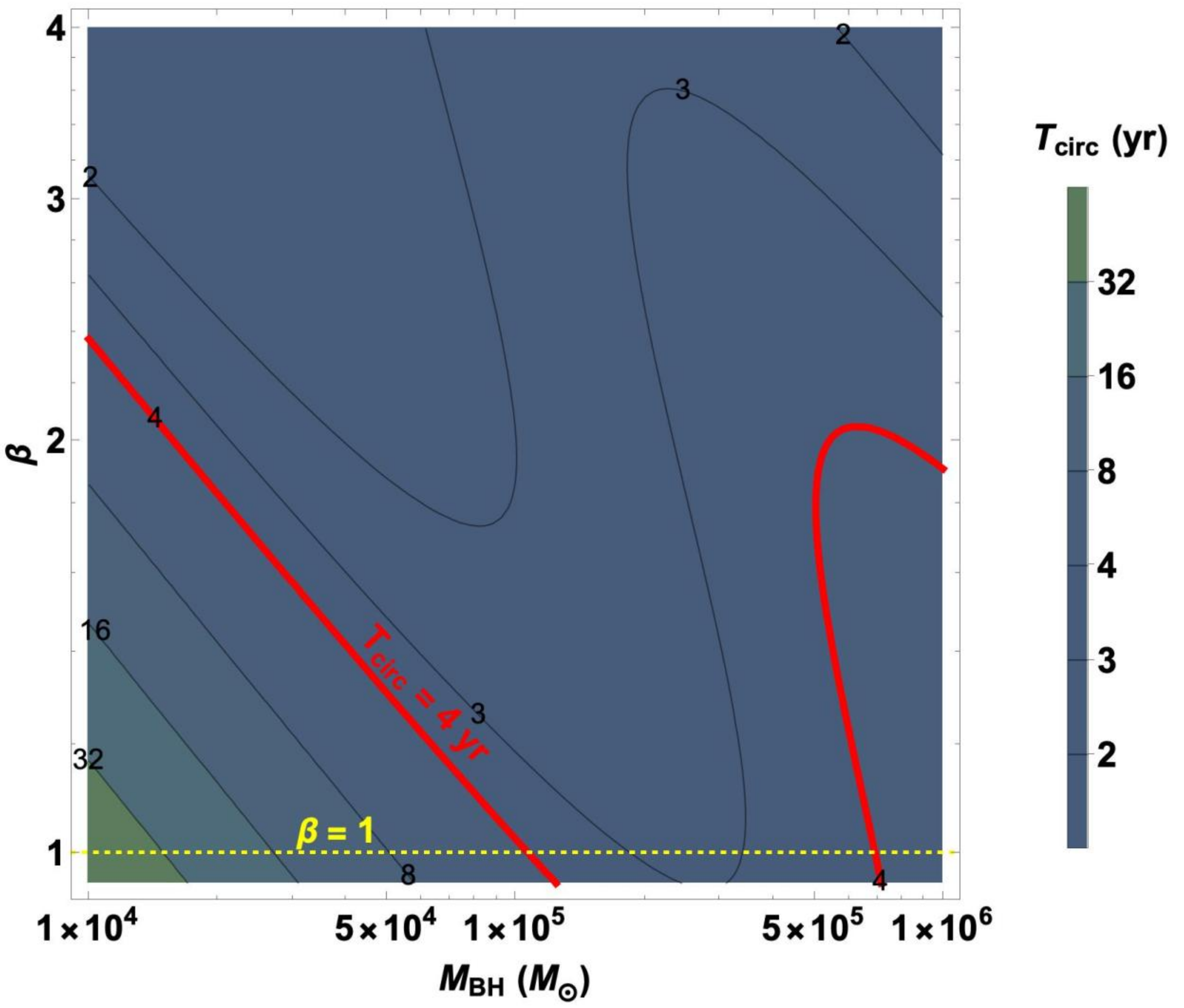
C. Jin et al. arXiv: 2501.09580

Reprocessing Model



C. Jin et al. arXiv: 2501.09580

Circularization Timescale

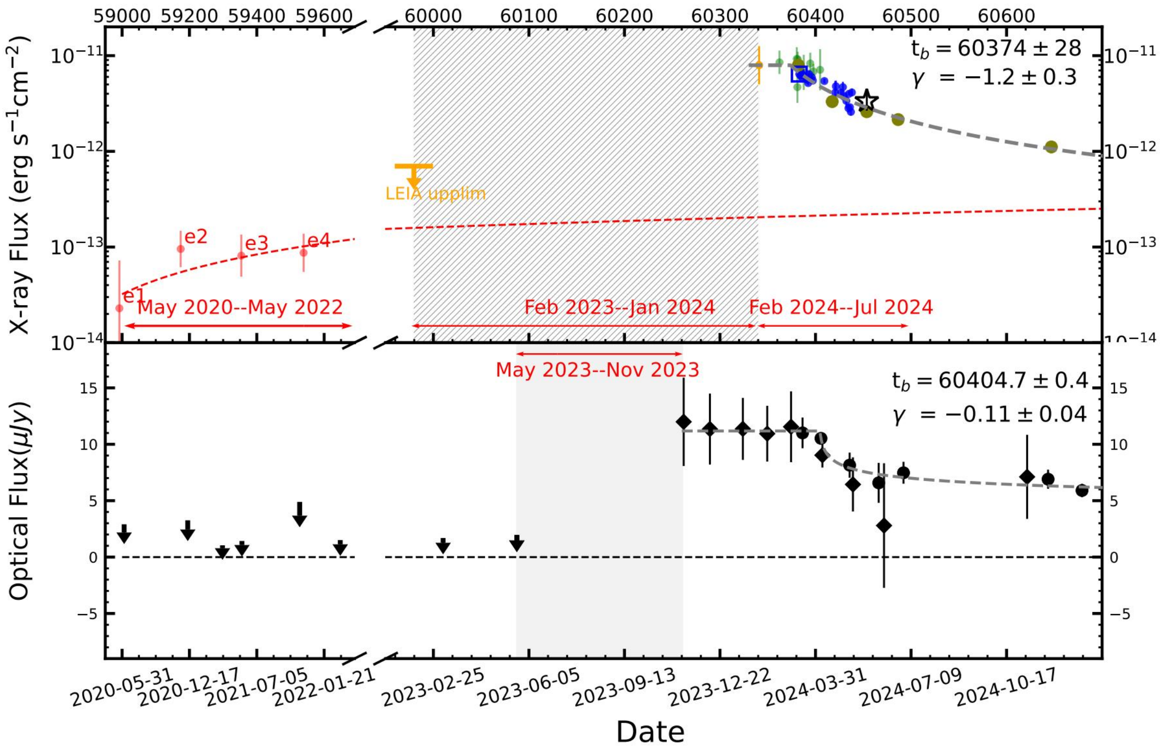


C. Jin et al. arXiv: 2501.09580

Other Possibilities?

➢ **X-rays:**

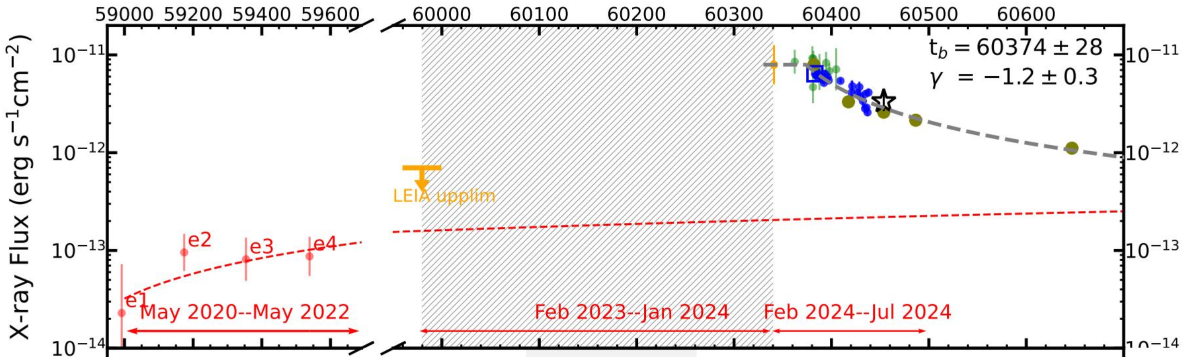
- **AGN –> not likely**
- **rpTDE –> ?**
- **pTDE + fTDE –> ?**
- **two distinct TDEs –> not likely**

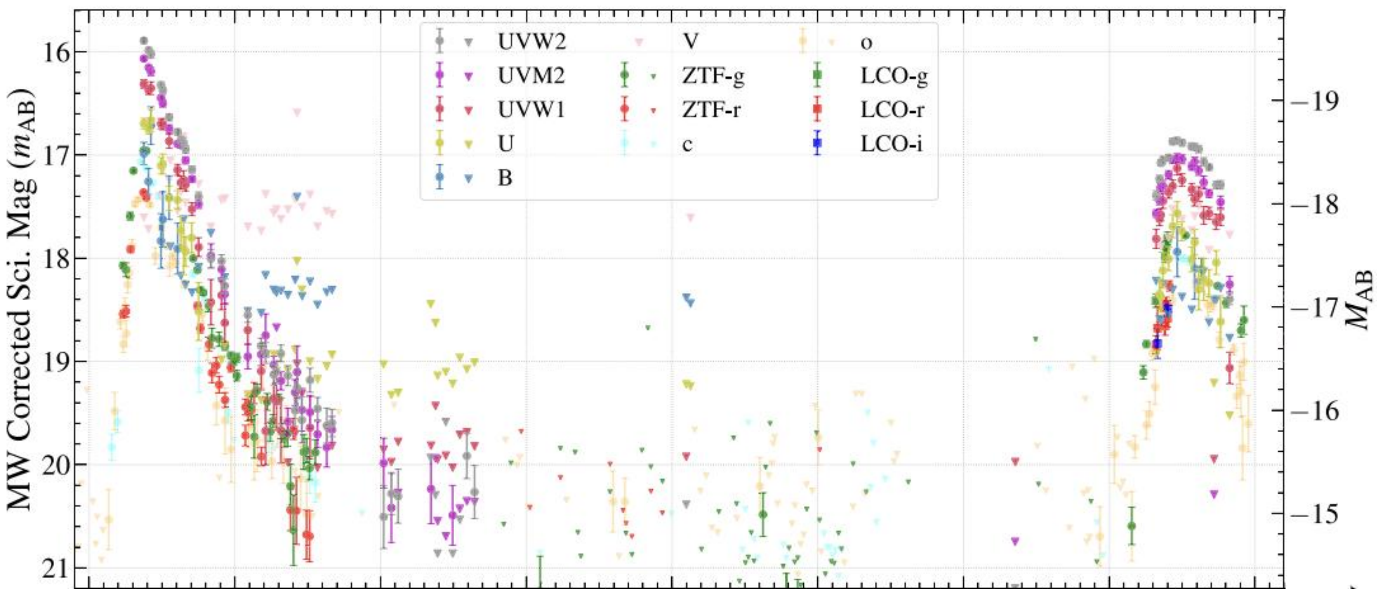


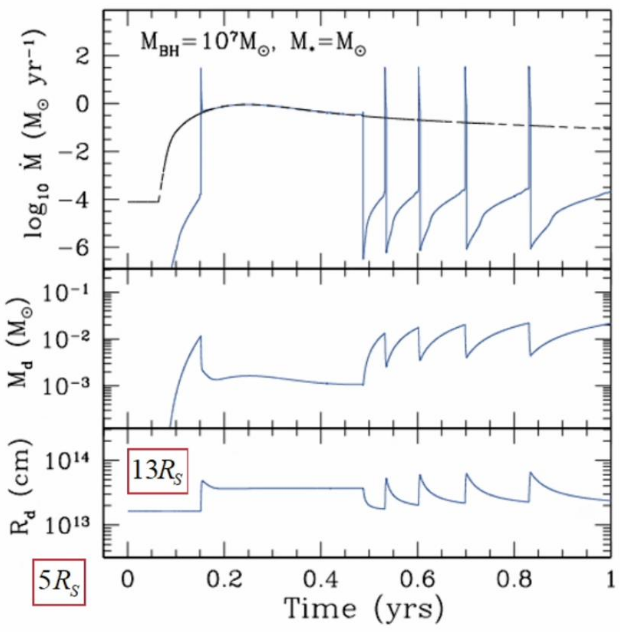
W. Li et al. arXiv: 2512.02147

EP240222a Model

Other Possibilities?







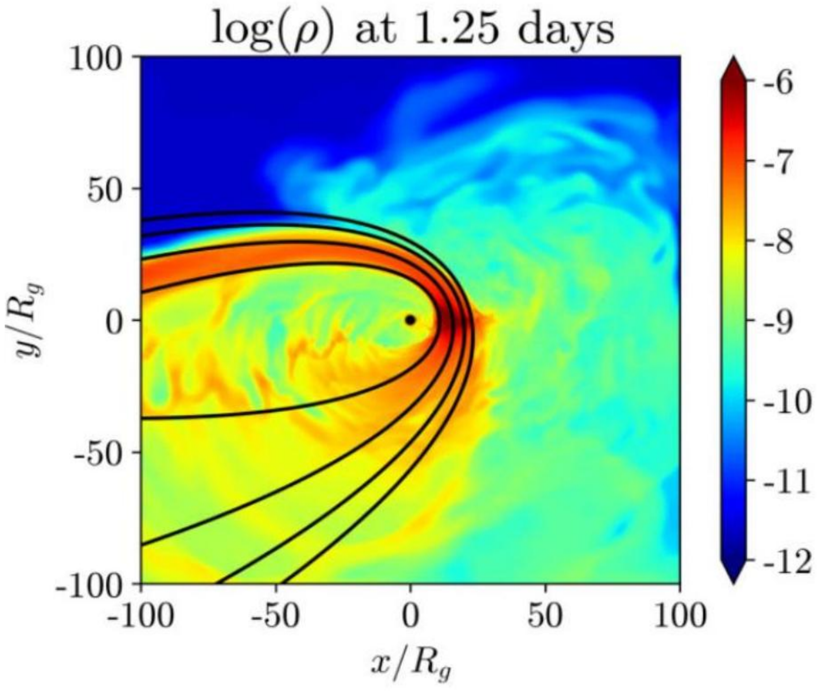
Repeating Partial TDE
Z. Lin et al. 2024

Radiation Pressure Instability
A. Piro & B. Mockler 2024

R_eff >> R_star? In Fact R_eff Should ~R_p

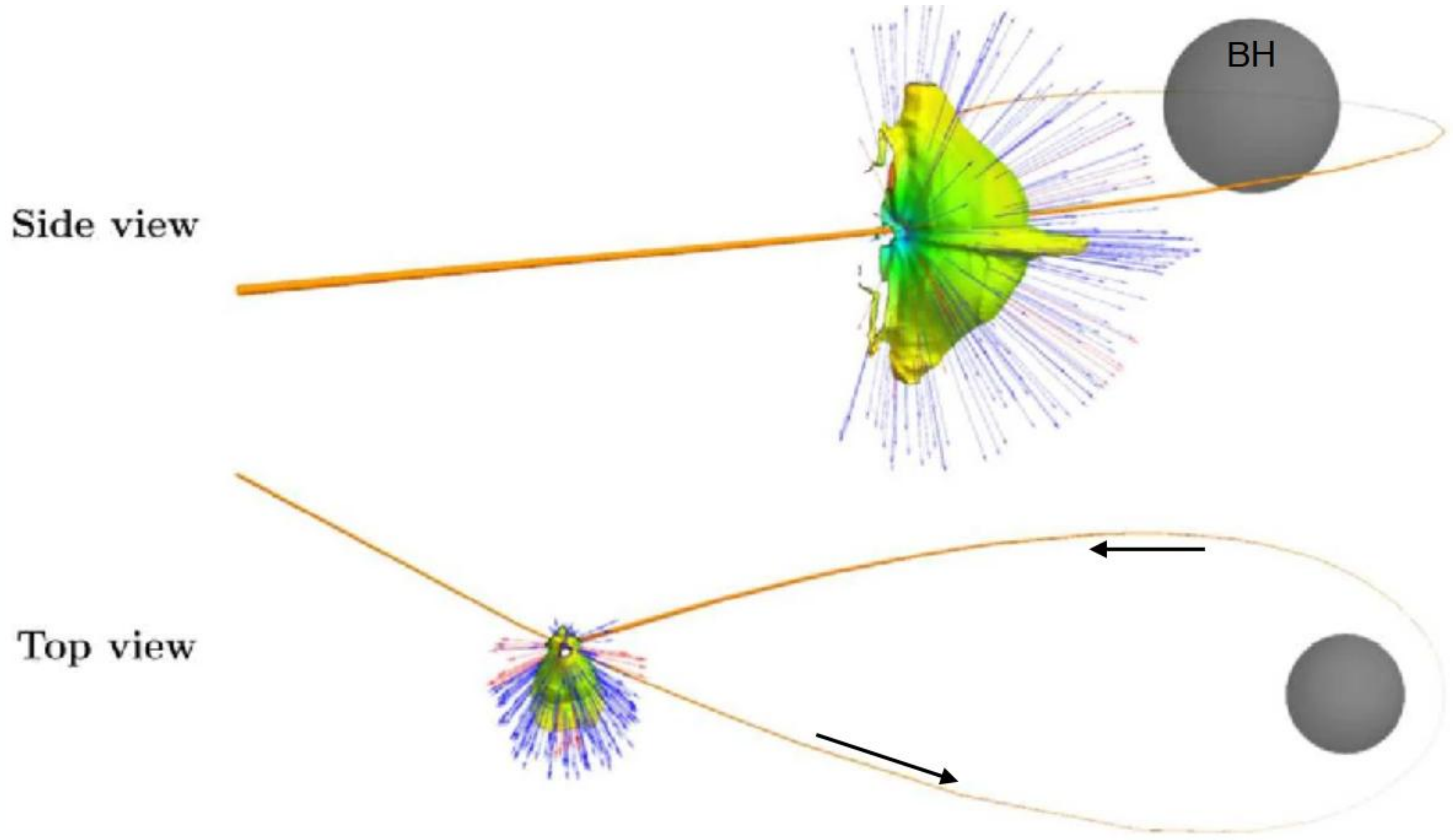
$$t_{\rm diff} \simeq \kappa_{\rm es} \frac{\Delta M}{4\pi c a_{\rm s}} \left(\frac{h_{\rm s}}{w_{\rm s}} \right)$$
$$\simeq 1.4 \times 10^{-2} \left(\frac{\Delta M}{0.01\ M_\odot} \right) \left(\frac{h_{\rm s}}{w_{\rm s}} \right)$$
$$\times \left(\frac{a_{\rm s}}{a_0} \right)^{-5/2} \beta^5 M_6^{-7/6} r_*^{-5/2} m_*^{5/3}\ t_{\rm s}$$

Diffusion Timescale
J. Chen & R. Shen 2021



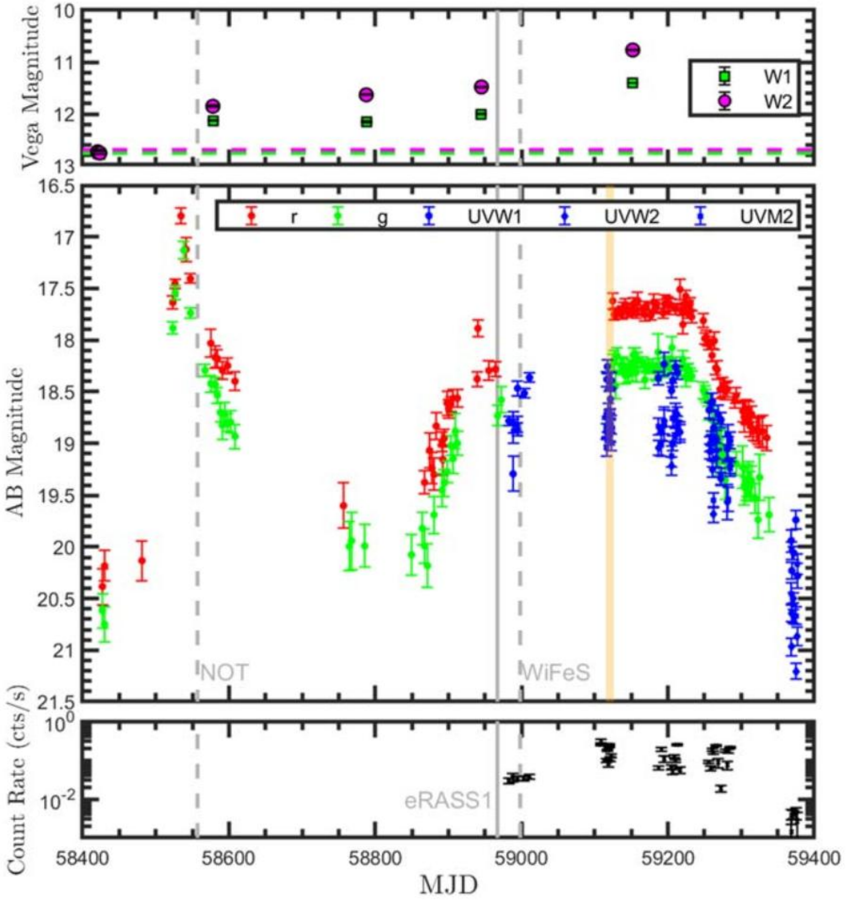
Differential Precession
Z. Andalman et al. 2022

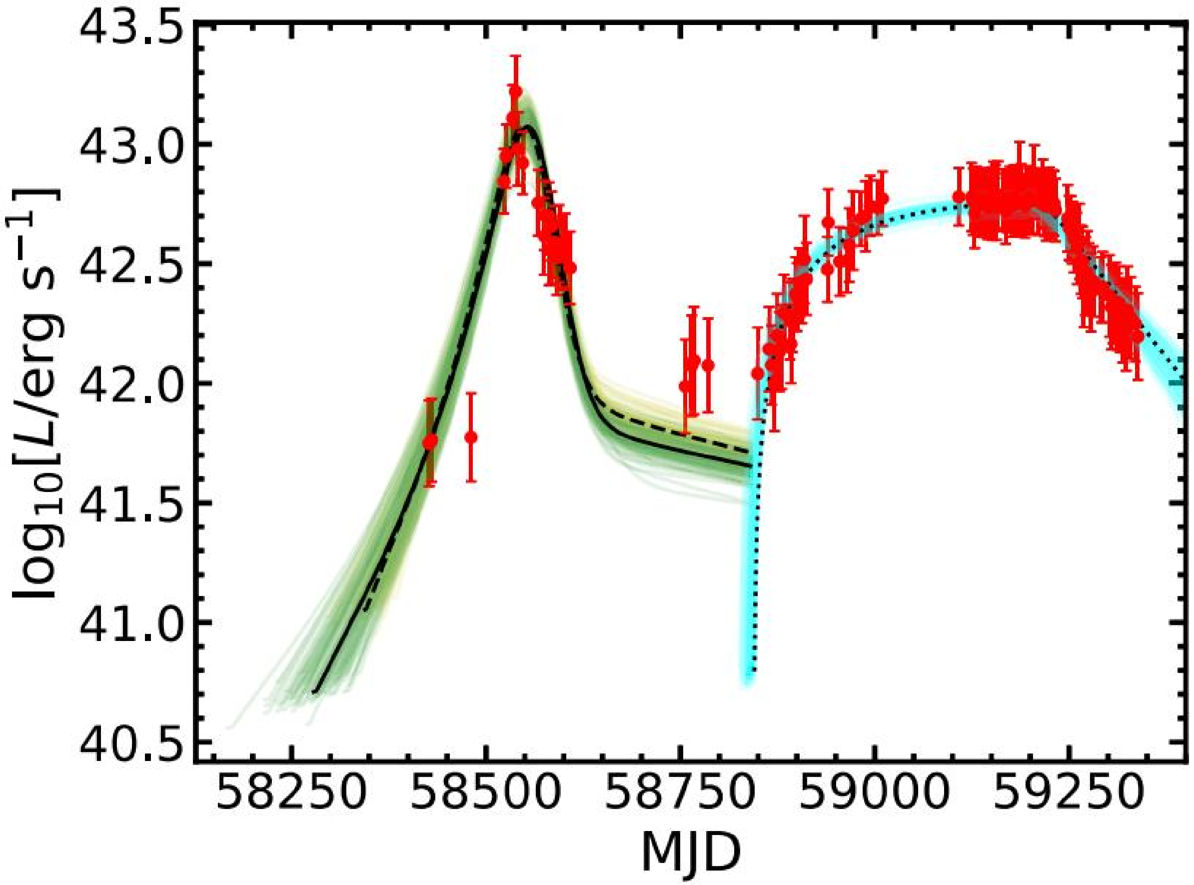
Self-Intersection Induce Outflow



Y. Jiang et al. 2014

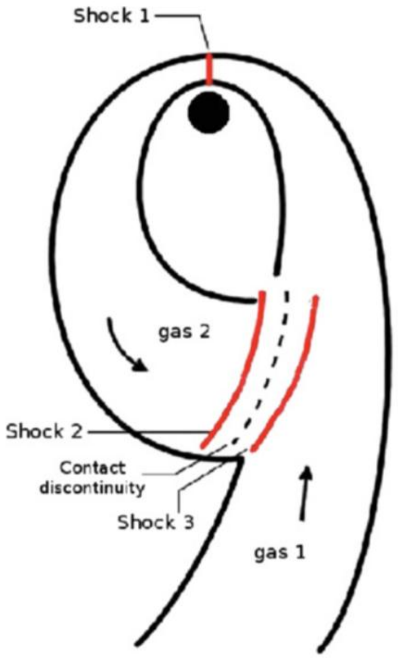
Two Steps Model

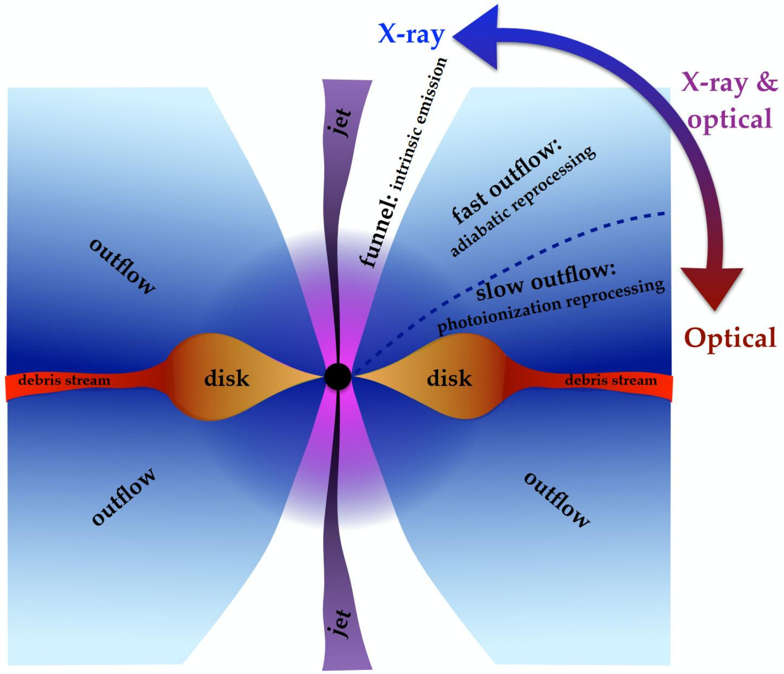


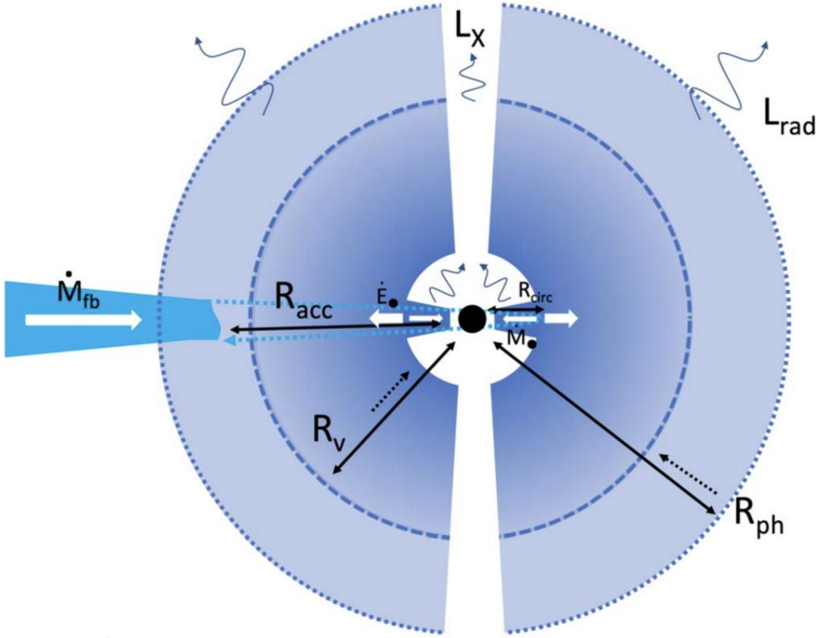


J. Chen & R. Shen 2018

Optical Emission Model





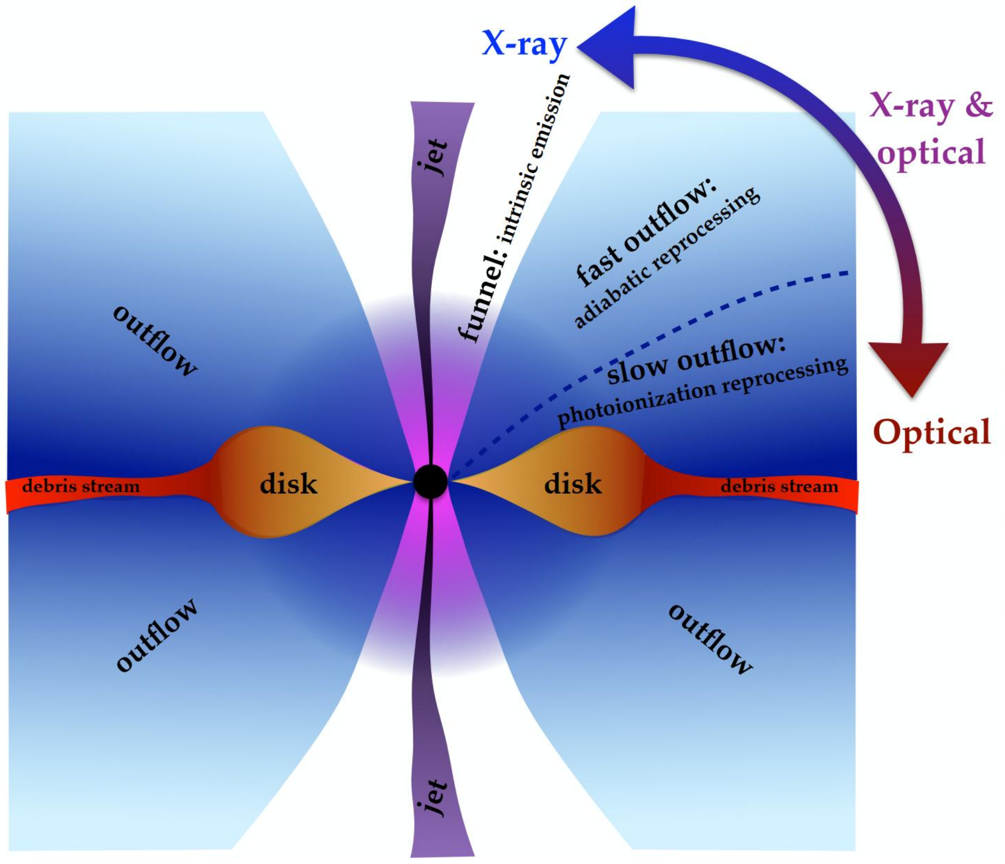


T. Piran et al. 2015

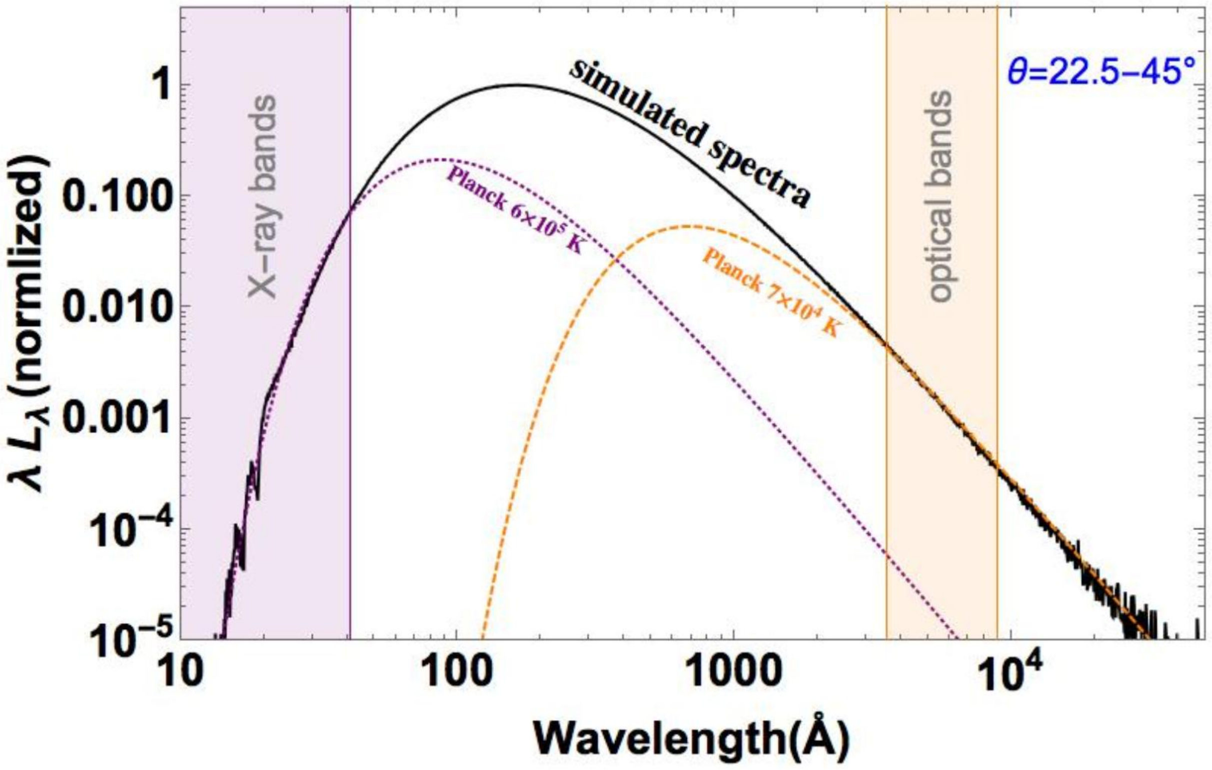
L. Dai et al. 2018

B. Metzger 2022

Unified Model

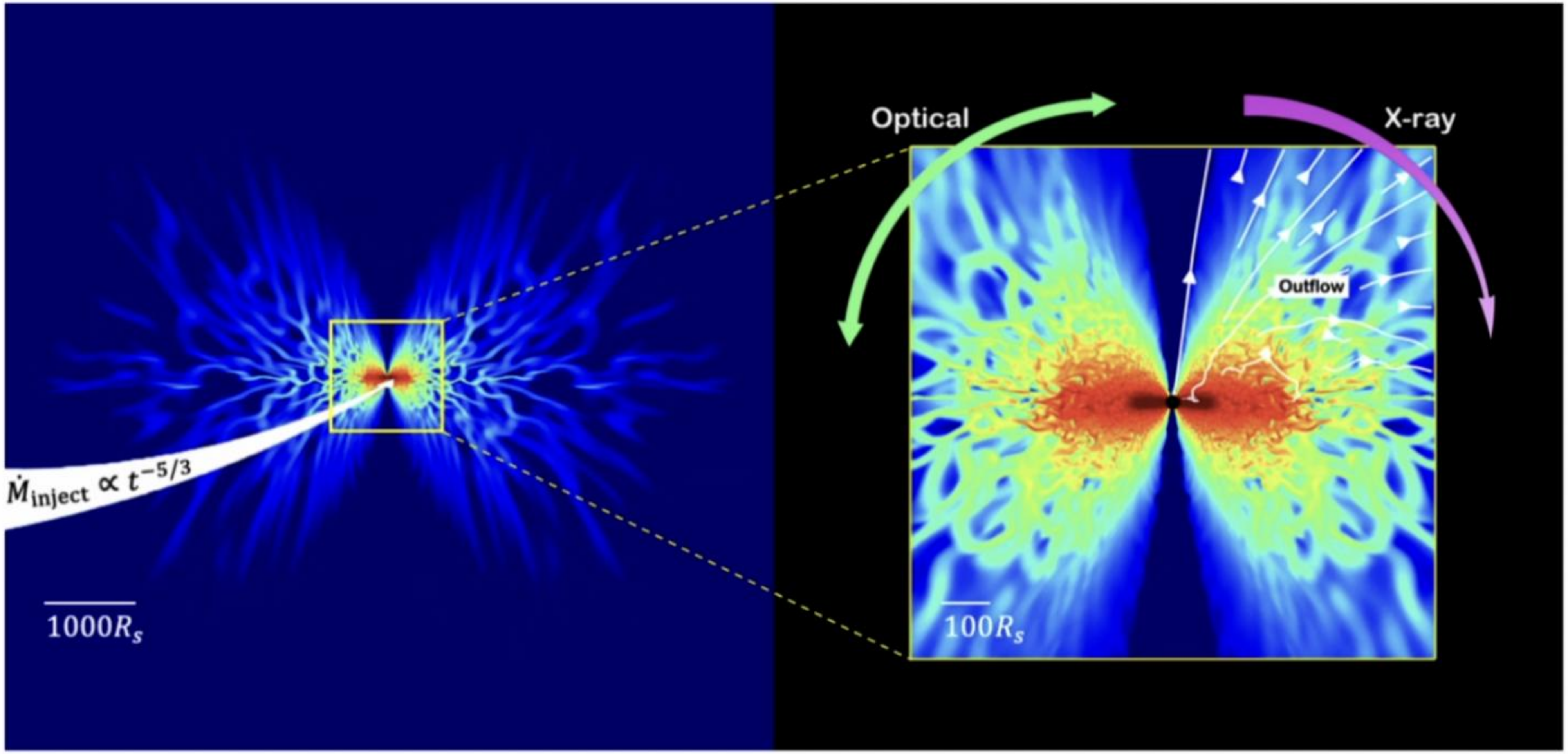


Bighter than black body assumption in UV



L. Dai et al. 2018

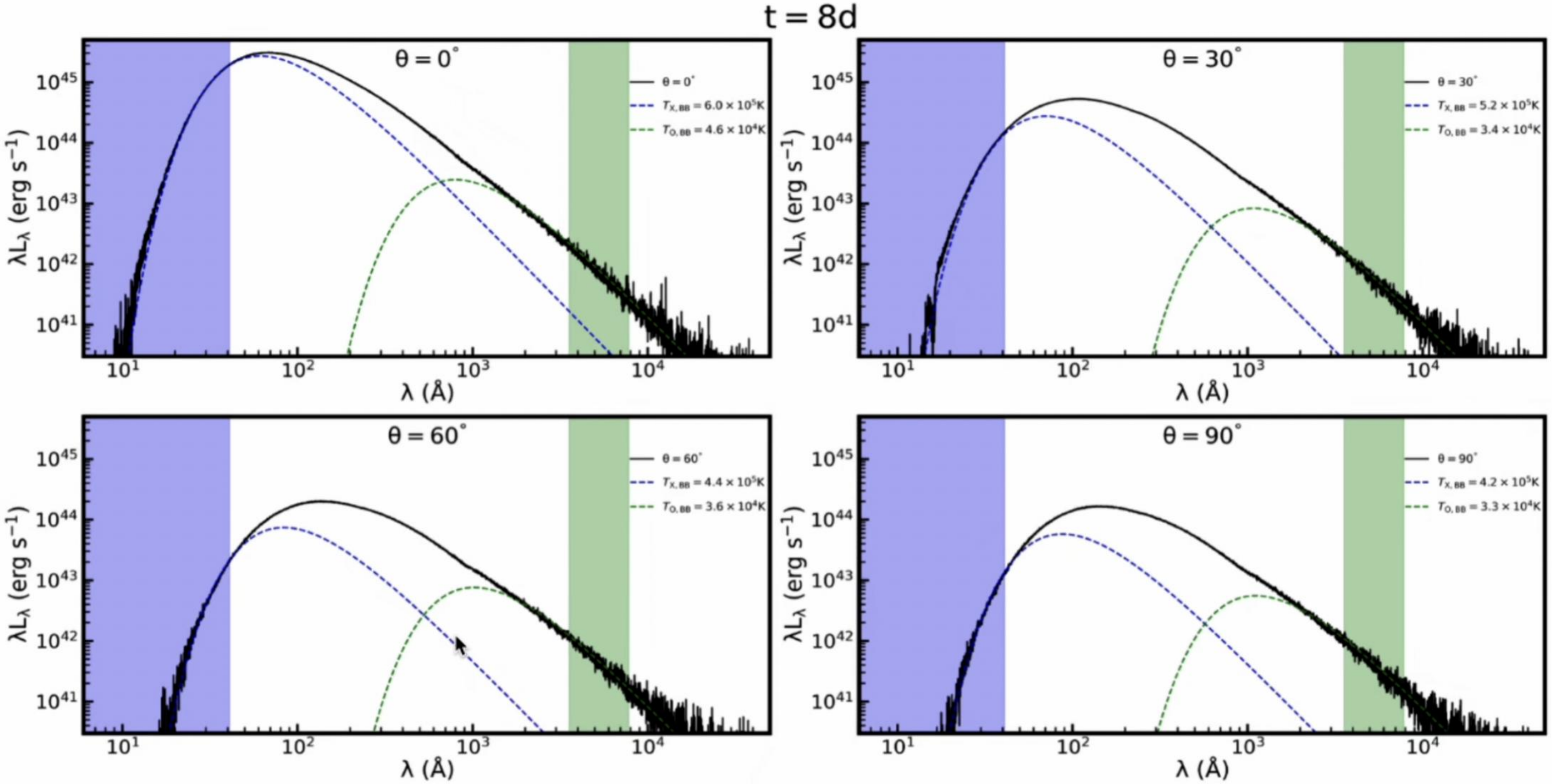
Unified Model



E. Qiao et al. 2025

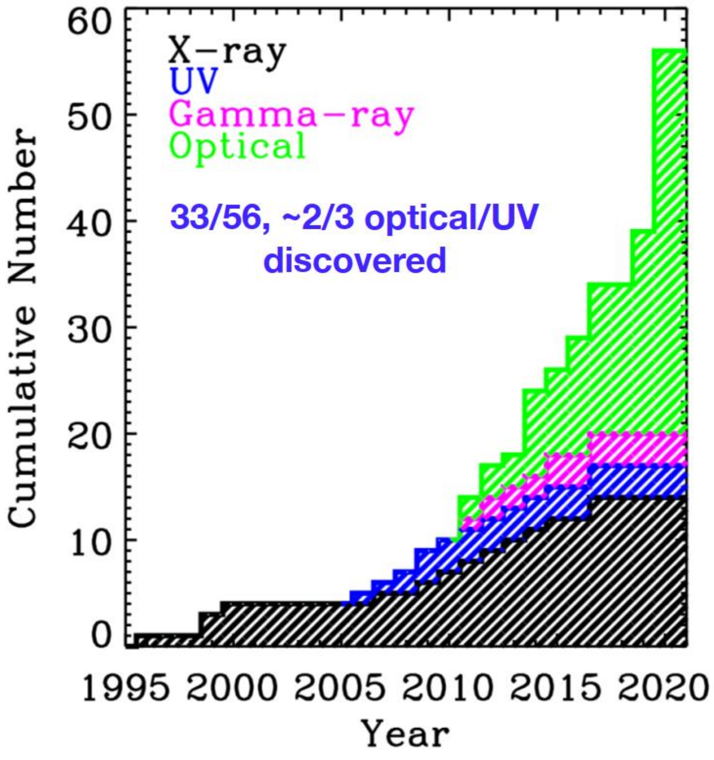
Unified Model

Two blackbody fitting to the spectra



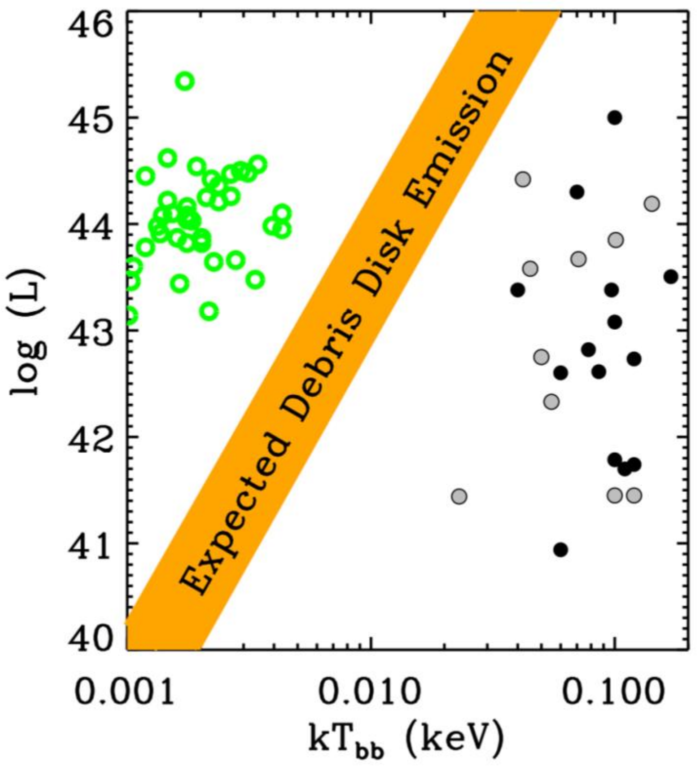
E. Qiao et al. 2025

Optical vs X-ray



Gezari 2021, ARA&A, 59, 21

Thermal TDEs



$$L = 4\pi R^2 \sigma T_{\rm bb}^4$$

credit: E. Qiao